



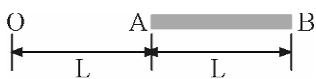
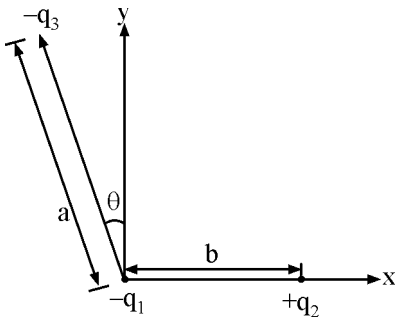
IIT JEE | MEDICAL | FOUNDATION

Final Step | Part - B | Physics

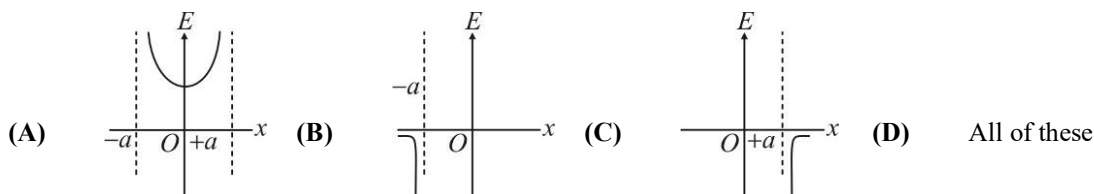
Electrostatics

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED '*' MAY HAVE MORE THAN ONE CORRECT OPTION.

- A simple pendulum of length l has a bob of mass m , with a charge q on it. A vertical sheet of charge, with surface charge density σ passes through the points of suspension. At equilibrium, the string makes an angle θ with the vertical, then :
 (A) $\tan \theta = \frac{\sigma q}{2\epsilon_0 mg}$ (B) $\tan \theta = \frac{\sigma q}{\epsilon_0 mg}$ (C) $\cot \theta = \frac{\sigma q}{2\epsilon_0 mg}$ (D) $\cot \theta = \frac{\sigma q}{\epsilon_0 mg}$
- A charged particle of mass m and charge q is released from rest in an electric field of constant magnitude E . the kinetic energy of the particle after time t will be :
 (A) $\frac{2E^2 t^2}{mq}$ (B) $\frac{Eq^2 m}{2t^2}$ (C) $\frac{E^2 q^2 t^2}{2m}$ (D) $\frac{Eqm}{2t}$
- A charge Q is distributed over two concentric hollow spheres of radius r and R ($> r$) such that the surface densities are equal. The potential at the common centre is:
 (A) $\frac{Q(R^2 + r^2)}{4\pi\epsilon_0(R + r)}$ (B) $\frac{Q}{R + r}$ (C) Zero (D) $\frac{Q(R + r)}{4\pi\epsilon_0(R^2 + r^2)}$
- * If $\oint_s \vec{E} \cdot d\vec{S} = 0$ over a surface, then :
 (A) The electric field inside the surface and on it is zero
 (B) The electric field inside the surface is necessarily uniform
 (C) The number of flux lines entering the surface must be equal to the number of flux lines leaving it
 (D) Net charge, if any, must necessarily be outside the surface
- The ratio of the forces between two small spheres with same charges when they are in air to when they are in a medium of dielectric constant K is :
 (A) $1 : K$ (B) $K : 1$ (C) $1 : K^2$ (D) $K^2 : 1$
- Two point charges $2q$ and $8q$ are placed at a distance r apart. Where should a third charge $-q$ be placed between them so that the electrical potential energy of the system is a minimum :
 (A) At a distance of $r/3$ from $2q$ (B) At a distance of $2r/3$ from $2q$
 (C) At a distance of $r/16$ from $2q$ (D) None of these
- A spherical conductor A of radius r placed concentrically inside a conducting shell B of radius R ($R > r$). A charge Q is given to A, and then A is joined to B by a metal wire. The charge flowing from A and B will be :
 (A) $Q\left(\frac{R}{R+r}\right)$ (B) $Q\left(\frac{r}{R+r}\right)$ (C) Q (D) Zero

8. A charge Q is uniformly distributed over a long rod AB of length L as shown in the figure. The electric potential at the point O lying at distance L from the end A is :
- 
- (A) $\frac{Q}{8\pi\epsilon_0 L}$ (B) $\frac{3Q}{4\pi\epsilon_0 L}$ (C) $\frac{Q}{4\pi\epsilon_0 L \ln 2}$ (D) $\frac{Q \ln 2}{4\pi\epsilon_0 L}$
- *9. If a conductor has a potential $V \neq 0$ and there are no charges anywhere else outside, then :
- (A) There must be charges on the surface or inside itself
 (B) There cannot be any charge in the body of the conductor
 (C) There must be charges only on the surface (D) There must be charges inside the surface
10. A non-conducting ring of radius R has charge Q distributed uniformly over it. If rotates with an angular velocity ω , the equivalent current will be :
- (A) Zero (B) $Q\omega$ (C) $Q \frac{\omega}{2\pi}$ (D) $Q \frac{\omega}{2\pi R}$
11. Four equal charge Q are placed at the four corners of a square of side a . the work done in removing a charge $-Q$ from the centre of the square to infinity is :
- (A) Zero (B) $\frac{\sqrt{2}Q^2}{4\pi\epsilon_0 a}$ (C) $\frac{\sqrt{2}Q^2}{\pi\epsilon_0 a}$ (D) $\frac{Q^2}{2\pi\epsilon_0 a}$
12. Two charges each equal to q , are kept at $x = -a$ and $x = a$ on the x -axis. A particle of mass m and charge $q_0 = q/2$ is placed at the origin. If charge q_0 is given, a small displacement y ($y \ll a$) along the y -axis, the net force acting on the particle is proportional to :
- (A) y (B) $-y$ (C) $1/y$ (D) $-1/y$
13. Suppose the electrostatic potential at some points in space are given by $V = (x^2 - 2x)$. the electrostatic field strength at $x = 1$ is :
- (A) Zero (B) -2 (C) 2 (D) 4
- *14. In a region of constant potential :
- (A) The electric field is uniform (B) The electric field is zero
 (C) There can be no charge inside the region
 (D) The electric field shall necessarily change if a charge is placed outside the region
15. An electric dipole is placed at an angle of 30° to a non-uniform electric field. The dipole will experience
- (A) A translational force only in the direction of the field.
 (B) A translational force only in the direction normal to the direction of the field.
 (C) A torque as well as a translational force.
 (D) A torque only
16. Three charges $-q_1, +q_2$ and $-q_3$ are placed as shown in the figure. The x -component of the force on $-q_1$ is proportional to :
- 
- (A) $\frac{q_2}{b^2} - \frac{q_3}{a^2} \cos \theta$
 (B) $\frac{q_2}{b^2} + \frac{q_3}{a^2} \sin \theta$
 (C) $\frac{q_2}{b^2} + \frac{q_3}{a^2} \cos \theta$
 (D) $\frac{q_2}{b^2} - \frac{q_3}{a^2} \sin \theta$

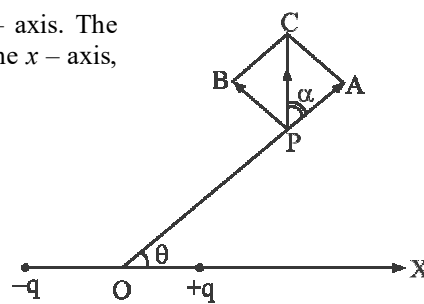
17. A point charge q and a charge $-q$ are placed at $x = -a$ and $x = +a$, respectively, which of the following represents a part of E - x graph ?



18. An electric dipole of common $\vec{P}$ is placed at the origin along the x -axis. The electric field at a point P , whose position vector makes an angle θ with the x -axis, will make an angle :

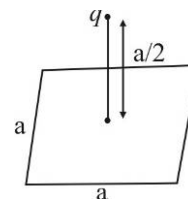
- (A) α
(B) θ
(C) $\theta + \alpha$
(D) $2\theta + \alpha$

with the x -axis, where $\tan \alpha = \frac{1}{2} \tan \theta$



19. A charge q is placed at a distance $a/2$ above the centre of a horizontal square surface of edge a as shown in figure. The electric flux through the square surface is :

- (A) $Q/2\epsilon_0$ (B) Q/ϵ_0
(C) $Q/6\epsilon_0$ (D) $Q/8\epsilon_0$



20. A thin spherical conducting shell of radius R has a charge q . Another charge Q is placed at the centre of the shell. The electrostatic potential at a point P at a distance $R/2$ from the centre of the shell is :

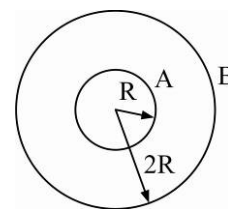
- (A) $\frac{2Q}{4\pi\epsilon_0 R}$ (B) $\frac{2Q}{4\pi\epsilon_0 R} - \frac{2q}{4\pi\epsilon_0 R}$ (C) $\frac{2Q}{4\pi\epsilon_0 R} + \frac{q}{4\pi\epsilon_0 R}$ (D) $\frac{(q+Q)}{4\pi\epsilon_0} \frac{2}{R}$

- *21. The work done to move a charge along an equipotential from A to B :

- (A) cannot be defined as $-\int_A^B E \cdot d\ell$ (B) must be defined as $-\int_A^B E \cdot d\ell$
(C) is zero (D) can have a non-zero value

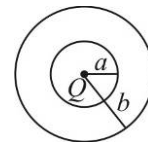
22. In figure shown, conducting shells A and B have charges Q and $2Q$ distributed uniformly over A and B . The value of $V_A - V_B$ is :

- (A) $\frac{Q}{4\pi\epsilon_0 R}$ (B) $\frac{Q}{8\pi\epsilon_0 R}$
(C) $\frac{3Q}{4\pi\epsilon_0 R}$ (D) $\frac{3Q}{8\pi\epsilon_0 R}$

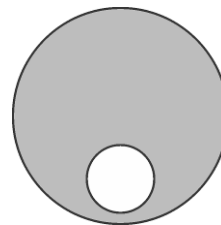


23. The region between two concentric spheres of radii ' a ' and ' b ', respectively (see figure), has volume charge density $\rho = \frac{A}{r}$ where A is a constant and r is the distance from the centre. At the centre of the spheres is a point charge Q . The value of A such that the electric field in the region between the spheres will be constant, is :

- (A) $\frac{Q}{2\pi a^2}$ (B) $\frac{Q}{2\pi(b^2 - a^2)}$ (C) $\frac{2Q}{\pi(a^2 - b^2)}$ (D) $\frac{2Q}{\pi a^2}$



24. A spherical portion has been removed from a solid sphere having a charge distributed uniformly in its volume as shown in the figure. The electric field inside the emptied space is:



- (A) Zero everywhere
(B) Non-zero and uniform
(C) Non-uniform
(D) Zero only at its centre

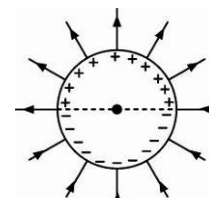
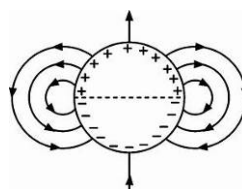
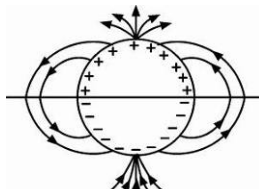
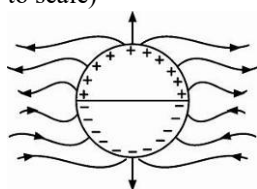
25. Within a spherical charge distribution of charge density $\rho(r)$, N equipotential surfaces of potential $V_0, V_0 + \Delta V, V_0 + 2\Delta V, \dots, V_0 + N\Delta V (\Delta V > 0)$, are drawn and have increasing radii $r_0, r_1, r_2, \dots, r_N$, respectively. If the difference in the radii of the surfaces is constant for all values of V_0 and ΔV then :

- (A) $\rho(r) = \text{constant}$ (B) $\rho(r) \propto \frac{1}{r^2}$ (C) $\rho(r) \propto \frac{1}{r}$ (D) $\rho(r) \propto r$

- *26. If there were only one type of charge in the universe, then :

- (A) $\oint_s E \cdot dS \neq 0$ on any surface (B) $\oint_s E \cdot dS = 0$ if the charge is outside the surface
(C) $\oint_s E \cdot dS$ could not be defined (D) $\oint_s E \cdot dS = \frac{q}{\epsilon_0}$ if charges of magnitude q were inside the surface

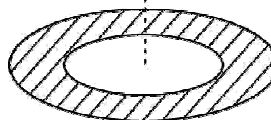
27. A long cylindrical shell carries positive surface charge σ in the upper half and negative surface charge $-\sigma$ in the lower half. The electric field lines around the cylinder will look like figure given in : (Figures are schematics and not drawn to scale)



- (A) (B) (C) (D)

28. A thin disc of radius $b = 2a$ has a concentric hole of radius a in it (see figure).

It carries uniform surface charge σ on it. If the electric field on its axis at height $h (h \ll a)$ from its centre is given as Ch then value of C is :



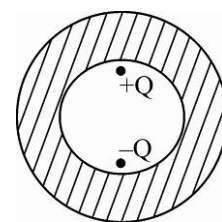
- (A) $\frac{\sigma}{a\epsilon_0}$ (B) $\frac{\sigma}{2a\epsilon_0}$ (C) $\frac{\sigma}{4a\epsilon_0}$ (D) $\frac{\sigma}{8a\epsilon_0}$

29. A charged particle q is shot towards another charged particle Q which is fixed with a speed v . It approaches Q upto a closest distance r and then returns. If q was given a speed $2v$, the closest distance of approach would be :



- (A) r (B) $2r$ (C) $r/2$ (D) $r/4$

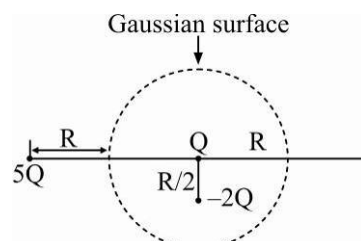
30. Shown in the figure are two point charges $+Q$ and $-Q$ inside the cavity of a spherical shell. The charges are kept near the surface of the cavity on opposite sides of the centre of the shell. If σ_1 is the surface charge on the inner surface and Q_1 net charge on it and σ_2 the surface charge on the outer surface and Q_2 net charge on it then



- (A) $\sigma_1 \neq 0, Q_1 \neq 0; \sigma_2 \neq 0, Q_2 \neq 0$ (B) $\sigma_1 \neq 0, Q_1 = 0; \sigma_2 \neq 0, Q_2 = 0$
(C) $\sigma_1 \neq 0, Q_1 = 0; \sigma_2 = 0, Q_2 = 0$ (D) $\sigma_1 = 0, Q_1 = 0; \sigma_2 = 0, Q_2 = 0$

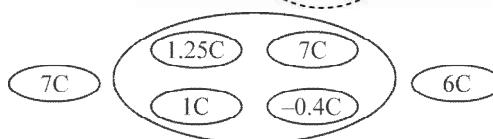
*31. Refer to the arrangement of charges in figure and a Gaussian surface of radius R with Q at the centre. Then :

- (A) Total flux through the surface of the sphere is $\frac{-Q}{\epsilon_0}$
 (B) Field on the surface of the sphere is $\frac{-Q}{4\pi\epsilon_0 R^2}$
 (C) Flux through the surface of sphere due to $5Q$ is zero
 (D) Field on the surface of sphere due to $-2Q$ is same everywhere



32. What is the electric flux linked with closed surface?

- (A) $10^{11} \text{ N-m}^2/\text{C}$ (B) $10^{12} \text{ N-m}^2/\text{C}$
 (C) $10^{10} \text{ N-m}^2/\text{C}$ (D) $8.86 \times 10^{13} \text{ N-m}^2/\text{C}$

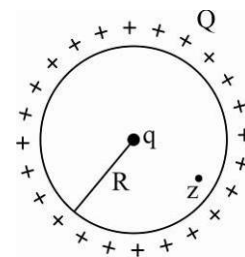


33. Two thin wire rings each having a radius R are placed at a distance d apart with their axes coinciding. The charges on the two rings are $+q$ and $-q$. The potential difference between the centres of the two rings is :

- (A) $\frac{qR}{4\pi\epsilon_0 d^2}$ (B) $\frac{q}{2\pi\epsilon_0} \left[\frac{1}{R} - \frac{1}{\sqrt{R^2 + d^2}} \right]$
 (C) Zero (D) $\frac{q}{4\pi\epsilon_0} \left[\frac{1}{R} - \frac{1}{\sqrt{R^2 + d^2}} \right]$

*34. A positive charge Q is uniformly distributed along a circular ring of radius R . A small test charge q is placed at the centre of the ring. Then :

- (A) If $q > 0$ and is displaced away from the centre in the plane of the ring, it will be pushed back towards the centre
 (B) If $q < 0$ and is displaced away from the centre in the plane of the ring, it will never return to the centre and will continue moving till it hits the ring
 (C) If $q < 0$, it will perform SHM for small displacement along the axis
 (D) q at the centre of the ring is in an unstable equilibrium within the plane of the ring for $q > 0$



35. The electric flux through a closed surface areas S enclosing charge Q is ϕ . If the surface area is doubled, then the flux is:

- (A) 2ϕ (B) $\frac{\phi}{2}$ (C) $\frac{\phi}{4}$ (D) ϕ

36. A point charge q is placed inside a conducting spherical shell of inner radius $2R$ and outer radius $3R$ at a distance of R from the centre of the shell. The electric potential at the centre of shell will be $\frac{1}{4\pi\epsilon_0}$ times :

- (A) $q/2R$ (B) $4q/3R$ (C) $5q/6R$ (D) $2q/3R$

For Assertion & Reason type Questions, please use following codes (wherever required in PHYSICS).

- (I) If both assertion and reason are true and reason is a correct explanation of the assertion
 (II) If both assertion and reason are true but the reason is not a correct explanation of assertion
 (III) If assertion is true but reason is false
 (IV) If assertion is false but reason is true (V) Both assertion and reason are false

37. **Assertion :** Metallic shield in the form of a hollow shell, can be built to block an electric field.

Reason : In a hollow spherical shell, the electric field inside it is zero at every point.

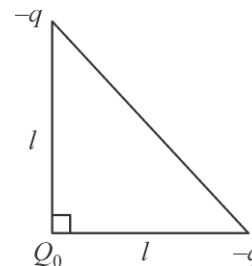
- (A) I (B) II (C) III (D) IV

38. **Assertion :** Free electrons move always from a region of higher potential to a region of lower potential.
Reason : Electron has a negative charge.
 (A) I (B) II (C) III (D) IV
39. **Assertion :** Two concentric conducting spherical shells are charged. The charge on the outer shell is varied keeping the charge on inner shell constant, as a result the electric potential difference between the two shells does not change.
Reason : In this type of situation potential difference depends on the charge of inner shell only
 (A) I (B) II (C) III (D) IV

INTEGER ANSWER TYPE QUESTIONS

The Answers to the following questions are positive integers of 1/2/3 digits or zero.

40. Assume that an electric field $\vec{E} = 30x^2\hat{i}$ exists in space. Then, the potential difference $V_A - V_0$, where V_0 is the potential at the origin and V_A is the potential at $x = 2m$, is n volt. Find n .
41. A charge Q is divided into two parts of magnitude q and $Q - q$. If the coulomb repulsion between them when they are separated at some distance is to be maximum, the ratio of Q/q should be .
42. A cylinder of radius R and length L is placed in a uniform electric field E parallel to the axis of cylinder the total flux through the curved surface of the cylinder is given by .
43. The magnitude of electric intensity at a distance x from a charge q is E . An identical charge is placed at a distance $2x$ from it. Then the magnitude of the force it experience is nqE . Find n .
44. The electric potential V at any point x, y, z (all in meters) in space is given by $V = 4x^2$ volts. The electric field (in V/m) at the point $(1\text{ m}, 0, 2\text{ m})$ is $n\hat{i}$. Find n .
45. Force between two identical charges placed at a distance of r in vacuum is F . Now a slab of dielectric constant 4 is inserted between these two charges. If the thickness of the slab is $r/2$, then the force between the charges will become nF . Find n .
46. An electric field $\vec{E} = (25\hat{i} + 30\hat{j})\text{ NC}^{-1}$ exists in a region of space. If the potential at the origin is taken to be zero then the potential at $x = 2m, y = 2m$ is $n\text{ V}$. Find n .
47. An electric field is given by $\vec{E} = (y\hat{i} + x\hat{j})\text{ N/C}$. The work done in moving a 1 C charge from $\vec{r}_A = (2\hat{i} + 2\hat{j})\text{ m}$ to $\vec{r}_B = (4\hat{i} + \hat{j})\text{ m}$ is J .
48. Two spherical conductors A and B of radii 1 mm and 2 mm are separated by a distance of 5 cm and are uniformly charged. If the spheres are connected by a conducting wire, then, in equilibrium position, the ratio of the magnitude of electric fields at the surface of the spheres A and B is .
49. The work done in moving an alpha particle between two points having potential difference 25 volts is $8 \times 10^x\text{ J}$. Find x .
50. Three charges $Q_0, -q$ and $-q$ are placed at the vertices of an isosceles right angle triangle as shown in the figure. The net electrostatics potential energy is zero, if Q_0 is equal to $\frac{2q}{\sqrt{n}}$. Find n .



DC Circuits

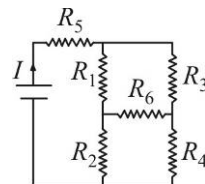
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1. If specific resistance of a wire is ρ , its volume is $3m^3$, and its resistance is 3 ohms , then its length will be :

(A) $\sqrt{\frac{1}{\rho}}$ (B) $\frac{3}{\sqrt{\rho}}$ (C) $\frac{1}{\rho}\sqrt{3}$ (D) $\rho\sqrt{\frac{1}{3}}$

2. In the given circuit, it is observed that the current I is independent of the value of the resistance R_6 . Then the resistance value must satisfy :

(A) $R_1 R_2 R_5 = R_3 R_4 R_6$ (B) $\frac{1}{R_5} + \frac{1}{R_6} = \frac{1}{(R_1 + R_2)} + \frac{1}{(R_3 + R_4)}$
 (C) $R_1 R_4 = R_2 R_3$ (D) $R_1 R_3 = R_2 R_4 = R_5 R_6$

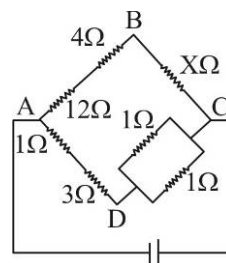


3. A set of n identical resistors each of resistance $R\Omega$, when connected in series, has an effective resistance of $x\text{ ohm}$. When the resistors are connected in parallel, the effective resistance is $y\text{ ohm}$. What is the relation between R , x , and y ?

(A) $R = \frac{xy}{(x+y)}$ (B) $R = (y-x)$ (C) $R = \sqrt{xy}$ (D) $R = (x+y)$

4. In the arrangement of resistances shown in figure, the potential difference between the points B and D will be zero when the unknown resistance X is :

- (A) 4Ω
 (B) 2Ω
 (C) 3Ω
 (D) EMF of the cell is needed to find out X .

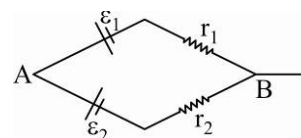


5. The supply voltage to room is 120 V . The resistance of the lead wire is 6Ω . A 60 W bulb is already switched on. What is the decrease of voltage across the bulb, when a 240 W heater is switched on in parallel to the bulb?

- (A) Zero (B) 2.9 V (C) 13.3 V (D) 10.04 V

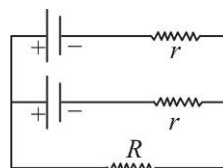
6. Two batteries of emf ε_1 and ε_2 ($\varepsilon_2 > \varepsilon_1$) and internal resistances r_1 and r_2 respectively are connected in parallel as shown in figure.

- (A) The equivalent emf ε_{eq} of the two cells is between ε_1 and ε_2 , i.e. $\varepsilon_1 < \varepsilon_{eq} < \varepsilon_2$.
 (B) The equivalent emf ε_{eq} is smaller than ε_1
 (C) The ε_{eq} is given by $\varepsilon_{eq} = \varepsilon_1 + \varepsilon_2$ always
 (D) ε_{eq} is independent of internal resistances r_1 and r_2



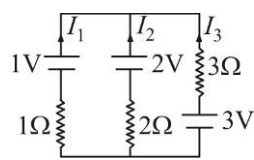
7. In the circuit shown, the EMF of each cell is E_m and internal resistance of each cell is r . The current through the resistor R is:

(A) $\frac{E_m}{2R+r}$ (B) $\frac{E_m}{2r+R}$
 (C) $\frac{2E_m}{R+2r}$ (D) $\frac{2E_m}{2R+r}$



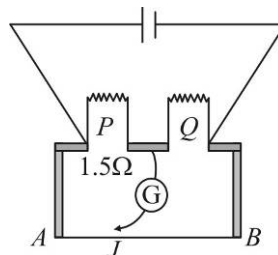
8. Current I_3 in the given circuit shown in figure is :

- (A) $\frac{5}{11}A$ (B) $\frac{7}{11}A$
(C) $\frac{2}{11}A$ (D) $\frac{9}{11}A$



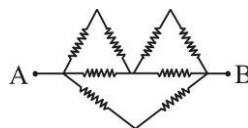
9. The meter bridge circuit shown in figure, is balanced when jockey J divides wire AB in two parts AJ and BJ in the ratio of 1 : 2 . The unknown resistance Q has value :

- (A) 1Ω
(B) 3Ω
(C) 4Ω
(D) 7Ω



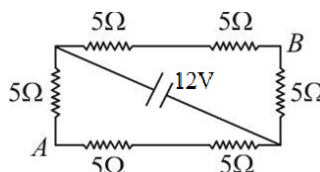
10. What is the equivalent resistance between A and B ? (Each resistor has resistance R) :

- (A) $\frac{4R}{3}$ (B) $\frac{5R}{3}$
(C) $\frac{4R}{5}$ (D) $\frac{3R}{4}$



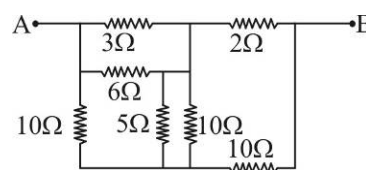
11. Which of the following characteristics of electrons determines the current in a conductor?
(A) Drift velocity alone (B) Thermal velocity alone
(C) Both drift velocity and thermal velocity (D) Neither drift nor thermal velocity
12. If EMF in a thermocouple is $\varepsilon = \alpha T + \beta T^2$, then the neutral temperature of the thermocouple is :
(A) $-\beta/(2\alpha)$ (B) $-2\beta/\alpha$ (C) $-\alpha/(2\beta)$ (D) $-2\alpha/\beta$
13. The potential difference between points A and B in the following circuit diagram will be :

- (A) $8V$ (B) $6V$
(C) $4V$ (D) $2V$



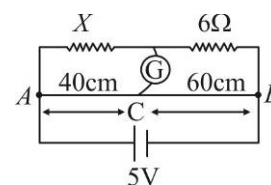
14. In the given circuit, the equivalent resistance between point A and B is :

- (A) $\frac{10}{3}\Omega$ (B) $\frac{5}{3}\Omega$
(C) $\frac{24}{57}\Omega$ (D) $\frac{57}{24}\Omega$



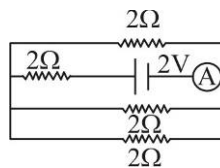
15. In the circuit shown, a meter bridge is in its balanced state, the meter bridge wire has a resistance $0.1\Omega/cm$. the value of unknown resistance X and the current drawn from the battery of negligible resistance is :

- (A) $6\Omega, 5A$ (B) $4\Omega, 0.1A$
(C) $4\Omega, 1.0A$ (D) $12\Omega, 0.5A$



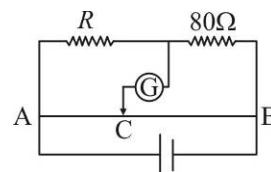
16. The reading of the ammeter in figure shown is :

(A) $\frac{1}{8} A$ (B) $\frac{3}{4} A$
(C) $\frac{1}{2}$ (D) $2A$



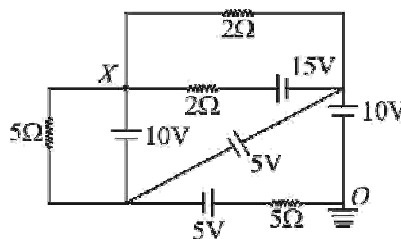
17. AB is wire of uniform resistance. The galvanometer G shows zero current when the length $AC = 20cm$ and $CB = 80cm$. The resistance R is equal to :

(A) 2Ω (B) 8Ω
(C) 20Ω (D) 40Ω



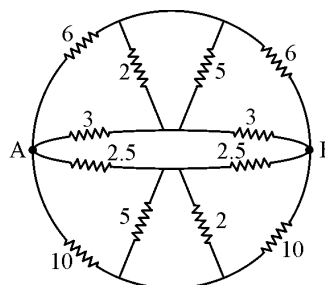
18. In the circuit shown, if point O is earthed, the potential of point X is equal to :

(A) $10V$
(B) $15V$
(C) $25V$
(D) $12.5V$



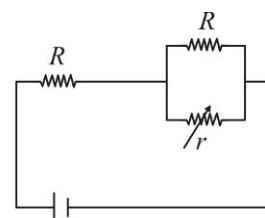
19. Find equivalent resistance between points A and B. Values of resistances in ohms are as shown.

(A) 2Ω
(B) 4Ω
(C) 3Ω
(D) None of these



20. In the circuit shown the resistance r is a variable resistance. If for $r = f R$, the heat generation in r is maximum then the value of f is :

(A) $\frac{1}{2}$ (B) 1
(C) $\frac{1}{4}$ (D) $\frac{3}{4}$



21. The resistance of the series combination of two resistance is S . When they are joined in parallel, the total resistance is P . If $S = nP$, then the minimum possible value of n is:

(A) 4 (B) 3 (C) 2 (D) 1

22. The resistance of an electrical toaster has a temperature dependence given by $R(T) = R_0 [1 + \alpha(T - T_0)]$ in its range of operation. At $T_0 = 300K$, $R = 100\Omega$ and at $T = 500K$, $R = 120\Omega$. The toaster is connected to a voltage source at $200V$ and its temperature is raised at a constant rate from 300 to $500 K$ in $30s$. The total work done in raising the temperature is :

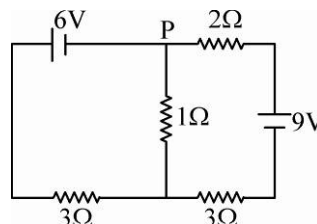
(A) $400 \ln \frac{5}{6} J$ (B) $200 \ln \frac{2}{3} J$ (C) $300 J$ (D) None of these

23. When $5 V$ potential difference is applied across a wire of length $0.1 m$, the drift speed of electrons is $2.5 \times 10^{-4} ms^{-1}$. If the electron density in the wire is $8 \times 10^{28} m^{-3}$, the resistivity of the material is close to :

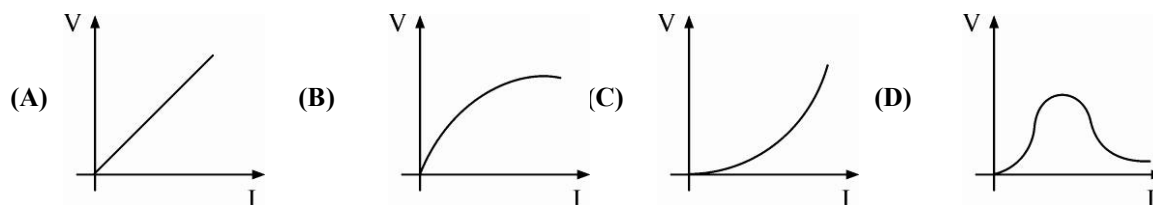
(A) $1.6 \times 10^{-6} \Omega m$ (B) $1.6 \times 10^{-5} \Omega m$ (C) $1.6 \times 10^{-8} \Omega m$ (D) $1.6 \times 10^{-7} \Omega m$

24. In the circuit shown, the current in the 1Ω resistor is

(A) 0.13 A , from Q to P
 (B) 0.13 A , from P to Q
 (C) 0.3 A , from P to Q
 (D) 0 A

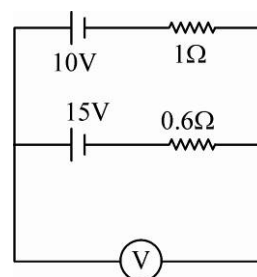


25. Suppose the drift velocity v_d in a material varied with the applied electric field E as $v_d \propto \sqrt{E}$. Then V - I graph for a wire made of such a material is best given by :

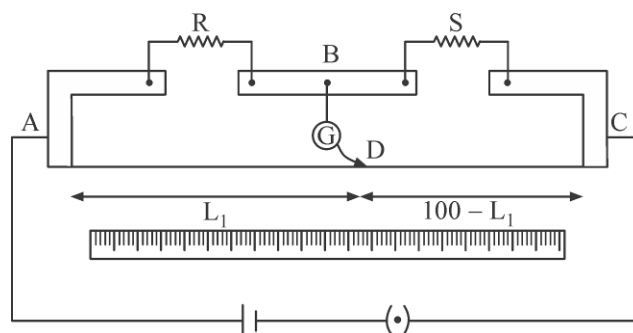


26. A 10 V battery with internal resistance 1Ω and a 15 V battery with internal resistance 0.6Ω are connected in parallel to a voltmeter (see figure). The reading in the voltmeter will be closed to :

(A) 11.9 V
 (B) 12.5 V
 (C) 13.1 V
 (D) 24.5 V

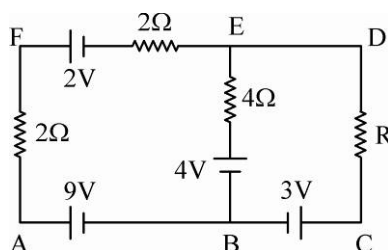


- *27. In a meter bridge experiment, the galvanometer shows no deflection when the jockey is touched at the point D . The lengths AD and DC are marked in the figure. If $R = 20\Omega$ and $S = 40\Omega$, L_1 is equal to:



(A) $\frac{100}{3}\text{ cm}$ (B) $\frac{50}{3}\text{ cm}$ (C) 25 cm (D) $\frac{25}{2}\text{ cm}$

28. In the electric network shown, no current flows through the 4Ω resistor in the arm EB . The potential difference between the points A and D will be :

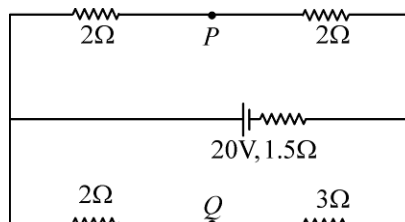


(A) 3 V (B) 4 V (C) 5 V (D) 6 V

29. The resistance of a bulb filament is 100Ω at a temperature of 100°C . If its temperature coefficient of resistance is 0.005 per $^\circ\text{C}$, its resistance will become 200Ω at a temperature of:
 (A) 300°C (B) 400°C (C) 500°C (D) 200°C

- *30. In the circuit shown:

- (A) The current through battery is greater than 5 A
 (B) P and Q are at same potential
 (C) P is at higher potential than Q
 (D) P is at lower potential than Q



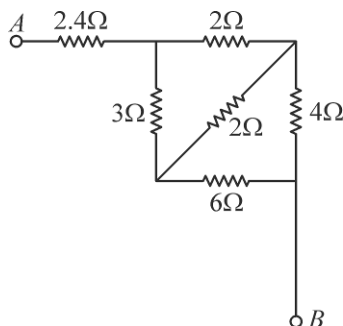
31. The resistance of a wire is 5Ω at 50°C and 6Ω at 100°C . The resistance of the wire at 0°C will be:
 (A) 2Ω (B) 1Ω (C) 4Ω (D) 3Ω

INTEGER ANSWER TYPE QUESTIONS

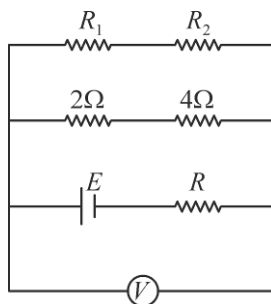
The Answers to the following questions are positive integers of 1/2/3 digits or zero.

32. Two cylindrical conductors, of diameter $2d$ and d , length L and $2L$, and different materials of resistivity ρ_1 and ρ_2 are connected in series with a battery. If the heat generated per unit time in the conductors is equal, $\frac{\rho_1}{\rho_2} = \underline{\hspace{2cm}}$.

33. The equivalent resistance between points A and B is $R_{AB} = \underline{\hspace{2cm}} \Omega$.

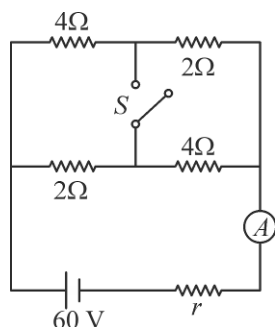


34. A battery of unknown EMF and unknown internal resistance is connected in series with a variable resistance R and an ideal voltmeter is connected across the terminals of the battery. For $R = 3\Omega$, the voltmeter reads 8 V , and for $R = 6\Omega$, the voltmeter reads 9.6 V . The EMF of the battery is $\underline{\hspace{2cm}}\text{ V}$.
35. In the circuit shown, the ideal voltmeter reads 12 V . Then, the rate of heat dissipation in the 4Ω resistance is $\underline{\hspace{2cm}}\text{ W}$.

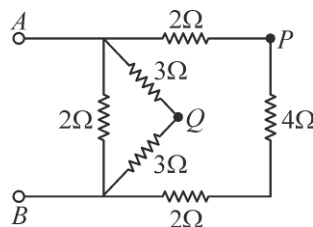


36. A galvanometer has range $0-10\text{ mA}$ and internal resistance 15Ω . To convert this galvanometer into a voltmeter of range $0-3\text{ V}$, a resistance $R = \underline{\hspace{2cm}}\Omega$ must be connected in parallel with it.

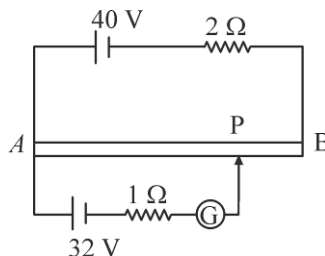
37. In the circuit shown, the ideal ammeter reads 18 A when the switch S is open. If the switch is closed, the ammeter reads _____ A .



38. A wire of length $l = 2\text{ m}$ is made of a material whose charge carrier density is $n = 8 \times 10^{28}\text{ m}^{-3}$ and resistivity is $\rho = 1.5 \times 10^{-8}\text{ }\Omega\text{ m}$. If a potential difference 96 V is applied across the ends of the wire, the drift velocity of charge carriers in the wire will be $v_d =$ _____ mm/s . [Charge on each carrier, $e = 1.6 \times 10^{-19}\text{ C}$]
39. In the given network, if an ideal battery of EMF 48 V is connected between the points A and B, the potential difference between the points P and Q will be _____ V .

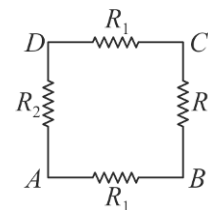


40. The resistance of a wire is $R_1 = 20.00\text{ }\Omega$ at temperature 300 K and $R_2 = 20.07\text{ }\Omega$ at temperature 310 K . The temperature coefficient of resistance for wire is $\alpha =$ _____ $\times 10^{-3}$
41. Two resistances R_1 and R_2 are connected in series with a battery. The time taken for the resistances to produce total heat H is t_1 . Now, the resistances are connected in parallel to each other and then connected to the same battery. Now, the time taken to produce total heat H is t_2 . If $\frac{t_2}{t_1} = \frac{12}{169}$, the value of $\frac{R_2}{R_1}$ is _____.
42. Two cylindrical conductors A and B, of the same diameter, length L and $2L$, and different materials of resistivity ρ and $\frac{5}{4}\rho$ are connected in series with a battery of EMF 28 V . If $L = 25\text{ cm}$, the magnitude of electric field in conductor A is $E_A =$ _____ N/C .
43. In a potentiometer experiment, a wire AB of length 1.2 m is used. For the circuit shown, the current through galvanometer becomes zero when the jockey is touched at point P. If $AP = 1.0\text{ m}$, the resistance per unit length of the wire AB is _____ Ω/m .

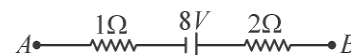


44. In the given network, if an ideal battery is connected between points A and B , a current $i_1 = 5.6A$ flows through the battery. If the same battery is connected between points B and C , a current $i_2 = 10.5A$ flows through the battery. The value of

$$10\left(\frac{R_1}{R_2}\right) = \underline{\hspace{2cm}}.$$



45. In the figure shown, the potential difference between A and B is $V_A - V_B = 4V$ and a current $2A$ flows from A to B. The internal resistance of the battery is $\underline{\hspace{2cm}} \Omega$.

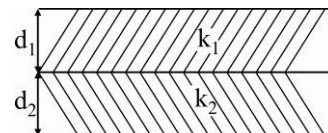


46. A bulb A has voltage rating $220V$ and power rating $40W$, and a bulb B has voltage rating $220V$ and power rating $120W$. If the two bulbs are connected in series with a DC voltage source of $220V$, the potential difference across bulb B is $\underline{\hspace{2cm}} V$.

Capacitors

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED '*' MAY HAVE MORE THAN ONE CORRECT OPTION.

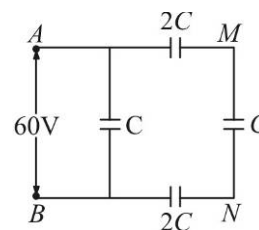
1. A parallel plate capacitor is made of two dielectric blocks in series. One of the blocks has thickness d_1 and dielectric constant k_1 and the other has thickness d_2 and dielectric constant k_2 as shown in figure. This arrangement can be thought as a dielectric slab of thickness $d (= d_1 + d_2)$ and effective dielectric constant k . The k is :



- (A) $\frac{k_1 d_1 + k_2 d_2}{d_1 + d_2}$ (B) $\frac{k_1 d_2 + k_2 d_1}{d_1 + d_2}$ (C) $\frac{k_1 k_2 (d_1 + d_2)}{k_1 d_2 + k_2 d_1}$ (D) $\frac{2k_1 k_2}{k_1 + k_2}$

2. In the circuit shown, a potential difference of 60 V is applied across AB. The potential difference between the points M and N is :

- (A) 10 V
(B) 15 V
(C) 20 V
(D) 30 V



3. A capacitor is connected to a battery. The force of attraction between the plates when the separation between them is halved:

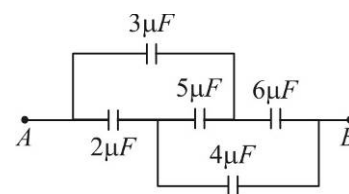
- (A) remains the same (B) becomes eight times
(C) becomes four times (D) becomes double

4. A parallel plate capacitor is made of two circular plates separated by a distance of 5 mm with a dielectric of dielectric constant 2.2 between them. When the electric field in the dielectric is $3 \times 10^4 \text{ V/m}$, the charge density of the positive plate will be close to :

- (A) $6 \times 10^{-7} \text{ C/m}^2$ (B) $3 \times 10^{-7} \text{ C/m}^2$ (C) $3 \times 10^4 \text{ C/m}^2$ (D) $6 \times 10^4 \text{ C/m}^2$

5. In the circuit shown, the equivalent capacitance between the points A and B is :

- (A) $\frac{10}{3} \mu\text{F}$ (B) $\frac{15}{4} \mu\text{F}$
(C) $\frac{12}{5} \mu\text{F}$ (D) $\frac{25}{6} \mu\text{F}$

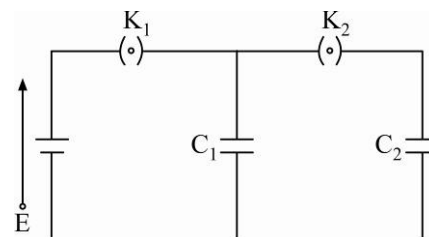


6. Capacity of a spherical capacitor is C_1 when inner sphere is charged and outer sphere is earthed and C_2 when inner sphere is earthed and outer sphere is charged. Then C_1/C_2 is : (a = radius of inner sphere, b = radius of outer sphere)

- (A) 1 (B) $\frac{a}{b}$ (C) $\frac{b}{a}$ (D) $\frac{a+b}{a-b}$

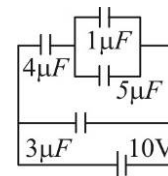
- *7. In the circuit shown in figure initially key K_1 is closed and key K_2 is open. Then K_1 is opened and K_2 is closed (order is important). When the circuit reaches steady state, let the final charge on C_1 and C_2 be Q_1 and Q_2 and the potential difference across them be V_1 and V_2 . Then:

- (A) $V_1 = V_2$
(B) $Q_1 = Q_2$
(C) $C_1 V_1 + C_2 V_2 = C_1 E$
(D) $\frac{Q_1}{C_1} = \frac{Q_2}{C_2}$



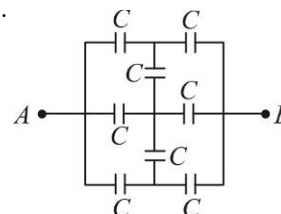
8. For the circuit shown in the adjoining figure, the charge on $4\mu F$ capacitor is :

(A) $30\mu C$ (B) $40\mu C$
(C) $24\mu C$ (D) $54\mu C$



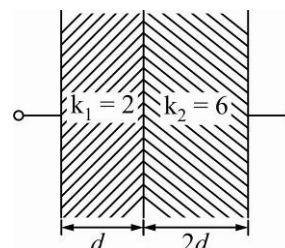
9. In the given figure, find the equivalent capacitance between A and B .

(A) $\frac{2C}{3}$ (B) $\frac{5C}{4}$
(C) $\frac{3C}{2}$ (D) $\frac{4C}{5}$



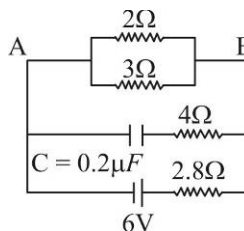
10. A parallel plate capacitor has two layers of dielectrics as shown in figure. This capacitor is connected across a battery, then the ratio of potential difference across the dielectric layers is :

(A) $4/3$
(B) $1/2$
(C) $1/3$
(D) $3/2$



11. In figure, the steady state current in 2Ω resistance is :

(A) $1.5A$
(B) $0.9A$
(C) $0.6A$
(D) zero

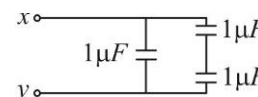


12. The plates of a parallel plate capacitor are 2 cm apart. A potential difference of 10 V is applied between them. The electric field between the plates is:

(A) 20 N/C (B) 500 N/C (C) 5 N/C (D) 250 N/C

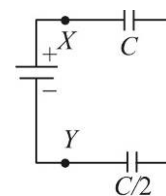
13. The effective capacitance between the points x and y in figure, will be :

(A) $1\mu F$ (B) $1.5\mu F$
(C) $2\mu F$ (D) $4\mu F$



14. The equivalent capacitance between the points X and Y in figure, will be :

(A) $2C/3$
(B) $C/3$
(C) $3C/2$
(D) $3C$



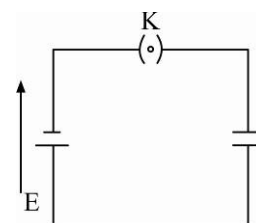
- *15. A parallel plate capacitor is connected to a battery as shown in figure. Consider two situations.

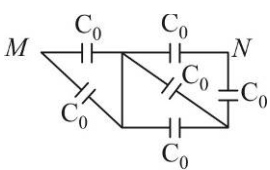
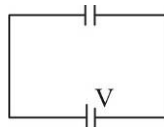
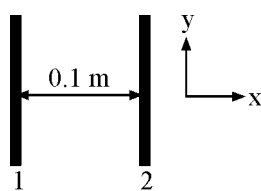
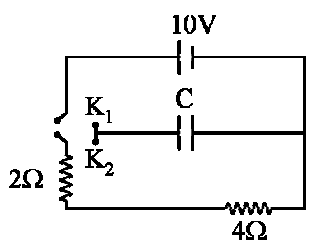
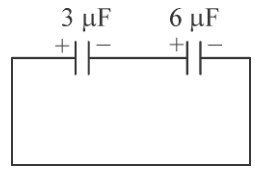
A : Key K is kept closed and plates of capacitors are moved apart using insulating handle.

B : Key K is opened and plates of capacitors are moved apart using insulating handle.

Choose the correct option(s).

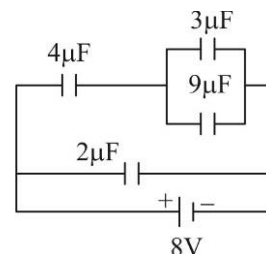
(A) In A : Q remains same but V changes
(B) In B : V remains same but Q changes
(C) In A : V remains same and Q changes
(D) In B : Q remains same and V changes



16. The effective capacitance of two capacitors of capacitances C_1 and C_2 (with $C_2 > C_1$) connected in parallel is $\frac{25}{6}$ times the effective capacitance when they are connected in series. The ratio C_2 / C_1 is :
- (A) $\frac{3}{2}$ (B) $\frac{4}{3}$ (C) $\frac{5}{3}$ (D) $\frac{25}{6}$
17. The equivalent capacitance between points M and N is :
- (A) $\frac{10}{11}C_0$ (B) $2C_0$ (C) C_0 (D) $\frac{5}{8}C_0$
- 
18. A capacitor of capacitance C is completely charged to a potential difference V . Now, with the battery connected, a dielectric slab of dielectric constant $K = 3$ is slowly inserted into the capacitor. The work done by the battery during the insertion of the slab is :
- (A) $\frac{1}{2}CV^2$ (B) CV^2 (C) $2CV^2$ (D) $\frac{3}{2}CV^2$
- 
19. Two insulating plates are both uniformly charged in such a way that the potential difference between them is $V_2 - V_1 = 20V$ (i.e., plate 2 is at a higher potential). The plates are separated by $d = 0.1\text{ m}$ and can be treated as infinitely large. An electron is released from rest on the inner surface of plate 1. What is its speed when it hits plate 2? ($e = 1.6 \times 10^{-19}\text{ C}$, $m_0 = 9.11 \times 10^{-31}\text{ kg}$)
- (A) $2.65 \times 10^6\text{ ms}^{-1}$ (B) $7.02 \times 10^{12}\text{ ms}^{-1}$ (C) $1.87 \times 10^6\text{ ms}^{-1}$ (D) $32 \times 10^{-19}\text{ ms}^{-1}$
- 
20. A parallel plate capacitor of capacitance C is connected to a battery of $emf\ V$. If a dielectric slab is completely inserted between the plates of the capacitor and battery remains connected, then electric field between plates :
- (A) Decreases (B) increases (C) remains constant. (D) may be increase or may be decrease.
21. A capacitor of capacitance $3\mu\text{F}$ is first charged by connecting it across a 10 V battery by closing the key K_1 . Then it is allowed to get discharged through 2Ω and 4Ω resistors by closing key K_2 . The total energy dissipated in 4Ω resistor is equal to :
- (A) $50\mu\text{J}$ (B) $100\mu\text{J}$ (C) $150\mu\text{J}$ (D) $225\mu\text{J}$
- 
22. Two capacitors of capacitance $3\mu\text{F}$ and $6\mu\text{F}$ are charged to a potential of $12V$ each. They are now connected as shown. Then, at steady state:
- (A) Potential difference across $3\mu\text{F}$ is zero (B) Potential difference across $3\mu\text{F}$ is $4V$ (C) Charge on $3\mu\text{F}$ is zero (D) Charge on $3\mu\text{F}$ is $10\mu\text{C}$
- 

23. Capacitance of a capacitor becomes $\frac{4}{3}$ times its original value if a dielectric slab of thickness $t = d/2$ is inserted between the plates (d = separation between the plates). The dielectric constant of the slab is:
 (A) 2 (B) 4 (C) 8 (D) 16

24. A combination of capacitors is set up as shown in the figure. The magnitude of the electric field due to a point charge Q (having a charge equal to the sum of the charges on the $4\mu F$ and $9\mu F$ capacitors), at a point distance $30m$ from it, is equal to:



- (A) $240 N/C$ (B) $360 N/C$
 (C) $420 N/C$ (D) $480 N/C$

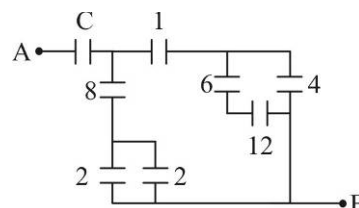
25. Let C be the capacitance of a capacitor discharging through a resistor R . Suppose t_1 is the time taken for the energy stored in the capacitor to reduce to half its initial value and t_2 is the time taken for the charge to reduce to one-fourth its initial value. Then, the ratio t_1/t_2 will be :

- (A) 1 (B) $1/2$ (C) $1/4$ (D) 2

26. Three capacitors each of $4\mu F$ are to be connected in such a way that the effective capacitance is $6\mu F$. This can be done by connecting them.

- (A) All in series (B) All in parallel
 (C) Two in parallel and one in series (D) Two in series and one in parallel

27. Figure shows a network of capacitors where the numbers indicates capacitances in micro Farad. The value of capacitance C if the equivalent capacitance between point A and B is to be $1\mu F$ is :



- (A) $\frac{32}{23}\mu F$ (B) $\frac{31}{23}\mu F$ (C) $\frac{33}{23}\mu F$ (D) $\frac{34}{23}\mu F$

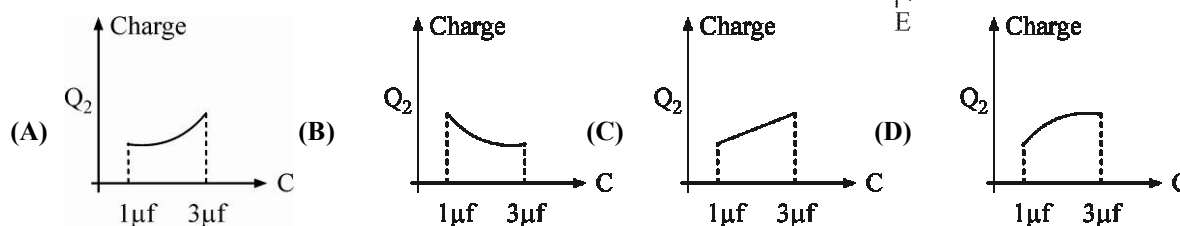
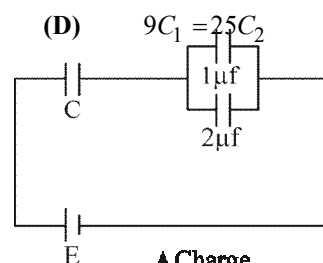
28. The work done against electric forces in increasing the potential difference of a capacitor from $20V$ to $40V$ is W . The work done in increasing its potential difference from $40V$ to $50V$ will be:

- (A) $4W$ (B) $3W/4$ (C) $2W$ (D) $W/2$

29. Two capacitors C_1 and C_2 are charged to $120 V$ and $200 V$, respectively. It is found that by connecting them together, the potential on each one can made zero. Then :

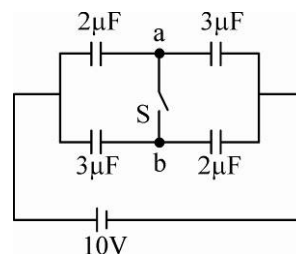
- (A) $5C_1 = 3C_2$ (B) $3C_1 = 5C_2$ (C) $25C_1 = 9C_2$ (D) $9C_1 = 25C_2$

30. In the given circuit, charge Q_2 on the $2\mu F$ capacitor changes as C is varied from $1\mu F$ to $3\mu F$. Q_2 as a function of ' C ' is given properly by (figures are drawn schematically and are not to scale)

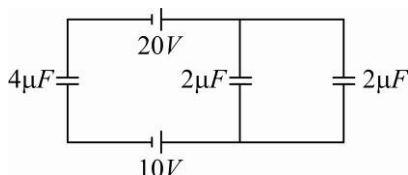


31. In the figure is shown a system of four capacitors connected across a 10 V battery. Charge that will flow from switch S when it is closed is :

(A) $5\mu\text{C}$ from b to a
 (B) $20\mu\text{C}$ from a to b
 (C) $5\mu\text{C}$ from a to b
 (D) zero



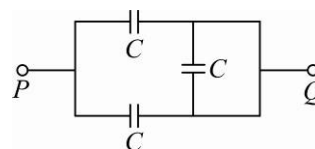
32. In the circuit shown in figure, charge stored in $4\mu\text{F}$ capacitor is :



(A) $20\mu\text{C}$ (B) $40\mu\text{C}$ (C) $10\mu\text{C}$ (D) $120\mu\text{C}$

33. The equivalent capacitance between points P and Q is:

(A) C (B) $2C$
 (C) $\frac{3}{2}C$ (D) $\frac{C}{2}$



34. A charged capacitor C with initial charge q_0 is connected to a coil of self inductance L at $t = 0$. The time at which the energy is stored equally between the electric and the magnetic fields is:

(A) $\frac{\pi}{4}\sqrt{LC}$ (B) $2\pi\sqrt{LC}$ (C) $\sqrt{LC}$ (D) $\pi\sqrt{LC}$

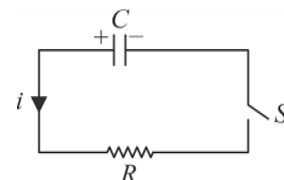
INTEGER ANSWER TYPE QUESTIONS

The Answers to the following questions are positive integers of 1/2/3 digits or zero.

35. A capacitor of capacitance $C = 5\mu\text{F}$ is charged and then connected with a resistor $R = 1\text{ k}\Omega$. Just after the key S is closed, the current through the resistor is $i_0 = 25\mu\text{A}$.

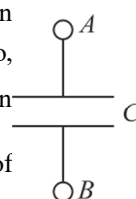
Then, the time after which the current becomes $i = \frac{i_0}{2}$ is $t = \underline{\hspace{1cm}}$ ms.

(e is the base of nature logarithm)



36. Two identical uncharged capacitors A and B are taken and a dielectric slab of di-electric constant $K = 3$ is inserted into capacitor B . Now, the two capacitors are connected in series with a battery of EMF V_0 and a resistance R . After a long time, the potential difference across capacitor B is V_B . The ratio $\frac{V_0}{V_B} = \underline{\hspace{1cm}}$

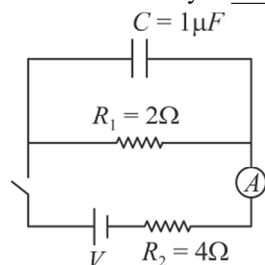
37. A battery of EMF V is connected across the terminals A and B of an uncharged capacitor C_1 . After current through the battery has become zero, the battery is removed and an uncharged capacitor C_2 is connected between



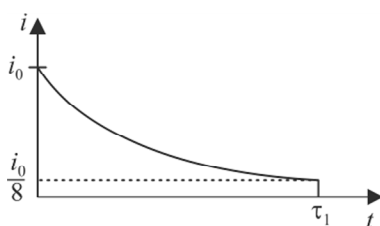
A and B . If the final potential difference across C_1 is $\frac{V}{6}$, the ratio of

capacitances $\frac{C_2}{C_1} = \underline{\hspace{1cm}}$.

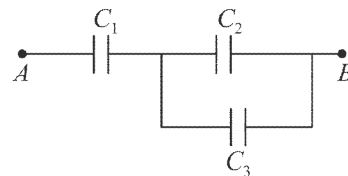
38. An ideal ammeter and an uncharged capacitor are connected in a circuit as shown. A long time after the key is closed, the reading of the ammeter is $i = 3A$. The EMF of the battery is _____ V .



39. A charged capacitor is allowed to discharge. The current flowing in the circuit is plotted against time as shown. The time instant when the current is $\frac{i_0}{16}$ is $t = \tau_2$. Then $\frac{\tau_2 + \tau_1}{\tau_2 - \tau_1} = \underline{\hspace{2cm}}$.



40. Three identical capacitors C_1, C_2 and C_3 are connected as shown. The capacitance between points A and B is C . Now, a dielectric slab of dielectric constant K is inserted inside C_2 . Now, the capacitance between points A and B is C' . If $\frac{C'}{C} = \frac{5}{4}$, the value of $K = \underline{\hspace{2cm}}$.



41. The dielectric inside a capacitor breaks down if the electric field in it exceeds $E = 10 \text{ N/C}$. If the separation between the plates is $d = 0.2 \text{ mm}$, the break down voltage of the capacitor is _____ mV .
42. An uncharged capacitor of capacitance $C = 10\mu F$ and a resistance $R = 20k\Omega$ are connected in series with an ideal battery of EMF 16 V , and the switch is closed at $t = 0$. The energy stored in the capacitor between $t_1 = 140 \text{ ms}$ and $t_2 = 280 \text{ ms}$ is $E = \underline{\hspace{2cm}} \mu J$. [$\log_e 2 = 0.7$]
43. The two plates of a parallel plate capacitor of capacitance C are given charge $+\frac{Q}{3}$ and $+\frac{2Q}{3}$. The potential difference between the plates becomes V . Then, $\frac{Q}{CV} = \underline{\hspace{2cm}}$.
44. Two identical capacitors A and B of capacitance C are taken and the separation between the plates of capacitor B is increased to 3 times the original separation. The two capacitors are then connected in series with a battery and charged. The electrostatic force between the plates of capacitor A is F_A and the electrostatic force between the plates of capacitor B is F_B . Then, $\frac{F_1}{F_2} = \underline{\hspace{2cm}}$.
45. An initially uncharged capacitor is connected in series with an ideal battery of EMF 12 V and a resistance R and the switch is closed at $t = 0$. The charge on the positive plate of the capacitor is given as a function of time by: $q(t) = (1.2 \times 10^{-4})e^{-500t}$, where all quantities are in SI units. The resistance $R = \underline{\hspace{2cm}} \Omega$.

46. Two capacitors of capacitance C and $2C$ are charged to the same potential. Let the total potential energy stored in the capacitors be U_i . The capacitors are now connected in series such that plates carrying opposite charge are connected to each other. After the current has become zero, the total potential energy stored in both capacitors is U_f . Then, $10\left(\frac{U_i}{U_f}\right) = \underline{\hspace{2cm}}$.
47. A capacitor charged to a potential V has potential energy U . It is connected in series with a resistance and discharged. Until the time the potential across the capacitor reduces to $\frac{V}{N}$, the heat dissipated in the resistance is $H = \frac{15}{16}U$. Then, $N = \underline{\hspace{2cm}}$.
48. A parallel plate capacitor of capacitance $C = 2\mu F$, plate separation d and area of the plates A initially has vacuum between the plates. Now, two dielectric slabs of area of cross-section A , thickness $\frac{d}{3}$ and $\frac{2d}{3}$, and dielectric constant $K_1 = 2$ and $K_2 = 4$ are inserted in the space between the plates. The capacitance of the capacitor becomes $C' = \underline{\hspace{2cm}}\mu F$.
49. Three identical capacitors are connected in series with a battery. At steady state, the total potential energy stored in the capacitors is U_S . Now, the capacitors are discharged and then connected in parallel with the same battery. At steady state, the total potential energy stored in the capacitors is U_P . Then, $\frac{U_P}{U_S} = \underline{\hspace{2cm}}$.

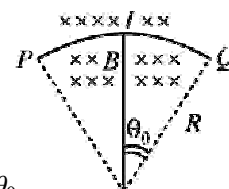
Magnetic Effect of Current

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED '*' MAY HAVE MORE THAN ONE CORRECT OPTION.

1. Two charged particles traverse identical helical paths in a completely opposite sense in a uniform magnetic field $B = B_0 \hat{k}$. v_1 and v_2 are velocity components of the two particles parallel to magnetic field.

- (A) They have equal z -components of momenta
 (B) They must have equal charges
 (C) They necessarily represent a particle-antiparticle pair
 (D) The charge to mass ratio satisfy : $\frac{m_1 V_1}{q_1} + \frac{m_2 V_2}{q_2} = 0$

2. A wire carrying current I is tied between points P and Q and is in the shape of a circular arc of radius R due to a uniform magnetic field B (perpendicular to the plane of the paper, shown by XXX) in the vicinity of the wire. If the wire subtends an angle $2\theta_0$ at the centre of the circle (of which it forms an arc) then the tension in the wire is :



- (A) IBR (B) $\frac{IBR}{\sin \theta_0}$ (C) $\frac{IBR}{2 \sin \theta_0}$ (D) $\frac{IBR \theta_0}{\sin \theta_0}$

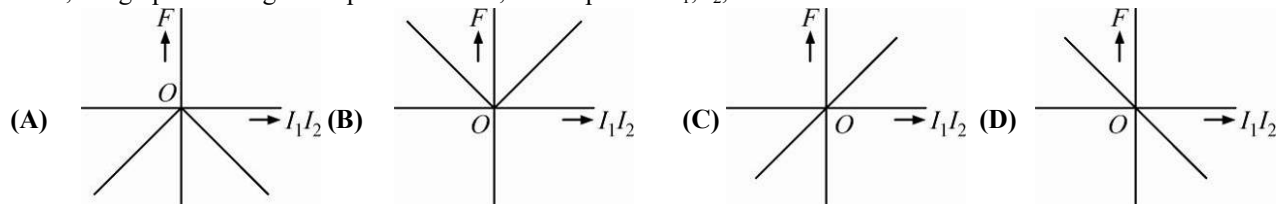
3. A charged particle with charge q enters a region of constant, uniform and mutually orthogonal fields $\vec{E}$ and $\vec{B}$ with a velocity $\vec{v}$ perpendicular to both $\vec{E}$ and $\vec{B}$, and comes out without any change in magnitude or direction of $\vec{v}$. Then :

- (A) $\vec{v} = \vec{E} \times \frac{\vec{B}}{B^2}$ (B) $\vec{v} = \vec{B} \times \frac{\vec{E}}{B^2}$ (C) $\vec{v} = \vec{E} \times \frac{\vec{B}}{E^2}$ (D) $\vec{v} = \vec{B} \times \frac{\vec{E}}{E^2}$

4. Biot-Savart law indicates that the moving electrons (velocity $\vec{v}$) produce a magnetic field $\vec{B}$ such that

- (A) $\vec{B} \perp \vec{v}$
 (B) $\vec{B} \parallel \vec{v}$
 (C) it obeys inverse cube law
 (D) it is along the line joining the electron and point of observation

5. Two long straight parallel wires, carrying (adjustable) current I_1 and I_2 , are kept at a distance d apart. If the force F between the two wires is taken as positive when the wires repel each other and negative when the wires attract each other, the graph showing the dependence of F , on the product I_1, I_2 , would be :



6. Two long conductors, separated by a distance d carry currents I_1 and I_2 and in the same direction. They exert a force F on each other. Now the current in one of them is increased to two times and its direction is reversed. The distance is also increased to $3d$. The new value of the force between them is :

- (A) $-2F$ (B) $-\frac{2F}{3}$ (C) $\frac{F}{3}$ (D) $-\frac{F}{3}$

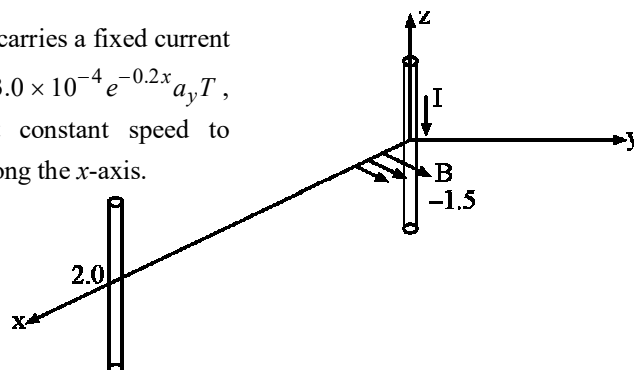
7. A current carrying circular loop of radius R is placed in the x - y plane with centre at the origin. Half of the loop with $x > 0$ is now bent so that it now lies in the y - z plane.
- (A) The magnitude of magnetic moment now diminishes
 (B) The magnetic moment does not change (C) The magnitude of B at $(0, 0, z), z \gg R$ increases
 (D) The magnitude of B at $(0, 0, z), z \gg R$ is unchanged

8. A magnetic needle lying parallel to a magnetic field requires W unit of work to turn it through 60° . The torque needed to maintain the needle in this position will be :

- (A) $\left(\frac{\sqrt{3}}{2}\right)W$ (B) $\sqrt{3}W$ (C) W (D) $2W$

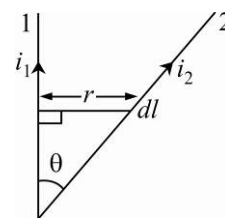
9. A conductor lies along the z -axis at $-1.5 \leq z < 1.5$ m and carries a fixed current of 10.0 A in $-a_z$ direction (see figure). For a field $B = 3.0 \times 10^{-4} e^{-0.2x} a_y$ T, find the power required to move the conductor at constant speed to $x = 2.0$ m, $y = 0$ in 5×10^{-3} s. Assume parallel motion along the x -axis.

- (A) 1.57 W
 (B) 2.97 W
 (C) 14.85 W
 (D) 29.7 W



10. Wire 1 and 2 carrying currents I_1 and I_2 respectively are inclined at an angle θ to each other. What is the force on a small element dl of wire 2 at a distance r from wire 1 (as shown in figure) due to the magnetic field of wire 1?

- (A) $\frac{\mu_0}{2\pi r} I_1 I_2 dl \tan \theta$ (B) $\frac{\mu_0}{2\pi r} I_1 I_2 dl \sin \theta$
 (C) $\frac{\mu_0}{2\pi r} I_2 I_2 dl \cos \theta$ (D) $\frac{\mu_0}{4\pi r} I_1 I_2 dl \sin \theta$



11. An electron is projected with velocity v_0 in a uniform electric field E perpendicular to the field. Again it is projected with velocity v_0 perpendicular to a uniform magnetic field B . If r_1 is initial radius of curvature just after entering in the electric field and r_2 is initial radius of curvature just after entering in magnetic field then the ratio r_1/r_2 is equal to :

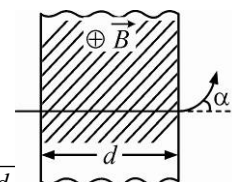
- (A) $\frac{Bv_0^2}{E}$ (B) $\frac{B}{E}$ (C) $\frac{Ev_0}{B}$ (D) $\frac{Bv_0}{E}$

12. An electron is projected with uniform velocity along the axis of a current carrying long solenoid. Which of the following is true?

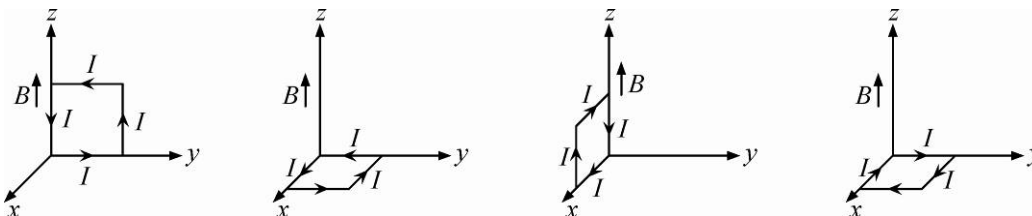
- (A) The electron will be accelerated along the axis
 (B) The electron path will be circular about the axis
 (C) The electron will experience a force at 45° to the axis and hence execute a helical path
 (D) The electron will continue to move with uniform velocity along the axis of the solenoid

13. A proton (mass m) accelerated by a potential difference V flies through a uniform transverse magnetic field occupies a region of space by width d . If α be the angle of deviation of proton from initial direction of motion (see figure), the value of $\sin \alpha$ will be :

- (A) $\frac{B}{2} \sqrt{\frac{q}{mV}}$ (B) $\frac{B}{d} \sqrt{\frac{q}{2mV}}$ (C) $Bd \sqrt{\frac{q}{2mV}}$ (D) $qV \sqrt{\frac{Bd}{2m}}$



14. A rectangular loop of sides 10 cm and 5 cm carrying a current I of 12 A is placed in different orientations as shown in the figure below :

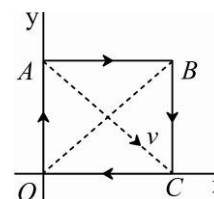


If there is a uniform magnetic field of 0.3 T in the positive z direction, in which orientations the loop would be in (i) stable equilibrium and (ii) unstable equilibrium.

- (A) (2) and (4), respectively (B) (2) and (3), respectively
(C) (1) and (2), respectively (D) (1) and (3), respectively

15. $OABC$ is a current carrying square loop. An electron is projected from the centre of loop along its diagonal AC as shown. Unit vector in the direction of initial acceleration will be :

- (A) $\hat{k}$ (B) $-\left(\frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$
(C) $-\hat{k}$ (D) $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$



16. If in a circular coil A of radius R , current I is flowing and in another coil B of radius $2R$ a current $2I$ is flowing, then the ratio of the magnetic fields, B_A and B_B produced by them will be :

- (A) 1 (B) 2 (C) 1/2 (D) 4

17. A long wire carries a steady current. It is bent into a circle of one turn and the magnetic field at the centre of the coil is B . It is then bent into a circular loop of n turns. The magnetic field at the centre of the coil will be :

- (A) nB (B) n^2B (C) $2nB$ (D) $2n^2B$

18. Magnetic field at the centre of a circular loop of area A is B . Then magnetic moment of the loop will be :

- (A) $\frac{BA^2}{\mu_0\pi}$ (B) $\frac{BA}{\mu_0}\sqrt{A}$ (C) $\frac{BA\sqrt{A}}{\mu_0\pi}$ (D) $\frac{2BA}{\mu_0}\sqrt{\frac{A}{\pi}}$

19. In a cyclotron, a charged particle

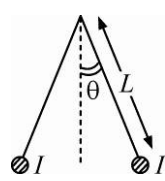
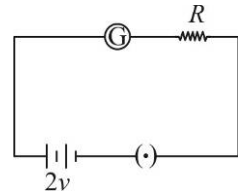
- (A) Undergoes acceleration all the time
(B) Speeds up between the dees because of the magnetic field
(C) Speeds up in a dee
(D) Slows down within a dee and speeds up between dees

20. A magnetic needle is kept in a non-uniform magnetic field. It may experience :

- (A) A torque but not a force (B) Neither a force nor a torque
(C) A force and a torque (D) A force but not a torque

21. A helium ion and a Hydrogen ion are accelerated from rest through a potential difference of V to velocities of U_{He} and U_H respectively. What will be the ratio of U_{He} to U_H :

- (A) 1/4 (B) $1/\sqrt{2}$ (C) $\sqrt{2}$ (D) 2

22. A long straight wire of radius a carries a steady current I . The current is uniformly distributed across its cross-section. The ratio of the magnetic field at distances $\frac{a}{2}$ and a from the axis is :
- (A) $1/4$ (B) 4 (C) 1 (D) $1/2$
23. Two long current carrying thin wires, both with current I in opposite direction, are held by insulating threads of length L and are in equilibrium as shown in the figure, with threads making an angle θ with the vertical. If wire have mass λ per unit length then the value of I is : (g = gravitational acceleration)
- 
- (A) $2\sqrt{\frac{\pi g L}{\mu_0}} \tan \theta$ (B) $\sqrt{\frac{\pi \lambda g L}{\mu_0}} \tan \theta$ (C) $\sin \theta \sqrt{\frac{\pi \lambda g L}{\mu_0 \cos \theta}}$ (D) $2 \sin \theta \sqrt{\frac{\pi \lambda g L}{\mu_0 \cos \theta}}$
24. A particle of mass M and charge Q moving with velocity $\vec{v}$ describes a circular path of radius R when subjected to a uniform transverse magnetic field of induction B . The work done by the field when the particle completes one full circle is :
- (A) $\left(\frac{Mv^2}{R}\right)2\pi R$ (B) Zero (C) $BQ2\pi R$ (D) $BQv2\pi R$
25. A circular current loop of magnetic moment M is in an arbitrary orientation in an external magnetic field B . The work done to rotate the loop by 30° about an axis perpendicular to its plane is :
- (A) MB (B) $\sqrt{3}\frac{MB}{2}$ (C) $\frac{MB}{2}$ (D) zero
26. A galvanometer has a 50 division scale. Battery has no internal resistance. It is found that there is deflection of 40 divisions when $R = 2400\Omega$. Deflection becomes 20 divisions when resistance taken from resistance box is 4900Ω . Then we can conclude.
- 
- (A) Current sensitivity of galvanometer is $20\mu A/\text{division}$
 (B) Resistance of galvanometer is 200Ω .
 (C) Resistance required on R.B for a deflection of 10 divisions is 9800Ω .
 (D) Full scale deflection current is $2mA$
27. A 50Ω resistance is connected to a battery of $5V$. A galvanometer of resistance 100Ω is to be used as an ammeter to measure current through the resistance, for this resistance r_s is connected to the galvanometer. Which of the following connections should be employed if the measured current is within 1% of the current without the ammeter in the circuit?
- (A) $r_s = 0.5\Omega$ in series with the galvanometer (B) $r_s = 1\Omega$ in series with galvanometer
 (C) $r_s = 1\Omega$ in parallel with it galvanometer (D) $r_s = 0.5\Omega$ in parallel with the galvanometer
28. At a specific instant emission of radioactive compound is deflected in a magnetic field. The compound can emit :
- I. Electrons II. Protons III. He^{2+} IV. Neutrons
- The emission at the instant can be :
- (A) I, II, III (B) I, II, III, IV (C) IV (D) II, III
29. A magnetic dipole is acted upon by two magnetic fields which are inclined to each other at an angle of 75° . One of the fields has a magnitude of $15 mT$. The dipole attains stable equilibrium at an angle of 30° with this field. The magnitude of the other field (in mT) is close to :
- (A) 1 (B) 11 (C) 36 (D) 1060

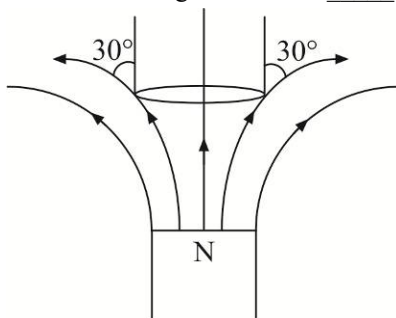
- *30.** A cubical region of space is filled with some uniform electric and magnetic fields. An electron enters the cube across one of its faces with velocity v and a positron enters via opposite face with velocity $-v$. At this instant.
- (A) The electric forces on both the particles cause identical accelerations
 (B) The magnetic forces on both the particles cause equal accelerations in magnitude
 (C) Both particles gain or loose energy at the same rate
 (D) The motion of the centre of mass (CM) is determined by B alone
- *31.** A charged particle would continue to move with a constant velocity in a region wherein,
- (A) $E = 0, B \neq 0$ (B) $E \neq 0, B \neq 0$ (C) $E \neq 0, B = 0$ (D) $E = 0, B = 0$
- 32.** A particle of charge $-16 \times 10^{-18} C$ moving with velocity 10 ms^{-1} along the x -axis enters a region where a magnetic field of induction B is along the y -axis and an electric field of magnitude 10^4 Vm^{-1} is along the negative z -axis. If the charged particle continues moving along the x -axis, the magnitude of B is :
- (A) 10^{-3} Wb/m^2 (B) 10^5 Wb/m^2 (C) 10^{16} Wb/m^2 (D) 10^3 Wb/m^2
- 33.** Two identical wires A and B , each of length ' l ' carry the same current I . Wire A is bent into a circle of radius R and wire B is bent to form a square of side ' a '. If B_A and B_B are the values of magnetic field at the centres of the circle and square respectively, then the ratio $\frac{B_A}{B_B}$ is :
- (A) $\frac{\pi^2}{8}$ (B) $\frac{\pi^2}{16\sqrt{2}}$ (C) $\frac{\pi^2}{16}$ (D) $\frac{\pi^2}{8\sqrt{2}}$
- 34.** A galvanometer having a coil resistance of 100Ω gives a full scale deflection, when a current of 1 mA is passed through it. The value of the resistance, which can convert this galvanometer into ammeter giving a full scale deflection for a current of 10 A , is :
- (A) 0.01Ω (B) 2Ω (C) 0.1Ω (D) 3Ω
- 35.** A charged particle of mass m and charge q travels on a circular path of radius r that is perpendicular to a magnetic field B . The time taken by the particle to complete one revolution is :
- (A) $\frac{2\pi m q}{B}$ (B) $\frac{2\pi q^2 B}{m}$ (C) $\frac{2\pi q B}{m}$ (D) $\frac{2\pi m}{q B}$
- 36.** To know the resistance G of a galvanometer by half deflection method, a battery of emf V_E and resistance R is used to deflect the galvanometer by angle θ . If a shunt of resistance S is needed to get half deflection then G , R and S are related by the equation.
- (A) $S(R + G) = RG$ (B) $2S(R + G) = RG$
 (C) $2G = S$ (D) $2S = G$
- 37.** Two concentric coils each of radius equal to $2\pi \text{ cm}$ are placed at right angles to each other. $3A$ and $4A$ are the currents flowing in each coil respectively. The magnetic induction in Wb/m^2 at the centre of the coils will be $\left(\mu_0 = 4\pi \times 10^{-7} \text{ Wb/A m}\right)$
- (A) 12×10^{-5} (B) 10^{-5} (C) 5×10^{-5} (D) 7×10^{-5}

INTEGER ANSWER TYPE QUESTIONS

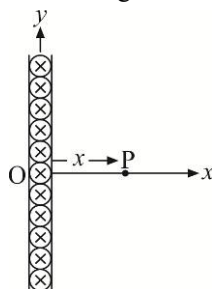
The Answers to the following questions are positive integers of 1/2/3 digits or zero.

- 38.** The maximum current a 2.0 mm diameter wire can carry if the magnetic field is not to exceed 0.08 T at its surface is _____ A.

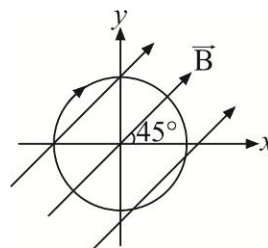
39. The magnitude of the Earth's magnetic field at the pole is approximately $7 \times 10^{-5} T$. Assuming that this field is produced by a current loop around the equator, the current that would generate such a field is $2 \times 10^n A$. The value of n is _____. Take radius of the earth as $6.4 \times 10^6 m$.
40. A solenoid of radius R is made of a long piece of wire of radius r , length $L (\gg R)$ and resistivity ρ . The magnetic field at the axis of the solenoid if the wire is connected to a battery of emf E is given by $\frac{\mu_0 E r^2}{k \rho L R}$. The value of k is _____.
41. In a hydrogen atom, the electron moves in an orbit of radius $0.5 \text{ \AA}$ making 10^{16} rev/s . The magnetic moment of the electron is given by $12.56 \times 10^{-n} \text{ Am}^2$.
42. A ring of mass 40 g radius 2 cm and carrying a current of 10 A floats in the magnetic field of a bar magnet when placed co-axially with the magnet. The magnitude of the magnetic field is $\underline{\quad} \underline{a} \underline{b} \times 10^{-2} T$. ($g = 10 \text{ m/s}^2$)



43. A large number of infinitely long thin wires, each carrying the same current, are kept along z -axis. The value of magnetic field ' B ' at a point P a distance x from the origin varies as $B \propto x^\lambda$. The value of λ is _____.



44. A circular loop of radius $\frac{2}{\sqrt{\pi}} m$ is placed in a uniform magnetic field of magnitude $100 T$ as shown in the figure. The loop carries a current of $1 A$. The magnetic of the torque acting on the loop is _____ Nm .



45. A steady current I gives through a wire loop PQR having shape of a right-angled triangle with $PQ = 3x$, $PR = 4x$ and $QR = 5x$. If the magnetic field at P due to this loop is $\frac{7\mu_0 I}{k\pi x}$, the value of k is:
46. A potential difference of $600 V$ is applied across the plates of a parallel plate capacitor. The separation between the plates is 3 mm . An electron projected parallel to the plates with a speed of $2 \times 10^4 \text{ m/s}$ passes undeviated. The magnitude of magnetic field within the plates is given by $10x T$. The value of x is:

47. A charge Q is distributed uniformly on a rod of length L . The rod rotates about an axis passing through one its ends and perpendicular to the rod. In this case, the magnetic moment of the rod is given by $\frac{Q\omega L^2}{\lambda}$. Find the value of λ .
48. A horizontal long wire carries a current of 200 A. Another wire of linear density $20 \times 10^{-3} \text{ kg m}^{-1}$ hangs below the long wire a distance of 2cm from it. Take $g = 10 \text{ m/s}^2$ and find the current in the second wire.
49. In a galvanometer, 5% of the total current passes through it. If the resistance of the galvanometer is G , the shunt resistance is $\frac{G}{k}$. The value of k is _____.
50. A long straight wire of radius a carries a current which is uniformly distributed over its cross-section. The ratio of magnetic fields at distances $\frac{a}{2}$ and $2a$ from the centre of the wire is $1:\lambda$. Find λ .
51. An α -particle (mass $= 6.4 \times 10^{-27} \text{ kg}$), having been accelerated by a potential difference of 10^4 V , enters normally a region of thickness 0.1 m having a transverse magnetic field of 0.1 T . In the field, it deviates by an angle θ° . Find the value of θ .
52. Two concentric circular coils A and B have radii 25 cm and 15 cm and carry currents 10 A and 15 A respectively in opposite directions A has 24 turns and B has 18 turns. The magnetic field at the centre is $a \underline{b} \underline{c} \mu_0 \text{ T}$. The sum $a + b + c$ is k . Find k .

Electromagnetic Induction

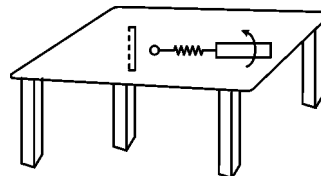
CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED '*' MAY HAVE MORE THAN ONE CORRECT OPTION.

1. A square of side L meters lies in the x - y plane in a region, where the magnetic field is given by $B = B_0(2\hat{i} + 3\hat{j} + 4\hat{k})T$, where B_0 is constant. The magnitude of flux passing through the square is :

(A) $2B_0L^2 \text{ Wb}$ (B) $3B_0L^2 \text{ Wb}$ (C) $4B_0L^2 \text{ Wb}$ (D) $\sqrt{29} B_0L^2 \text{ Wb}$

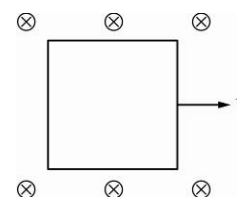
2. A metallic rod of length ℓ is tied to a string of length 2ℓ and made to rotate with angular speed ω on a horizontal table with one end of the string fixed. If there is a vertical magnetic field B in the region, the emf induced across the ends of the rod is :

(A) $\frac{2B\omega\ell^3}{2}$ (B) $\frac{3B\omega\ell^3}{2}$
(C) $\frac{4B\omega\ell^2}{2}$ (D) $\frac{5B\omega\ell^2}{2}$



3. A conducting square loop of side L and resistance R moves in its plane with a uniform velocity v perpendicular to one of its sides. A magnetic induction B , constant in time and space, pointing perpendicular to and into the plane of loop exists everywhere. The current induced in the loop is :

(A) BLv/R clockwise (B) BLv/R anticlockwise
(C) $2BLv/R$ anticlockwise (D) zero



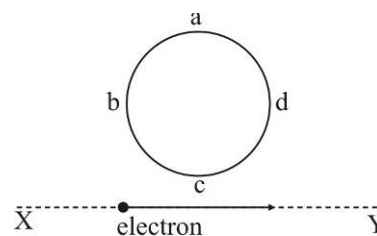
4. A rectangular loop of size $(2m \times 1m)$ is placed in x - y plane. A uniform but time varying magnetic field of strength $\vec{B} = (20t\hat{i} + 10t^2\hat{j} + 50\hat{k})T$ where t is time elapsed in second. The magnitude of induced emf (in V) at time t is :

(A) $20 + 20t$ (B) 20 (C) $20t$ (D) Zero

5. An electron moves on a straight line path XY as shown.

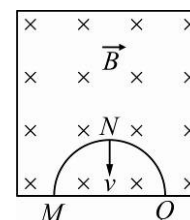
The $abcd$ is a coil adjacent to the path of electron. What will be the direction of current, if any induced in the coil ?

(A) The current will reverse its direction as the electron goes past the coil
(B) No current induced
(C) $abcd$ (D) $adcb$



6. A thin semicircular conducting ring of radius R is falling with its plane vertical in a horizontal magnetic induction $\vec{B}$. At the position MNQ the speed of the ring is v and the potential difference developed across the ring is :

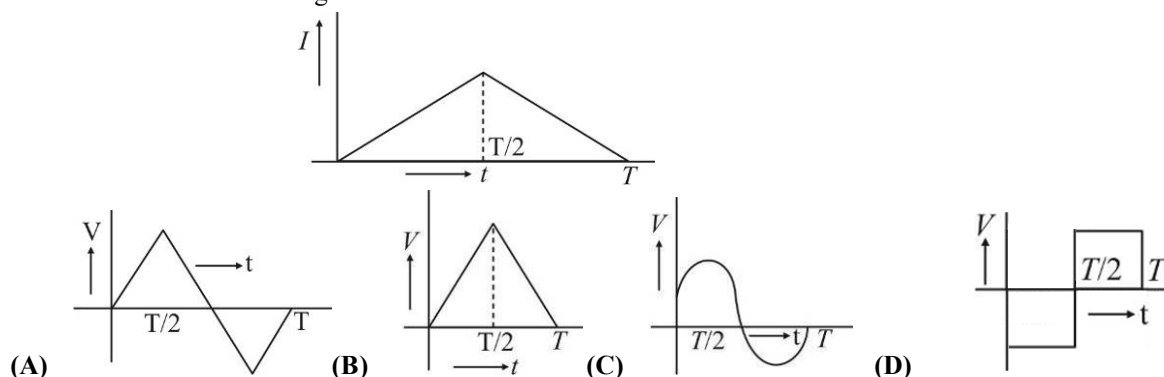
(A) Zero
(B) $Bv\pi R^2/2$ and M is at higher potential
(C) $2BRv$ and Q is at higher potential
(D) πBRv and Q is at higher potential



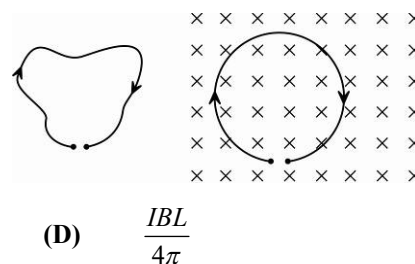
7. An ideal inductor of $10H$ is connected in series with a resistance of 5Ω and a battery of $5V$. $2s$ after the connection is made, the current flowing (in ampere) in the circuit is :

(A) $(1-e)$ (B) e (C) e^{-1} (D) $(1-e^{-1})$

- *8. A circular coil expands radially in region of magnetic field and no electromotive force is produced in the coil. This can be because
- The magnetic field is constant
 - The magnetic field is in the same plane as the circular coil and it may or may not vary
 - The magnetic field has a perpendicular (to the plane of the coil) component whose magnitude is decreasing suitably
 - There is a constant magnetic field in the perpendicular (to the plane of the coil) direction
9. A metal rod moves at a constant velocity in a direction perpendicular to its length. A constant uniform magnetic field exists in space in a direction perpendicular to the rod as well as its velocity. Select the correct statement(s) from the following.
- The entire rod is at the same electric potential
 - There is an electric field in the rod
 - The electric potential is highest at the centre of the rod and decrease towards its ends
 - The electric potential is lowest at the centre of the rod and increase towards its ends
- *10. The mutual inductance M_{12} of coil 1 with respect to coil 2
- Increases when they are brought nearer
 - Depends on the current passing through the coils
 - Increases when 1 of them is rotate about an axis
 - Is the same as M_{21} of coil 2 with respect to coil 1
11. The current (I) in the inductance is varying with time according to the plot shown in figure. Which one of the following the correct variation of voltage with time in the coil ?

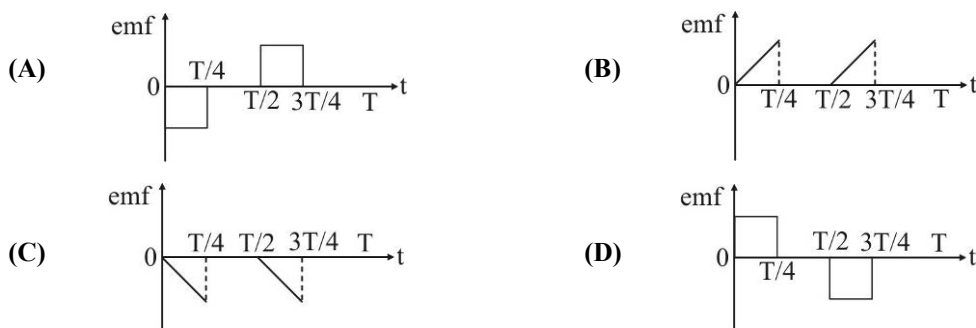
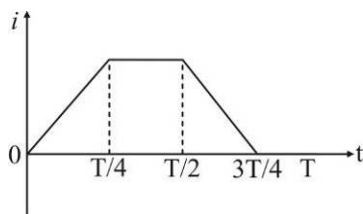


12. A choke coil should have :
- High inductance and high resistance
 - Low inductance and low resistance
 - High inductance and low resistance
 - Low inductance and high resistance
13. A thin flexible wire of length L is connected to two adjacent fixed point and the carries a current I in the clockwise direction, as shown in the figure. When the system is put in a uniform magnetic field of strength B going into the plane of the paper, the wire takes the shape of the circle. The tension in the wire is :



- IBL
- $\frac{IBL}{\pi}$
- $\frac{IBL}{2\pi}$
- $\frac{IBL}{4\pi}$

14. The current i in a coil varies with time as shown in the figure. The variation of induced emf with time would be :



15. In an LR circuit current at $t = 0$ is 20 A . After 2 s it reduces to 18 A . The time constant of the circuit is : (in second)

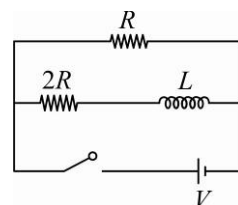
- (A) $\ln\left(\frac{10}{9}\right)$ (B) 2 (C) $\frac{2}{\ln\left(\frac{10}{9}\right)}$ (D) $2\ln\left(\frac{10}{9}\right)$

16. Dimensions of $\frac{\text{magnetic flux}}{\text{electric flux}}$ are :

- (A) $[LT^{-1}]$ (B) $[TL^{-1}]$ (C) $[L^3T^2A^{-2}]$ (D) $[M^0L^0T^0]$

17. The ratio of time constants during current growth and current decay of the circuit shown in figure is :

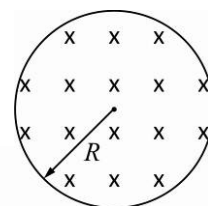
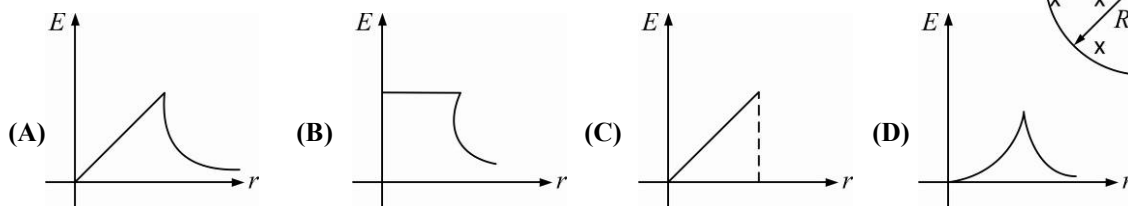
- (A) $1:1$ (B) $3:2$
(C) $2:3$ (D) $1:3$



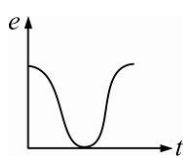
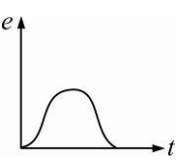
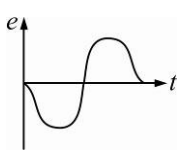
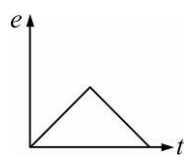
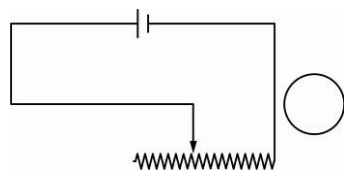
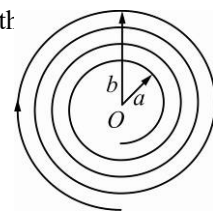
18. The self inductance L of a solenoid of length l and area of cross section A , with a fixed number of turns N increases as :

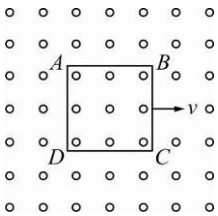
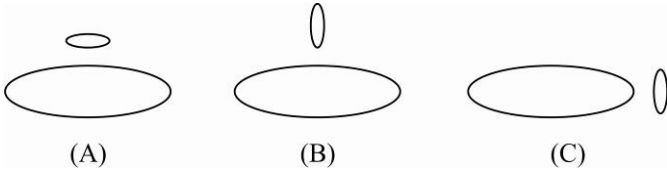
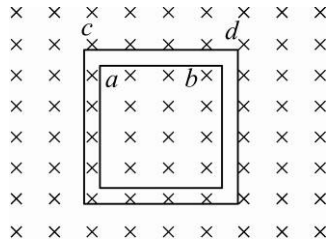
- (A) l and A increase (B) l decreases and A increases
(C) l increases and A decreases (D) both l and A decreases

19. A cylindrical space of radius R is filled with a uniform magnetic induction B parallel to the axis of the cylinder. If B changes at a constant rate, the graph showing the variation of induced electric field with distance r from the axis of cylinder is :



- *20. A metal plate is getting heated. It can be because
 (A) A direct current is passing through the plate (B) It is placed in a time varying magnetic field
 (C) It is placed in a space varying magnetic field, but does not vary with time
 (D) A current (either direct or alternating) is passing through the plate
21. **Assertion :** The SI unit of $\frac{T-m^2}{s}$ is equivalent to volts.
Reason : Both are the unit of potential difference.
 (A) A (B) B (C) C (D) D (E) E
22. **Assertion :** When number of turns in a coil is doubled, coefficient of self-inductance of the coil become 4 times.
Reason : This is because $L \propto N^2$.
 (A) A (B) B (C) C (D) D (E) E
- *23. An e.m.f. is produced in a coil, which is not connected to an external voltage source. This can be due to :
 (A) The coil being in a time varying magnetic field
 (B) The coil moving in time varying magnetic field
 (C) The coil moving in a constant magnetic field
 (D) The coil is stationary in external spatially varying magnetic field, which does not change with time
24. **Assertion :** A bulb connected in series with a solenoid is connected to AC source. If a soft iron core is introduced in the solenoid, the bulb will glow brighter.
Reason : On introducing soft iron core in the solenoid, the inductance increases.
 (A) A (B) B (C) C (D) D (E) E
25. The variation of induced emf (e) with time (t) in a coil if a

 Short bar magnet is moved along its axis with a constant velocity is best represented as :
 (A)  (B)  (C)  (D) 
26. **Assertion :** When resistance of rheostat is increased, clockwise current is induced in the ring.
Reason : Magnetic flux through the ring is out of the page and decreasing.
 (A) A (B) B (C) C
 (D) D (E) E

27. A long insulated copper wire is closely wound as a spiral of ' N ' turns. The spiral has inner radius ' a ' and outer radius ' b '. The spiral lies in the $X-Y$ plane and a steady current I flows through the wire. The Z -component of the magnetic field at the centre of the spiral is:
 (A) $\frac{\mu_0 N I}{2(b-a)} \ell n\left(\frac{b}{a}\right)$ (B) $\frac{\mu_0 N I}{2b} \ell n\left(\frac{b}{a}\right)$
 (C) $\frac{\mu_0 N I}{2(b-a)} \ell n\left(\frac{b+a}{b-a}\right)$ (D) $\frac{\mu_0 N I}{2b} \ell n\left(\frac{b+a}{b-a}\right)$


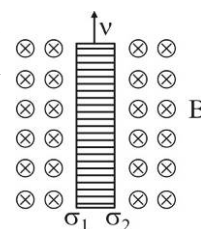
28. A metallic square loop $ABCD$ is moving in its own plane with velocity v in a uniform magnetic field perpendicular to its plane as shown in the figure. Electric field is induced :
- (A) in AD , but not in BC
 (B) in BC , but not in AD
 (C) neither in AD nor in BC
 (D) in both AD and BC
- 
29. In a uniform magnetic field of induction B , a wire in the form of semicircle of radius r rotates about the diameter of the circle with angular frequency ω . If the total resistance of the circuit is R , the mean power generated per period of rotation is:
- (A) $\frac{B\pi r^2 \omega}{2R}$ (B) $\frac{(B\pi r^2 \omega)^2}{8R}$ (C) $\frac{(B\pi r \omega)^2}{2R}$ (D) $\frac{(B\pi r \omega^2)^2}{8R}$
30. Two circular coils can be arranged in any of the three situations shown in the figure. Their mutual inductance will be :
- (A) Maximum in situation (A)
 (B) Maximum in situation (B)
 (C) Maximum in situation (C)
 (D) The same in all situations
- 
31. The figure shows certain wire segments joined together to form a coplanar loop. The loop is placed in a perpendicular magnetic field in the direction going into the plane of the figure. The magnitude of the field increases with time. I_1 and I_2 are the currents in the segments ab and cd . Then,
- (A) I_1 is in the direction ba and I_2 is in the direction dc
 (B) $I_1 < I_2$
 (C) I_1 is in the direction ba and I_2 is in the direction cd
 (D) $I_1 > I_2$
- 
32. Two identical circular loops of metal wire are lying on a table without touching each other. Loop A carries a current which increases with time. In response, the loop B :
- (A) remains stationary (B) is attracted by the loop A
 (C) is repelled by the loop A (D) rotates about its CM , with CM fixed
33. An infinitely long cylinder is kept parallel to a uniform magnetic field B direction along positive z -axis. The direction of induced current as seen from the z -axis will be :
- (A) Clockwise of the $+ve$ z -axis (B) Zero
 (C) Anticlockwise of the $+ve$ z -axis (D) Along the magnetic field
34. Two coaxial solenoids are made by winding thin insulated wire over a pipe of cross-sectional area $A = 10 \text{ cm}^2$ and length $= 20 \text{ cm}$. If one of the solenoids has 300 turns and the other 400 turns, their mutual inductance is : ($\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$)
- (A) $2.4 \pi \times 10^{-5} \text{ H}$ (B) $4.8 \pi \times 10^{-4} \text{ H}$ (C) $4.8 \pi \times 10^{-5} \text{ H}$ (D) $2.4 \pi \times 10^{-4} \text{ H}$
35. A conducting metal circular wire loop of radius r is placed perpendicular to a magnetic field which varies with time as $B = B_0 e^{-t/\tau}$, where B_0 and τ are constants, at time $t = 0$. If the resistance of the loop is R then the heat generated in the loop after a long time ($t \rightarrow \infty$) is :
- (A) $\frac{\pi^2 r^4 B_0^4}{2\tau R}$ (B) $\frac{\pi^2 r^4 B_0^2}{2\tau R}$ (C) $\frac{\pi^2 r^4 B_0^2 R}{\tau}$ (D) $\frac{\pi^2 r^4 B_0^2}{\tau R}$

36. A fighter plane of length 20 m , wing span (distance from tip of one wing to the tip of the other wing) of 15 m and height 5 m is flying towards east over Delhi. Its speed is 240 m s^{-1} . The earth's magnetic field over Delhi is $5 \times 10^{-5}\text{ T}$ with the declination angle $\sim 0^\circ$ and dip of θ such that $\sin \theta = 2/3$. If the voltage developed is V_B between the lower and upper side of the plane and V_w between the tips of the wings then V_B and V_w are close to :

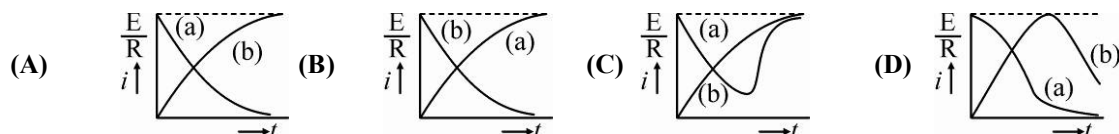
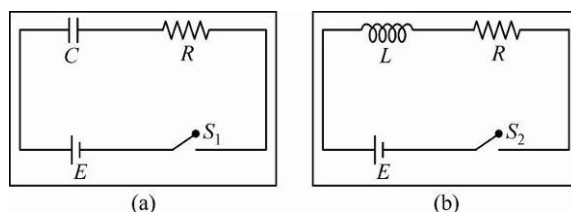
- (A) $V_B = 40\text{ mV}$; $V_w = 135\text{ mV}$ with left side of pilot at higher voltage
 (B) $V_B = 45\text{ mV}$; $V_w = 120\text{ mV}$ with right side of pilot at higher voltage
 (C) $V_B = 40\text{ mV}$; $V_w = 135\text{ mV}$ with right side of pilot at higher voltage
 (D) $V_B = 45\text{ mV}$; $V_w = 120\text{ mV}$ with left side of pilot at higher voltage

37. Consider a thin metallic sheet perpendicular to the plane of the paper moving with speed ' v ' in a uniform magnetic field B going into the plane of the paper (see figure). If charge densities σ_1 and σ_2 are induced on the left and right surfaces, respectively, of the sheet then (ignore fringe effects).

- (A) $\sigma_1 = \frac{-\epsilon_0 v B}{2}$, $\sigma_2 = \frac{\epsilon_0 v B}{2}$ (B) $\sigma_1 = \epsilon_0 v B$, $\sigma_2 = -\epsilon_0 v B$
 (C) $\sigma_1 = \frac{\epsilon_0 v B}{2}$, $\sigma_2 = \frac{-\epsilon_0 v B}{2}$ (D) $\sigma_1 = \sigma_2 = \epsilon_0 v B$



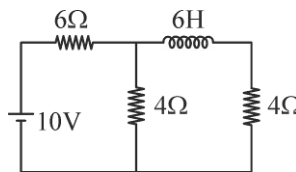
38. In the circuits (A) and (B) switches S_1 and S_2 are closed at $t = 0$ and are kept closed for a long time. The variation of currents in the two circuits for $t \geq 0$ are roughly shown by (figures are schematic and not drawn to scale)



INTEGER ANSWER TYPE QUESTIONS

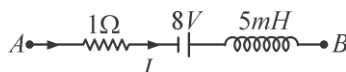
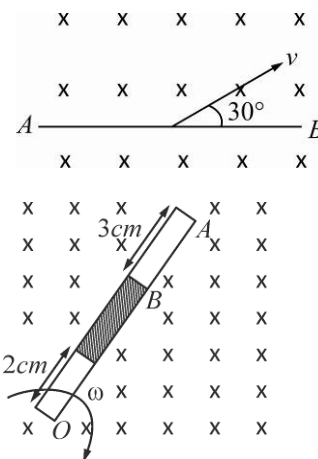
The Answers to the following questions are positive integers of 1/2/3 digits or zero.

39. A loop, made of straight edges has six corners at $A(0, 0, 0)$, $B(L, 0, 0)$, $C(L, L, 0)$, $D(0, L, 0)$, $E(0, L, L)$ and $F(0, 0, L)$. A magnetic field $B = B_0(\hat{i} + \hat{k})T$ is present in the region. The flux passing through the loop $ABCDEF$ (in that order) is $x B_0 L^2 Wb$. Find x .
40. In the given circuit if the ratio of i_1 to i_2 is x . Where i_1 is the initial (at $t = 0$) current, and i_2 is steady state (at $t = \infty$) current through the battery. Find $10x$.



41. A transformer (step up) with a $1:8$ turn ratio has 60 Hz , 120 V across the primary. The load in the secondary is 10^4 W . The current in the secondary is I (in Ampered). Find $10I$.

42. A uniformly wound solenoidal coil of self-inductance $1.8 \times 10^{-4} H$ and resistance 6Ω is broken up into two identical coils. These identical coils are then connected in parallel across a $15V$ battery of negligible resistance. The steady state current through the battery is? (in A)
43. A current of $2A$ is increasing at a rate of $4 A/s$ through a coil of inductance $2H$. The energy stored in the inductor per unit time at given instant is : (in J/s)
44. A coil of inductance $1 H$ and negligible resistance is connected to a source of DC supply, whose voltage is given by $V = 4t$. If the voltage is applied when $t = 0$, then find the energy stored in the coil in $4 s$ (in J) is:
45. A conducting rod AB of length $l = 1m$ is moving at a velocity $v = 4m/s$ making an angle 30° with its length. A uniform magnetic field $B = 2T$ exists in a direction perpendicular to the plane of motion. Then : $V_A - V_B = \underline{\hspace{2cm}} V$.
46. A rod of length $10 cm$ made up of conducting and non-conducting material (shaded part is non-conducting). The rod is rotated with constant angular velocity $10 rad/s$ about point O , in constant magnetic field of $2 T$ as shown in the figure. The induced emf between the points A and B of rod will be $\frac{x}{1000}$. Find x
47. The flux linked with a coil at any instant t is given by : $\phi = 10t^2 - 50t + 250$
The magnitude of induced emf at $t = 3 s$ is : (in Volts)
48. A coil of inductance $8.4mH$ and resistance 6Ω is connected to a $12V$ battery. The current in the coil is $1A$ at approximately the time : (in ms)
49. When current in a coil changes from $5A$ to $2A$ in $0.1s$, an average voltage of $50 V$ is produced. The self-inductance of the coil in Henry is L Find $100 L$.
50. In the circuit, what is potential difference $V_B - V_A$ (in V) when current I is $5 A$ and is decreasing at the rate of $10^3 A s^{-1}$?
51. A capacitor of capacity $2\mu F$ is charged to a potential difference of $12V$. It is then connected across an inductor of inductance $6\mu H$. What is the current (in A) in the circuit at a time when the potential difference across the capacitor is $6.0 V$?
52. Two coils, 1 and 2 have a mutual inductance $M = 5H$ and resistance $R = 10\Omega$ each. A current flows in coils as $I_1 = 2t^2$, where t is time. Find the total charge (in C) that has flow through coil 2 between $t = 0$ and $t = 2s$.
53. The current in a coil of self-induction 2.0 henry is increasing according to $i = 2 \sin t^2$ ampere. Find the amount of energy spent during the period when the current changes from 0 to 2 ampere.



AC Circuits

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED “*” MAY HAVE MORE THAN ONE CORRECT OPTION.

- If the *rms* current in a 50 Hz ac-circuit is 5 A, the value of the current 1/300 seconds after its value becomes zero is.
 (A) $5\sqrt{2} A$ (B) $5\sqrt{3/2} A$ (C) $5/6 A$ (D) $5/\sqrt{2} A$
- A resistance is connected to a capacitor in AC and the phase difference is $\frac{\pi}{4}$ between current and voltage. When the same resistance is connected to an inductor, phase difference becomes $\tan^{-1}(2)$. Power factor of the circuit when both capacitor and inductor are connected to the resistance will be :
 (A) 1 (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{2}$
- Which of the following combinations should be selected for better tuning of an LCR circuit used for communication?
 (A) $R = 20\Omega, L = 1.5 H, C = 35\mu F$ (B) $R = 25\Omega, L = 2.5 H, C = 45\mu F$
 (C) $R = 15\Omega, L = 3.5 H, C = 30\mu F$ (D) $R = 25\Omega, L = 1.5 H, C = 45\mu F$
- The output of a step-down transformer is measured to be 24 V when connected to a 12 watt light bulb. The value of the peak current is.
 (A) $1/\sqrt{2} A$ (B) $\sqrt{2} A$ (C) 2A (D) $2\sqrt{2} A$
- The figure shows an experimental plot discharging of a capacitor in an R-C circuit. The time constant τ of this circuit lies between :
 (A) 150 s and 200 s
 (B) 0 and 50 s
 (C) 50 s and 100s
 (D) 100 s and 150 s
- *6. As the frequency of an ac circuit increases, the current first increases and then decreases. What combination of circuit elements is most likely to comprise the circuit?
 (A) Inductor and capacitor (B) Resistor and inductor
 (C) Resistor and capacitor (D) Resistor, inductor and capacitor
- Alternating current can not be measured by DC ammeter because :
 (A) AC cannot pass through DC ammeter
 (B) AC changes direction
 (C) Average value of current for complete cycle is zero
 (D) DC ammeter will get damaged
- *8. In an alternating current circuit consisting of elements in series, the current increases on increasing the frequency of supply. Which of the following elements are likely to constitute the circuit?
 (A) Only resistor (B) Resistor and an inductor
 (C) Resistor and a capacitor (D) Only a capacitor

9. The core of any transformer is laminated so as to :
 (A) Reduce the energy loss due to eddy currents (B) Make it light weight
 (C) Make it robust and strong (D) Increase the secondary voltage
- *10. Electrical energy is transmitted over large distances at high alternating voltages. Which of the following statements is(are) correct?
 (A) For a given power level, there is a lower current
 (B) Lower current implies less power loss
 (C) Transmission lines can be made thinner
 (D) It is easy to reduce the voltage at the receiving end using step-down transformers
11. In an AC generator, a coil with N turns, all of the same area A and total resistance R , rotates with frequency ω in a magnetic field B . The maximum value of emf generated in the coil is :
 (A) $NABR\omega$ (B) NAB (C) $NABR$ (D) $NAB\omega$
- *12. For an LCR circuit, the power transferred from the driving source to the driven oscillator is $P = I^2 Z \cos \phi$.
 (A) Here, the power factor $\cos \phi \geq 0$, $P \geq 0$
 (B) The driving force can give no energy to the oscillator ($P = 0$) in some cases
 (C) The driving force cannot syphon out ($P < 0$) the energy out of oscillator
 (D) The driving force can take away energy out of oscillator
13. **Assertion :** Choke coil is preferred to a capacitor to reduce AC in a circuit.
Reason : Inductor acts as a leak-proof wattless resistor.
 (A) A (B) B (C) C (D) D (E) E
14. In an LCR circuit, capacitance is changed from C to $2C$. For the resonant frequency to remain unchanged, the inductance should be changed from L to ?
 (A) $4L$ (B) $2L$ (C) $L/2$ (D) $L/4$
15. **Assertion :** At series resonance, LCR circuit have a minimum current.
Reason : At resonance, in LCR circuit, the current and emf remain in phase with each other.
 (A) A (B) B (C) C (D) D (E) E
16. The phase difference between the alternating current and emf is $\pi/2$. Which of the following cannot be the constituent of the circuit?
 (A) C alone (B) L, C (C) R, L (D) L alone
- *17. The line that draws power supply to your house from street has
 (A) Zero average current
 (B) 220 V average voltage
 (C) Voltage and current out of phase by 90°
 (D) Voltage and current possibly differing in phase ϕ such that $|\phi| < \frac{\pi}{2}$
18. A resistance 'R' draws power 'P' when connected to an AC source. If an inductance is now placed in series with the resistance, such that the impedance of the circuit becomes 'Z', the power drawn will be :
 (A) $P\left(\frac{R}{Z}\right)$ (B) P (C) $P\left(\frac{R}{Z}\right)^2$ (D) $P\sqrt{\frac{R}{Z}}$
19. A transformer having efficiency of 90% is working on 200 V and 3kW power supply. If the current in the secondary coil 6A, the voltage across the secondary coil and the current in the primary coil respectively are :
 (A) 300V, 15 A (B) 450V, 15 A (C) 450V, 13.5A (D) 600V, 15A

20. In an electrical circuit R , L , C and ac voltage source are all connected in series. When L is removed from the circuit. The phase difference between the voltage and the current in the circuit is $\pi/3$ if instead, C is removed from the circuit, the phase difference is again $\pi/3$ the power factor of the circuit is :

(A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) 1 (D) $\frac{\sqrt{3}}{2}$

21. The transistor –oscillator using a resonant circuit with an inductor L (of negligible resistance) and a capacitor C in series produce oscillations of frequency f . If L is doubled and C is changed to $4C$, the frequency will be :

(A) $f/2$ (B) $f/4$ (C) $8f$ (D) $f/2\sqrt{2}$

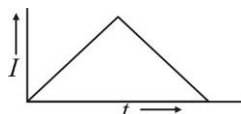
22. In a circuit L , C and R are connected in series with a alternating voltage source of frequency f . the current leads the voltage by 45° . The value of C is :

(A) $\frac{1}{\pi f(2\pi f L - R)}$ (B) $\frac{1}{2\pi f(2\pi f L - R)}$ (C) $\frac{1}{\pi f(2\pi f L + R)}$ (D) $\frac{1}{2\pi f(2\pi f L + R)}$

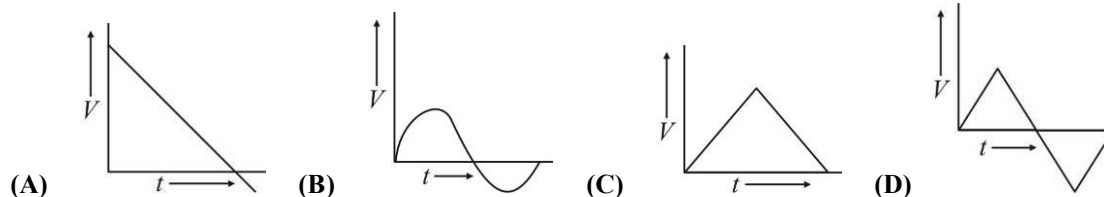
23. The power factor of an AC circuit with having resistance R and inductance L (connected in series) and an angular frequency ω is :

(A) $\frac{R}{\omega L}$ (B) $\frac{R}{(R^2 - \omega^2 L^2)^{1/2}}$ (C) $\frac{\omega L}{R}$ (D) $\frac{R}{(R^2 + \omega^2 L^2)^{1/2}}$

24. The current I in an A.C. circuit inductance coil varies with time according to the graph given below.



Which one of the following graphs gives the variation of voltage with time ?



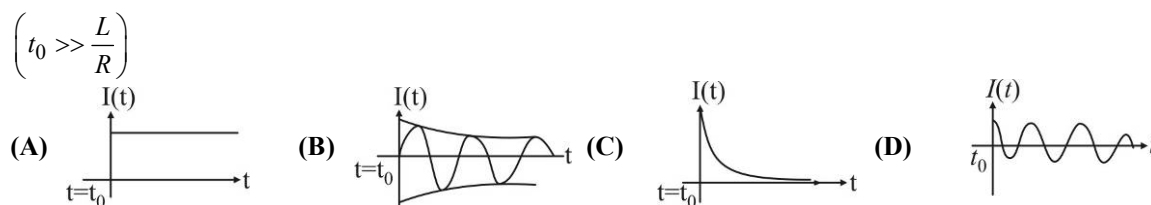
25. A condenser of capacity C is charged to a potential difference of V_1 . the plates of the condenser are then connected to an ideal inductor of inductance L . the current through the inductor when the potential difference across the condenser reduced to V_2 is :

(A) $\left(\frac{C(V_1 - V_2)^2}{L} \right)^{\frac{1}{2}}$ (B) $\frac{C(V_1^2 - V_2^2)}{L}$ (C) $\frac{C(V_1^2 + V_2^2)}{L}$ (D) $\left(\frac{C(V_1^2 - V_2^2)}{L} \right)^{\frac{1}{2}}$

26. An ac is given by equation $I = I_1 \cos \omega t + I_2 \sin \omega t$. The rms value of current is given by :

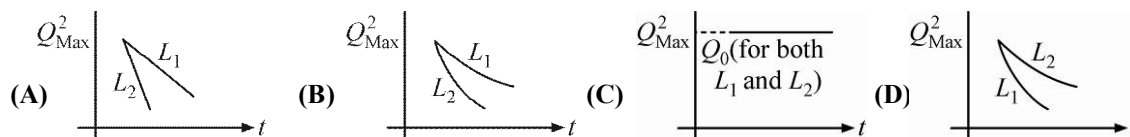
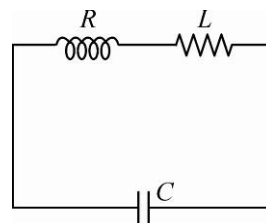
(A) $\frac{I_1 + I_2}{2}$ (B) $\frac{(I_1 + I_2)^2}{\sqrt{2}}$ (C) $\frac{1}{\sqrt{2}} \sqrt{I_1^2 + I_2^2}$ (D) $\frac{I_1^2 + I_2^2}{2}$

27. A series LR circuit is connected to a voltage source with $V(t) = V_0 \sin \omega t$. After very large time, current $I(t)$ behaves as



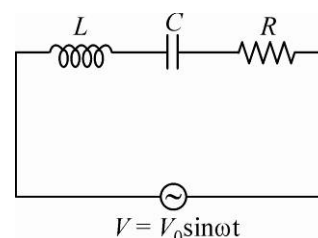
28. An LCR circuit is equivalent to a damped pendulum. In an LCR circuit the capacitor is charged to Q_0 and then connected to the L and R as shown here.

If a student plots graphs of the square of maximum charge (Q_{Max}^2) on capacitor with time (t) for two different value L_1 and L_2 ($L_1 > L_2$) of L then which of the following represents this graph correctly? (plots are schematic and not drawn to scale)



29. For the LCR circuit, shown here, the current is observed to lead the applied voltage. An additional capacitor C' , when joined with the capacitor C present in the circuit, makes the power factor of the circuit unity. The capacitor C' , must have been connected in :

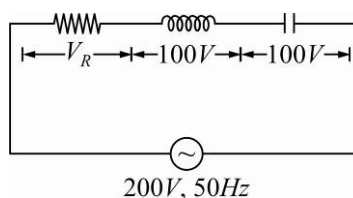
- (A) series with C and has a magnitude $\frac{1 - \omega^2 LC}{\omega^2 L}$
 (B) series with C and has a magnitude $\frac{C}{(\omega^2 LC - 1)}$
 (C) parallel with C and has a magnitude $\frac{C}{(\omega^2 LC - 1)}$
 (D) parallel with C and has a magnitude $\frac{1 - \omega^2 LC}{\omega^2 L}$



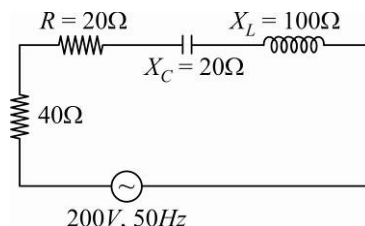
INTEGER ANSWER TYPE QUESTIONS

The Answers to the following questions are positive integers of 1/2/3 digits or zero.

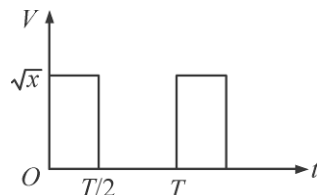
30. In an AC circuit the voltage applied is $E = E_0 \sin \omega t$. The resulting current in the circuit is $I = I_0 \sin \left(\omega t - \frac{\pi}{2} \right)$.
 The power consumption in the circuit is given by $x E_0 I$.
31. The self-inductance of the motor of an electric fans is $10 H$. In order to impart maximum power at $50 Hz$, it should be connected to a capacitance of : (in μF)
32. In a series resonant LCR circuit, the voltage across R is $100 V$ and $R = 1k\Omega$ with $C = 2\mu F$. The resonant frequency ω is $200 rad/s$. At resonance the voltage across L is : (in V)
33. In a transformer, number of turns in the primary are 140 and that in the secondary are 280 . If current in primary is $4 A$, then that in the secondary is: (in A)
34. By what percentage the impedance in an AC series circuit should be increased so that the power factor changes from $(1/2)$ to $(1/4)$ (when R is constant)?
35. In the circuit shown in figure value of V_R is : (in V)



36. The power factor of the circuit shown in the figure is P . Then Find $100 P$.



37. In an oscillating LC circuit the maximum charge on the capacitor is Q . The charge on the capacitor when the energy is stored equally between the electric and magnetic fields is $Q/\sqrt{x}$. Find x .
38. An AC series circuit has a resistance of 12Ω and an impedance of 15Ω . The power factor of the circuit will be P . Find $10 P$.
39. The *r.m.s* value of potential difference V shown in the figure is V_0 . Find x .



40. A sinusoidal alternating current of peak value I_0 passes through a heater of resistance R . Then the mean power output of heater is $\frac{I_0^2 R}{\sqrt{x}}$. Find x .
41. In an LCR series AC circuit, the voltage across each of the components L , C and R is $50V$. The voltage across the LC combination will be : (in V)
42. An ac voltage is represented by $E = 220\sqrt{2} \cos(50\pi)t$. How many times will the current becomes zero in the second?
43. In a series $L-C-R$ circuit $R = 200\Omega$ and the voltage and the frequency of the main supply is $220 V$ and $50 Hz$ respectively. On taking out the capacitance from the circuit the current lags behind the voltage by 30° . On taking out the inductor from the circuit the current leads the voltage by 30° . The power dissipated in the $L-C-R$ circuit is (in W)
44. An arc lamp requires a direct current of $10 A$ at $80 V$ to function. If it is connected to $220 V$ (rms), $50 Hz$ AC supply, the series inductor needed for it to work is close to $\frac{x}{1000} H$. Find x .

Light & Interference

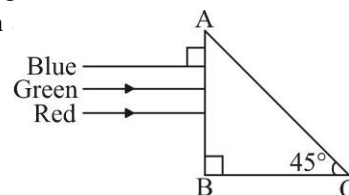
CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED “*” MAY HAVE MORE THAN ONE CORRECT OPTION.

1. A thin convex lens made from crown glass $\left(\mu = \frac{3}{2}\right)$ has focal length f . When it is measured in two different liquids having refractive indices $\frac{4}{3}$ and $\frac{5}{3}$, it has the focal length f_1 and f_2 , respectively. The correct relation between the focal lengths is :

- (A) $f_1 = f_2 < f$ (B) $f_1 > f$ and f_2 becomes negative
(C) $f_2 > f$ and f_1 become negative (D) f_1 and f_2 both become negative

2. A beam of light consisting of red, green and blue colours is incident on a right angled prism. The refractive index of the material of the prism for the above red, green blue wavelengths are 1.39, 1.44 and 1.47 respectively. The prism will :

- (A) Not separate the three colours at all
(B) Separate the red colour part from the green and blue colours
(C) Separate the blue colour part from the red and green colours
(D) Separate all the three colours from one another



3. At two point P and Q on screen in Young's double slit experiment, waves from slits S_1 and S_2 have a path difference 0 and $\frac{\lambda}{4}$ respectively. The ratio of intensities at P and Q will be :

- (A) 3 : 2 (B) 2 : 1 (C) $\sqrt{2} : 1$ (D) 4 : 1

4. At the first minimum adjacent to the central maximum of a single –slit diffraction pattern, the phase difference between the Huygen's wavelet from the edge of the slit and the wavelet from the midpoint of the slit is :

- (A) π radian (B) $\frac{\pi}{8}$ radian (C) $\frac{\pi}{4}$ radian (D) $\frac{\pi}{2}$ radian

5. In an astronomical telescope in normal adjustment a straight black line of length L is drawn on inside part of objective lens. The eye-piece forms a real image of this line. The length of this image is I .

The magnification of the telescope is

- (A) $\frac{L+I}{L-I}$ (B) $\frac{L}{I}$ (C) $\frac{L}{I} + 1$ (D) $\frac{L}{I} - 1$

6. A breaker contains water up to a height h_1 and kerosene of height h_2 above water so that the total height of (water + kerosene) is $(h_1 + h_2)$. Refractive index of water is μ_1 and that of kerosene is μ_2 . The apparent shift in the position of the bottom of the beaker when viewed from above is :

- (A) $\left(1 - \frac{1}{\mu_1}\right)h_2 + \left(1 - \frac{1}{\mu_2}\right)h_1$ (B) $\left(1 + \frac{1}{\mu_1}\right)h_1 + \left(1 + \frac{1}{\mu_2}\right)h_2$
(C) $\left(1 - \frac{1}{\mu_1}\right)h_1 + \left(1 - \frac{1}{\mu_2}\right)h_2$ (D) $\left(1 + \frac{1}{\mu_1}\right)h_2 - \left(1 + \frac{1}{\mu_2}\right)h_1$

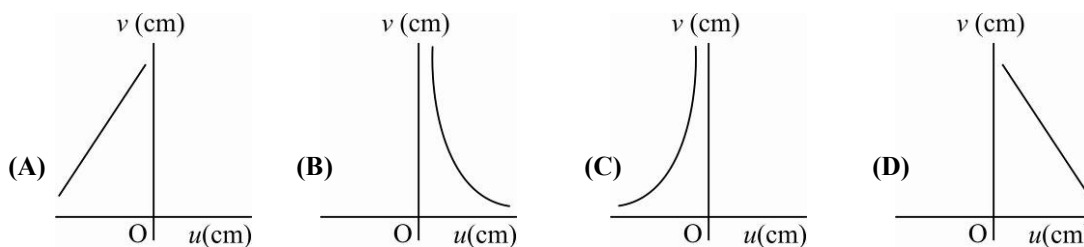
7. For a parallel beam of monochromatic light of wavelength ' λ ', diffraction is produced by a single slit whose width ' a ' is of the order of the wavelength of the light. If ' D ' is the distance of the screen from the slit, the width of the central maxima will be :
- (A) $\frac{Da}{\lambda}$ (B) $\frac{2Da}{\lambda}$ (C) $\frac{2D\lambda}{a}$ (D) $\frac{D\lambda}{a}$
8. Two identical thin plano-convex glass lenses (refractive index 1.5) each having radius of curvature of 20 cm are placed with their convex surface in contact at the centre. The intervening space is filled with oil of refractive index 1.7. The focal length of the combination is :
- (A) -50 cm (B) 50 cm (C) -20 cm (D) -25 cm
9. **Statement-I** On viewing the clear blue portion of the sky through a Calcite Crystal, the intensity of transmitted light varies as the crystal is rotated.
Statement-II The light coming from the sky is polarized due to scattering of sun light by particles in the atmosphere. The scattering is largest for blue light.
- (A) Statement I is false, Statement II is true
 (B) Statement I is true, Statement II is false
 (C) Statement I is true, Statement II is true Statement II is the correct explanation of Statement I
 (D) Statement I is true, Statement II is true; Statement II is not correct explanation of Statement I
10. The refracting angle of a prism is A , and refractive index of the material of the prism is $\cot(A/2)$. The angle of minimum deviation is :
- (A) $90^\circ - A$ (B) $180^\circ + 2A$ (C) $180^\circ - 3A$ (D) $180^\circ - 2A$
11. In a double slit experiment, the two slits are 1 mm apart and the screen is placed 1 m away. A monochromatic light of wavelength 500 nm is used. What will be the width of each slit for obtaining ten maxima of double slit within the central maxima of single slit pattern ?
- (A) 0.5 mm (B) 0.02 mm (C) 0.2 mm (D) 0.1 mm
12. A green light is incident from the water to the air-water interface at the critical angle (θ). Select the correct statement.
- (A) The entire spectrum of visible light will come out of the water at an angle of 90° to the normal
 (B) The spectrum of visible light whose frequency is less than that of green light will come out of the air medium
 (C) The spectrum of visible light whose frequency is more than that of green light will come out to the air medium.
 (D) The entire spectrum of visible light will come out of the water at various angles to the normal
13. In the Young's double slit experiment, the intensity of light at a point on the screen where the path difference λ is K , (λ being the wavelength of light used). The intensity at a point where the path difference is $\lambda/4$ will be :
- (A) K (B) $K/4$ (D) $K/2$ (D) zero
14. Let the $x-y$ plane be the boundary between two transparent media. Medium 1 in $z \geq 0$ has a refractive index of $\sqrt{2}$ and medium 2 with $z < 0$ has refractive index of $\sqrt{3}$. A ray of light in medium 1 given by the vector $\vec{A} = 6\sqrt{3}\hat{i} + 8\sqrt{3}\hat{j} - 10\hat{k}$ is incident on the plane of separation. The angle of refraction in medium 2 is :
- (A) 45° (B) 60° (C) 75° (D) 30°
15. If the focal length of objective lens is increased then magnifying power of
- (A) Microscope will increase but that of telescope decrease
 (B) Microscope and telescope both will increase.
 (C) Microscope and telescope both will decrease
 (D) Microscope will decrease but that of telescope will increase.

16. The angle of a prism is A . one of its refracting surface is silvered. Light rays falling at an angle of incidence $2A$ on the first surface returns back through the same path after suffering reflection at the silvered surface. The refractive index μ , of the prism is :
- (A) $2 \sin A$ (B) $2 \cos A$ (C) $\frac{1}{2} \cos A$ (D) $\tan A$
17. This question has paragraph followed by two statements, Statement I and Statement II. Of the given four alternatives after the statements, choose one that describes the statements.
A thin air film is formed by putting the convex surface of a plane-convex lens over a plane glass plate. With monochromatic light, this film gives an interference pattern due to light reflected from the top (convex) surface and the bottom (glass plate) surface of the film.
Statement 1 : When light reflects from the air-glass plate interface, the reflected wave suffers a phase change of π .
Statement 2 : The centre of the interference pattern is dark.
- (A) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1
(B) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1
(C) Statement 1 is false, Statement 2 is true
(D) Statement 1 is true, Statement 2 is false
18. A plano convex lens fits exactly into a plano concave lens. Their plane surfaces are parallel to each other, if lenses are made of different materials of refractive indices μ_1 and μ_2 and R is the radius of curvature of the curved surface of the lenses, then the focal length of the combination is :
- (A) $\frac{R}{(\mu_1 - \mu_2)}$ (B) $\frac{2R}{(\mu_2 - \mu_1)}$ (C) $\frac{R}{2(\mu_1 + \mu_2)}$ (D) $\frac{R}{2(\mu_1 - \mu_2)}$
19. For a normal eye, the cornea of eye provides a converging power of $40 D$ and the least converging power of the eye lens behind the cornea is $20 D$. using this information, the distance between the retina and the cornea-eye lens can be estimated to be :
- (A) 1.67 cm (B) 1.5 cm (C) 5 cm (D) 2.5 cm
20. A car is fitted with a convex side-view mirror of focal length 20 cm . A second car 2.8 m behind the first car is overtaking the first car at a relative speed of 15 m/s . The speed of the image of the second car as seen in the mirror of the first one is :
- (A) $\frac{1}{15} \text{ m/s}$ (B) 10 m/s (C) 15 m/s (D) $\frac{1}{10} \text{ m/s}$
21. A mixture of light, consisting of wavelength 590 nm and an unknown wavelength, illuminates Young's double slit and gives rise to two overlapping interference patterns on the screen. The central maximum of both lights coincide. Further, it is observed that the third bright fringe of known light coincides with the 4th bright fringe of the unknown light. From this data, the wavelength of the unknown light is :
- (A) 393.4 nm (B) 885.0 nm (C) 442.5 nm (D) 776.8 nm
22. A parallel beam of light of wavelength λ is incident normally on a narrow slit. A diffraction pattern formed on a screen placed perpendicular to the direction of the incident beam. At the second minimum of the diffraction pattern, the phase difference between the rays coming from the two edges of slit is :
- (A) 2π (B) 3π (C) 4π (D) $\pi\lambda$
23. A ray of light is incident at an angle of incidence I , on one face of a prism of angle A (assumed to be small) and emerges normally from the opposite face. If the refractive index of the prism is μ , the angle of incidence I , is nearly equal to :
- (A) μA (B) $\frac{\mu A}{2}$ (C) $\frac{A}{\mu}$ (D) $\frac{A}{2\mu}$

24. Two beams, A and B of plane polarized light with mutually perpendicular planes of polarization are seen through a Polaroid. From the position, when the beam A has maximum intensity (and beam B has zero intensity), a rotation of Polaroid through 30° makes the two beams appear equally bright. If the initial intensities of the two beams are I_A and I_B respectively, then I_A/I_B equals :

(A) 3 (B) $3/2$ (C) 1 (D) $1/3$

25. A student measures the focal length of a convex lens by putting an object pin at a distance u from the lens and measuring the distance v of the image pin. The graph between u and v plotted by the student should look like :



26. A concave mirror of focal length f_1 is placed at a distance of d from a convex lens of focal length f_2 . A beam of light coming from infinity and falling on this convex lens –concave mirror combination returns to infinity. The distance d must equal :

(A) $f_1 + f_2$ (B) $-f_1 + f_2$ (C) $2f_1 + f_2$ (D) $-2f_1 + f_2$

27. In an optics experiments, with the position of the object fixed, a student varies the position of a convex lens and for each position; the screen is adjusted to get a clear image of the object. A graph between the object distance u and the image distance v , from the lens, is plotted using the same scale for the two axes. A straight line passing through the origin and making an angle 135° with the x -axis meets the experimental curve at P . The magnitude of the coordinates of P will be:

(A) $(2f, 2f)$ (B) $\left(\frac{f}{2}, \frac{f}{2}\right)$ (C) (f, f) (D) $(4f, 4f)$

28. For the angle of minimum deviation of a prism to, be equal to its refracting angle, the prism must be made of a material whose refractive index:

(A) Lies between $\sqrt{2}$ and 1 (B) lies between 2 and $\sqrt{2}$
(C) is less than 1 (D) is greater than 2

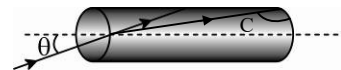
29. A biconvex lens has a radius of curvature of magnitude 20 cm . which one of the following options describe best the image formed of an object of height 2 cm placed 30 cm from the lens ?

(A) Virtual, upright, height = 1 cm (B) Virtual, upright, height = 0.5 cm
(C) Real, inverted, height = 4 cm (D) Real, inverted, height = 1 cm

30. A transparent solid cylinder rod has a refractive index of $\frac{2}{\sqrt{3}}$. It is surrounded by air. A light ray is incident at the mid point of one end of the rod as shown in the figure.

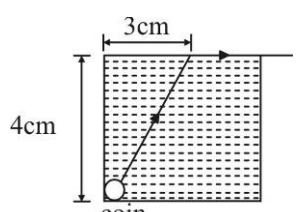
The incident angle θ for which the light ray grazes along the wall of the rod is :

(A) $\sin^{-1}\left(\frac{1}{2}\right)$ (B) $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$ (C) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (D) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

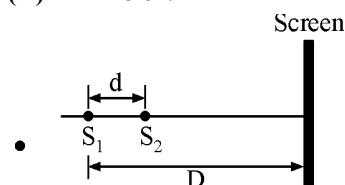
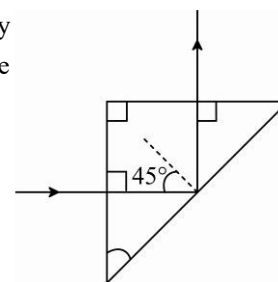
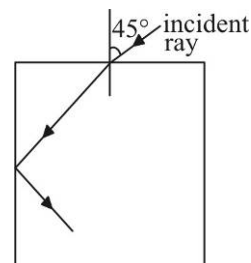


31. A thin prism of angle 15° made of glass of refractive index $\mu_1 = 1.5$ is combined with another prism of glass of refractive index $\mu_2 = 1.75$. The combination of the prisms produces dispersion without deviation. The angle of the second prism should be :

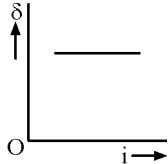
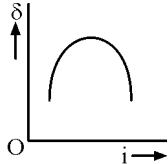
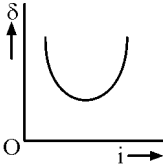
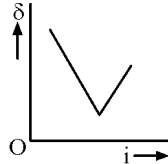
(A) 5° (B) 7° (C) 10° (D) 12°

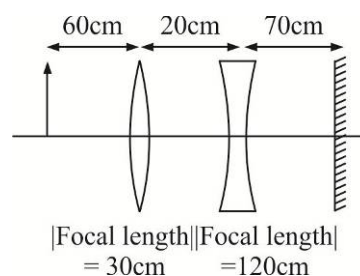
32. In a Young's double slit experiment the intensity at a point where the path difference is $\frac{\lambda}{6}$ (λ being the wavelength of the light used) is I . If I_0 denotes the maximum intensity, I/I_0 is equal to :
- (A) $\frac{1}{\sqrt{2}}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{1}{2}$ (D) $\frac{3}{4}$
33. A ray of light travelling in a transparent medium of refractive index μ , falls on a surface separating the medium from air at an angle of incidence of 45° . For which of the following value of μ the ray can undergo total internal reflection ?
- (A) $\mu = 1.33$ (B) $\mu = 1.40$ (C) $\mu = 1.50$ (D) $\mu = 1.25$
34. Two thin lenses of focal length f_1 and f_2 are in contact and coaxial. The power of the combination is :
- (A) $\frac{f_1 + f_2}{2}$ (B) $\frac{f_1 + f_2}{f_1 f_2}$ (C) $\sqrt{\frac{f_1}{f_2}}$ (D) $\sqrt{\frac{f_2}{f_1}}$
35. A Young's double slit experiment uses a monochromatic source. The shape of the interference fringe formed on a screen is :
- (A) Hyperbola (B) Circle (C) Straight line (D) Parabola
36. A small coin is resting on the bottom of a beaker filled with liquid. A ray of light from the coin travels upto the surface of the liquid and moves along its surface. How fast is the light travelling in the liquid ?
- (A) $2.4 \times 10^8 \text{ m/s}$ (B) $3.0 \times 10^8 \text{ m/s}$
(C) $1.2 \times 10^8 \text{ m/s}$ (D) $1.8 \times 10^8 \text{ m/s}$
- 
37. A fish looking up through the water sees the outside world, contained in a circular horizon. If the refractive index of water is $4/3$ and the fish is 12 cm below the water surface, the radius of this circle in cm is :
- (A) $36\sqrt{7}$ (B) $\frac{36}{\sqrt{7}}$ (C) $36\sqrt{5}$ (D) $4\sqrt{5}$
38. The angular resolution of a 10 cm diameter telescope at a wavelength of $5000 \text{ \AA}$ is of the order of :
- (A) 10^6 rad (B) 10^{-2} rad (C) 10^{-4} rad (D) 10^{-6} rad
39. If I_0 is the intensity of the principal maximum in the single slit diffraction pattern, then what will be its intensity when the slit width is doubled?
- (A) $2I_0$ (B) $4I_0$ (C) I_0 (D) $\frac{I_0}{2}$
40. A beam of unpolarised light of intensity I_0 is passed through a Polaroid A and then through another Polaroid B which is oriented so that its principle plane makes an angle of 45° relative to that of A . The intensity of the emergent light is
- (A) I_0 (B) $\frac{I_0}{2}$ (C) $\frac{I_0}{4}$ (D) $\frac{I_0}{8}$
41. The refractive index of the material of a prism is $\sqrt{2}$ and its refracting angle is 30° . One of the refracting surfaces of the prism is made a mirror inwards. A beam of monochromatic light entering the prism from the other face will retrace its path after reflection from the mirrored surface if its angle of incidence on the prism is :
- (A) 45° (B) 60° (C) 0 (D) 30°

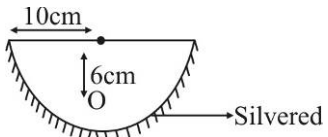
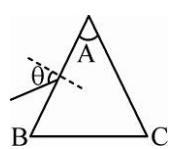
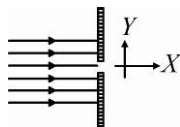
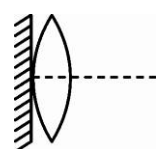
42. Two point white dots are 1 mm apart on a black paper. They are viewed by eye of pupil diameter 3 mm . Approximately, what is the maximum distance at which these dots can be resolved by the eye? [Take wavelength of light = 500 nm]
 (A) 5 m (B) 1 m (C) 6 m (D) 3 m
43. For the given incident ray as shown in figure, the condition of total internal refraction of this ray required refractive index of prism will be :
 (A) $\frac{\sqrt{3}+1}{2}$ (B) $\frac{\sqrt{2}+1}{2}$
 (C) $\sqrt{\frac{3}{2}}$ (D) $\sqrt{\frac{7}{6}}$
44. The refractive index of glass is 1.520 for red light and 1.525 for blue light. Let D_1 and D_2 be angles of minimum deviation for red and blue light respectively in a prism of this glass. Then :
 (A) $D_1 < D_2$ (B) $D_1 = D_2$
 (C) D_1 can be less than or greater than D_2 depending upon the angle of prism
 (D) $D_1 > D_2$
45. A light ray is incident perpendicular to one face of a 90° prism and totally internally reflected at the glass-air interface. If the angle of reflection is 45° , we conclude that for the refractive index n as :
 (A) $n < \frac{1}{\sqrt{2}}$ (B) $n > \sqrt{2}$
 (C) $n > \frac{1}{\sqrt{2}}$ (D) $n < \sqrt{2}$
46. A disc is placed on a surface of pond which has refractive index $5/3$. A source of light is placed 4 m below the surface of liquid. The minimum radius of disc needed so that light is not coming out is :
 (A) ∞ (B) 3 m (C) 6 m (D) 4 m
47. A plano-convex lens of refractive index 1.5 and radius of curvature 30 cm is silvered at the curved surface. Now, this lens has been used to form the image of an object. At what distance from this lens, an object be placed in order to have a real image of the size of the object?
 (A) 20 cm (B) 30 cm (C) 60 cm (D) 80 cm
48. A bubble in glass slab ($\mu = 1.5$) when viewed from one side appears at 5 cm and 2 cm from other side, then thickness of slab is :
 (A) 3.75 cm (B) 3 cm (C) 10.5 cm (D) 2.5 cm
49. Two coherent point source S_1 and S_2 are separated by a small distance d as shown. The fringes obtained on the screen will be :
 (A) Points (B) Straight line
 (C) Semi-circle (D) Concentric circles
50. A tall man of height 6 feet , want to see his full image. Then required minimum length of the mirror will be :
 (A) 12 feet (B) 3 feet (C) 6 feet (D) any length



51. When a un-polarised light of intensity I_0 is incident on a polarizing sheet, the intensity of the light which does not get transmitted is :
 (A) $\frac{1}{2}I_0$ (B) $\frac{1}{4}I_0$ (C) Zero (D) I_0
52. The angle of incidence at which reflected light is totally polarized for reflection from air to glass (refractive index n), is:
 (A) $\sin^{-1}(n)$ (B) $\sin^{-1}\left(\frac{1}{n}\right)$ (C) $\tan^{-1}\left(\frac{1}{n}\right)$ (D) $\tan^{-1}(n)$
53. A luminous object is placed at a distance of 30 cm from the convex lens of focal length 20 cm. on the other side of the lens, at what distance from the lens a convex mirror of radius of curvature 10 cm be placed in order to have an upright image of the object coincident with it ?
 (A) 50 cm (B) 30 cm (C) 12 cm (D) 60 cm
54. The maximum number of possible interference maxima for slit-separation equal to twice the wavelength in Young's double slit experiment, is :
 (A) Infinite (B) Five (C) Three (D) Zero
55. Light enters at an angle of incidence in transparent rod of refractive index n . for what value of the refractive index of the material of the rod the light once entered into it will not leave it through its lateral face whatsoever be the value of angle of incidence ?
 (A) $n = 1.1$ (B) $n = 1$ (C) $n > \sqrt{2}$ (D) $n = 1.3$
56. The image formed by an objective of a compound microscope is :
 (A) Virtual and Diminished (B) Real and Diminished
 (C) Real and Enlarged (D) Virtual and Enlarged
57. Light travels through a glass plate of thickness t and having a refractive index μ . If c is the velocity of light in vacuum, the time taken by light to travel this thickness of glass is :
 (A) $\frac{t}{\mu c}$ (B) $\frac{\mu t}{c}$ (C) $t\mu c$ (D) $\frac{tc}{\mu}$
58. Diameter of a plano-convex lens is 6 cm and thickness at the centre is 3mm. If speed of light in material of lens is $2 \times 10^8 \text{ m/s}$, the focal length of the lens is :
 (A) 15 cm (B) 20 cm (C) 30 cm (D) 10 cm
59. A lens is placed between a source of light and a wall. It forms images of area A_1 and A_2 on the wall, for its two different positions. The area of the source of light is :
 (A) $\frac{A_1 - A_2}{2}$ (B) $\frac{1}{A_1} + \frac{1}{A_2}$ (C) $\sqrt{A_1 A_2}$ (D) $\frac{A_1 + A_2}{2}$
60. An astronomical telescope has a large aperture to :
 (A) Reduce spherical aberration (B) Have high resolution
 (C) Increase span of observation (D) Have low dispersion
61. The earth radiates in the infra-red region of the spectrum. The spectrum is correctly given by :
 (A) Rayleigh Jeans law (B) Planck's law
 (C) Stefan's law of Radiation (D) Wien's law
62. In a fresnel biprism experiment, the two positions of lens given separation between the slits as 16 cm and 9 cm respectively. What is the actual distance of separation ?
 (A) 13 cm (B) 14 cm (C) 12.5 cm (D) 12 cm

63. Time taken by sunlight to pass through a window of thickness 4 mm whose refractive index is $\frac{3}{2}$ is :
 (A) $2 \times 10^{-4} \text{ s}$ (B) $2 \times 10^8 \text{ s}$ (C) $2 \times 10^{-11} \text{ s}$ (D) $2 \times 10^{11} \text{ s}$
64. The graph between angle of deviation (δ) and angle of incidence (i) for a triangular prism is represented by :
 (A)  (B)  (C)  (D) 
65. In Young's double slit experiment, the fringe width is found to be 0.4 mm . If the whole apparatus is immersed in water of refractive index $\frac{4}{3}$, without disturbing the geometrical arrangement, the new fringe width will be :
 (A) 0.30 mm (B) 0.40 mm (C) 0.53 mm (D) 450 microns
66. Unpolarized light of intensity I_0 is incident on surface of a block of glass at Brewster's angle. In that case, which one of the following statements is true?
 (A) transmitted light is partially polarized with intensity $I_0/2$
 (B) transmitted light is completely polarized with intensity less than $I_0/2$
 (C) reflected light is completely polarized with intensity less than $I_0/2$
 (D) reflected light is partially polarized with intensity $I_0/2$
67. Wavelength of light used in an optical instrument are $\lambda_1 = 4000 \text{ \AA}$ and $\lambda_2 = 5000 \text{ \AA}$, then ratio of their respective resolving powers (corresponding to λ_1 and λ_2) is :
 (A) $16 : 25$ (B) $9 : 1$ (C) $4 : 5$ (D) $5 : 4$
68. An observer looks at a distant tree of height 10 m with a telescope of magnifying power of 20. To the observer the tree appears :
 (A) 10 times taller (B) 10 times nearer (C) 20 times taller (D) 20 times nearer
69. The box of a pin hole camera, of length L , has a hole of radius a . It is assumed that when the hole is illuminated by a parallel beam of light of wavelength λ the spread of the spot (obtained on the opposite wall of the camera) is the sum of its geometrical spread and the spread due to diffraction. The spot would then have its minimum size (say $b_{\min}$) when :
 (A) $a = \frac{\lambda^2}{L}$ and $b_{\min} = \left(\frac{2\lambda^2}{L}\right)$ (B) $a = \sqrt{\lambda L}$ and $b_{\min} = \left(\frac{2\lambda^2}{L}\right)$
 (C) $a = \sqrt{\lambda L}$ and $b_{\min} = \sqrt{4\lambda L}$ (D) $a = \frac{\lambda^2}{L}$ and $b_{\min} = \sqrt{4\lambda L}$
70. A convex lens, of focal length 30 cm , a concave lens of focal length 120 cm , and a plane mirror are arranged as shown. For an object kept at a distance of 60 cm from the convex lens, the final image, formed by the combinations, is a real image, at a distance of :
 (A) 60 cm from the convex lens
 (B) 60 cm from the convex lens
 (C) 70 cm from the concave lens
 (D) 70 cm from the concave lens
71. To determine refractive index of glass slab using a travelling microscope, minimum number of readings required are :
 (A) Two (B) Four (C) three (D) five



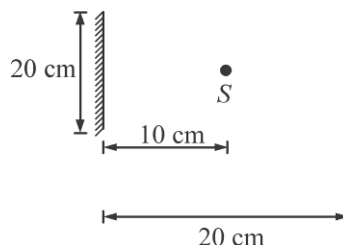
72. A hemispherical glass body of radius 10 cm and refractive index 1.5 is silvered on its curved surface. A small air bubble is 6 cm below the flat surface inside it along the axis. The position of the image of the air bubble made by the mirror is seen.
- (A) 14 cm below flat surface (B) 20 cm below flat surface
(C) 16 cm below flat surface (D) 30 cm below flat surface
- 
73. Two stars are 10 light years away from the earth. They are seen through a telescope of objective diameter 30 cm . The wavelength of light is 600 nm . To see the stars just resolved by the telescope, the minimum distance between them should be $(1\text{ light year} = 9.46 \times 10^{15}\text{ m})$ of the order of
- (A) 10^8 km (B) 10^{10} km (C) 10^{11} km (D) 10^6 km
74. Assuming human pupil to have a radius of 0.25 cm and a comfortable viewing distance of 25 cm , the minimum separation between two objects that human eye can resolve at 500 nm wavelength is :
- (A) $100\text{ }\mu\text{m}$ (B) $300\text{ }\mu\text{m}$ (C) $1\text{ }\mu\text{m}$ (D) $30\text{ }\mu\text{m}$
75. On a hot summer night, the refractive index of air is smallest near the ground and increases with height from the ground. When a light beam is directed horizontally, the Huygens' principle leads us to conclude that as it travels, the light beam:
- (A) bends downwards (B) bends upwards
(C) becomes narrower (D) goes horizontally without any deflection
76. Monochromatic light is incident on a glass prism of angle A . If the refractive index of the material of the prism is μ , a ray, incident at an angle θ , on the face AB would get transmitted through the face AC of the prism provided.
- (A) $\theta > \cos^{-1} \left[\mu \sin \left(A + \sin^{-1} \left(\frac{1}{\mu} \right) \right) \right]$ (B) $\theta < \sin^{-1} \left[\mu \sin \left(A + \sin^{-1} \left(\frac{1}{\mu} \right) \right) \right]$
(C) $\theta > \sin^{-1} \left[\mu \sin \left(A - \sin^{-1} \left(\frac{1}{\mu} \right) \right) \right]$ (D) $\theta < \cos^{-1} \left[\mu \sin \left(A - \sin^{-1} \left(\frac{1}{\mu} \right) \right) \right]$
- 
77. You are asked to design a shaving mirror assuming that a person keeps it 10 cm from his face and view the magnified image of the face at the closest comfortable distance of 25 cm . The radius of curvature of the mirror would then be:
- (A) 30 cm (B) 24 cm (C) 60 cm (D) -24 cm
78. A parallel beam of electrons travelling in x -direction falls on a slit of width d (see figure). If after passing the slit, an electron acquires momentum p_y in the y -direction then for a majority of electrons passing through the slit (h is plank's constant)
- (A) $|p_y| d = h$ (B) $|p_y| d > h$ (C) $|p_y| d < h$ (D) $|p_y| d \gg h$
- 
79. A telescope has an objective lens of focal length 150 cm and an eyepiece of focal length 5 cm . If a 50 m tall tower at a distance of 1 km is observed through this telescope in normal setting, the angle formed by the image of the tower is θ , than θ is closed to :
- (A) 1° (B) 15° (C) 30° (D) 60°
80. A thin convex lens of focal length f is put on a plane mirror as shown in the figure. When an object is kept at a distance a from the lens - mirror combination, its image is formed at a distance $a/3$ in front the combination. The value of a is:
- (A) f (B) $2f$ (C) $3f$ (D) $\frac{3}{2}f$
- 

81. In a Young's double slit experiment with light of wavelength λ the separation of slits is d and distance of screen is D such that $D \gg d \gg \lambda$. If the fringe width is β , the distance from point of maximum intensity to the point where intensity falls to half of maximum intensity on either side is :
- (A) $\frac{\beta}{2}$ (B) $\frac{\beta}{4}$ (C) $\frac{\beta}{3}$ (D) $\frac{\beta}{6}$

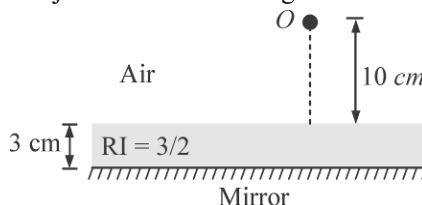
INTEGER ANSWER TYPE QUESTIONS

The Answers to the following questions are positive integers of 1/2/3 digits or zero.

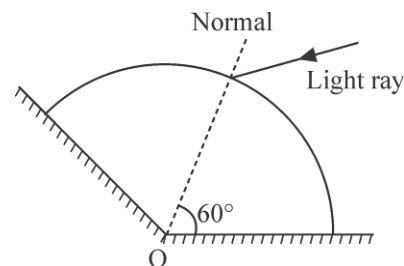
82. Two plane mirrors each 1.6 m long, are facing each other. The distance between the mirrors is 20 cm. A light incident on one end of one of the mirrors at an angle of incidence of 30° . How many times is the ray reflected before it reaches the other end?
83. A point source of light S is placed at a distance 10 cm in front of the centre of a mirror of width 20 cm suspended vertically on a wall. An insect walks with a speed 10 m/s in front of the mirror along a line parallel to the mirror at a distance 20 cm from it as shown in figure. Find the maximum time during which the insect can see the image of the source S in the mirror.



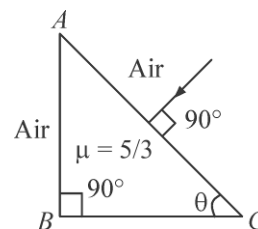
84. A point source of light S is placed on the major optical axis of the concave mirror at a distance of 60 cm. At what distance from the concave mirror should a flat mirror be placed for the rays to converge again at the point S having been reflected from the concave mirror and then from the flat one? The radius of the concave mirror is 80 cm.
85. Find the position of final image of an object. O as shown in figure



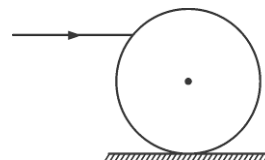
86. A fish is rising up vertically inside a pond with velocity 4 cm/s, and notices a bird, which is diving downward and its velocity appears to be 16 cm/s (to the fish). What is the real velocity of of diving bird, if refractive index of water is $4/3$?
87. Consider the situation as shown in figure. The point O is the centre. The light ray forms an angle of 60° with the normal. The normal makes an angle 60° with the horizontal and each mirror makes an angle 60° with the normal. The value of refractive index of that spherical portion so that light ray retraces its path is $\sqrt{n}$, then n is :



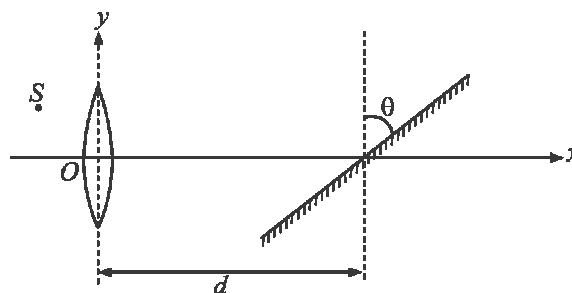
88. A ray of light makes normal incidence of the diagonal face of a right angled prism as shown in figure. If $\theta = 37^\circ$, then the angle of deviation after second step (from AB) is ($\sin 37^\circ = 3/5$)



89. A cylindrical glass rod of radius 0.1 m , and refractive index $\sqrt{3}$ lies on a horizontal plane mirror. A horizontal ray of light moving perpendicular to the axis of the rod is incident on it. At what height from the mirror should the ray be incident so that it leaves the rod at a height of 0.1 m above the plane mirror?

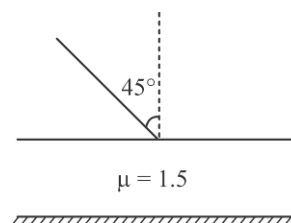


90. A thin converging lens of focal length $f = 1.5\text{ m}$ is placed along y -axis such that its optical centre coincides with the origin. A small light source S is placed at $(-2.0\text{ m}, 0.1\text{ m})$. Where should a plane mirror inclined at an angle θ , $\tan \theta = 0.3$ be placed such that y -coordinate of final image is 0.3 be placed such that y -coordinate of final image is 0.3 m , i.e. find d .



91. Two waves of equal frequencies have their amplitude in the ratio of $5 : 3$. They are superimposed on each other. Calculate the ratio of the maximum to minimum intensities of the resultant wave.
92. A Young's double slit apparatus has slits separated by 0.25 mm and a screen 48 cm away from the slits. The whole apparatus is immersed in water and the slits are illuminated by the red light ($\lambda = 700\text{ nm}$ in vacuum). Find the fringe width of the pattern formed on the screen. ($\mu_w = 4/3$) in mm

93. One side of a glass slab is silvered as shown in figure. A ray of light is incident on the other side at an angle of incidence $i = 45^\circ$. Refractive index of glass is given as 1.5 .



The deviation of the ray of light from its initial path when it comes out of the slab is:

94. A ball is dropped from a height of 20 m above the surface of water in a lake. The refractive index of water is $4/3$. A fish inside the lake, in the line of fall of the ball, is looking at the ball. At an instant, when the ball is 12.8 m above the water surface, the fish see the speed of ball as.
95. The magnification of an object placed in front of a convex lens of focal length 20 cm is $+2$. To obtain a magnification of -2 , the object has to be moved a distance equal to :

Modern Physics

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED '*' MAY HAVE MORE THAN ONE CORRECT OPTION.

1. The radiation corresponding to $3 \rightarrow 2$ transition of hydrogen atom falls on a metal surface to produce photoelectrons. These electrons are made to enter a magnetic field of $3 \times 10^{-4} \text{ T}$. If the radius of the largest circular path following by these electrons is 10.0 mm , the work function of the metal is close to :

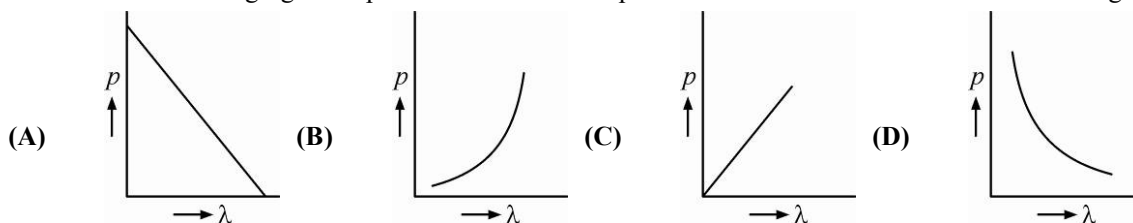
(A) 1.8 eV (B) 1.1 eV (C) 0.8 eV (D) 1.6 eV

2. A photoelectric surface is illuminated successively by monochromatic light of wavelength λ and if the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case, the work function of the surface of the material is :

(h = Planck's constant, c = speed of light)

(A) $\frac{2hc}{\lambda}$ (B) $\frac{hc}{3\lambda}$ (C) $\frac{hc}{2\lambda}$ (D) $\frac{hc}{\lambda}$

3. Which of the following figures represent the variation of particle momentum and the associated de-Broglie wavelength?



4. A certain metallic surface is illuminated with monochromatic light of wavelength, λ . The stopping potential for photoelectric current for this light is $3V_0$. If the same surface is illuminated with light of wavelength 2λ , the stopping potential is V_0 . The threshold wavelength for this surface for photo-electric effect is :

(A) $\frac{\pi}{4}$ (B) $\frac{\lambda}{6}$ (C) 6λ (D) 4λ

5. This question has Statement-I and Statement-II of the four choices given after the statements, choose the one that best describes the two statements.

Statement-1 A metallic surface is irradiated by a monochromatic light of frequency $\nu > \nu_0$ (the threshold frequency). The maximum kinetic energy and the stopping potential are $K_{\max}$ and V_0 respectively. If the frequency incident on the surface is doubled, both the $K_{\max}$ and V_0 are also doubled.

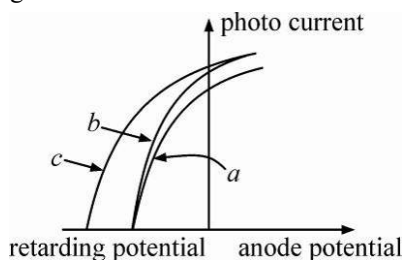
Statement-2 The maximum kinetic energy and stopping potential of photoelectrons emitted from a surface are linearly dependent on the frequency of incident light.

- (A) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1
 (B) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1
 (C) Statement 1 is false, Statement 2 is true
 (D) Statement 1 is true, Statement 2 is false

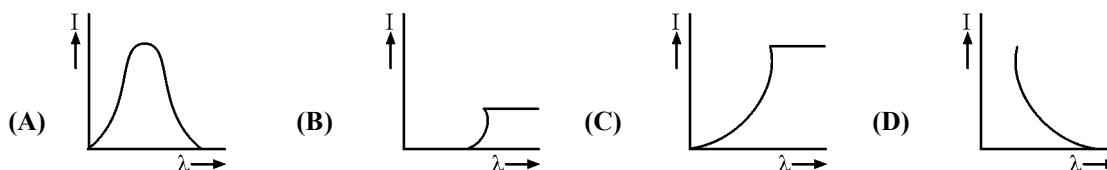
6. The de-Broglie wavelength of neutrons in thermal equilibrium at temperature T is :

(A) $\frac{3.08}{\sqrt{T}} \text{ \AA}$ (B) $\frac{0.308}{\sqrt{T}} \text{ \AA}$ (C) $\frac{0.0380}{\sqrt{T}} \text{ \AA}$ (D) $\frac{30.8}{\sqrt{T}} \text{ \AA}$

7. Electrons used in an electron microscope are accelerated by a voltage of 25 kV . If the voltage is increased to 100 kV then the de-Broglie wavelength associated with the electrons would :
- (A) increase by 2 times (B) decrease by 2 times
(C) decrease by 4 times (D) increase by 4 times
8. The surface of a metal is illuminated with the light of 400 nm . The kinetic energy of the ejected photoelectrons was found to be 1.68 eV . The work function of the metal is : ($hc = 1240\text{ eV}\cdot\text{nm}$)
- (A) 3.09 eV (B) 1.42 eV (C) 151 eV (D) 1.68 eV
9. A beam of cathode rays is subjected to crossed electric (E) and magnetic fields (B). The fields are adjusted such that the beam is not deflected. The specific charge of the cathode rays is given by :
- (A) $\frac{B^2}{2VE^2}$ (B) $\frac{2VB^2}{E^2}$ (C) $\frac{2VE^2}{B^2}$ (D) $\frac{E^2}{2VB^2}$
10. If the kinetic energy of a free electron doubles, its de-Broglie wavelength changes by the factor :
- (A) $1/2$ (B) 2 (C) $\frac{1}{\sqrt{2}}$ (D) $\sqrt{2}$
11. The figure shows a plot of photo current versus anode potential for a photo sensitive surface for three different radiations. Which one of the following is a correct statement :



- (A) Curves a and b represent incident radiations of same frequency but different intensities.
(B) Curves b and c represent incident radiations of different frequency and different intensities.
(C) Curves b and c represent incident radiations of same frequency having same intensity.
(D) Curves a and b represent incident radiations of different frequencies and different intensities.
12. Monochromatic light of frequency $6.0 \times 10^{14}\text{ Hz}$ is produced by a laser. The power emitted is $2 \times 10^{-3}\text{ W}$. The number of photons emitted, on an average, by the source per second is :
- (A) 5×10^{16} (B) 5×10^{17} (C) 5×10^{14} (D) 5×10^{15}
13. A photosensitive metallic surface has work function, $h\nu_0$. If photons of energy $2h\nu_0$ fall on this surface, the electrons come out with a maximum velocity $4 \times 10^6\text{ m/s}$. When the photon energy is increased to $5h\nu_0$, then maximum velocity of photoelectrons will be :
- (A) $2 \times 10^7\text{ m/s}$ (B) $2 \times 10^6\text{ m/s}$ (C) $8 \times 10^6\text{ m/s}$ (D) $8 \times 10^5\text{ m/s}$
14. The anode voltage of a photocells kept fixed. The wavelength λ of the light falling on the cathode is gradually changed. The plate current I of photocell varies as follows.



15. By photoelectric effect, Einstein proved:
 (A) $E = h\nu$ (B) $K.E. = \frac{1}{2}mv^2$ (C) $E = mc^2$ (D) $E = \frac{-Rhc^2}{n^2}$
16. In a photo-emissive cell, with exciting wavelength λ , the fastest electron has speed v . If the exciting wavelength is changed to $\frac{3\lambda}{4}$, the speed of the fastest emitted electron will be :
 (A) less than $v(4/3)^{1/2}$ (B) $v(3/4)^{1/2}$
 (C) $v(3/4)^{1/2}$ (D) greater than $v(4/3)^{1/2}$
17. According to Einstein's photoelectric equation the plot of the kinetic energy of the emitted photoelectrons from a metal V_s the frequency of the incident radiation gives a straight line whose slope :
 (A) Depends on the nature of the metal used
 (B) Depends on the intensity of the radiation
 (C) Depends both on the intensity of the radiation and the metal used
 (D) Is the same for all metals and independent of the intensity of the radiation
18. An electron beam has a kinetic energy equal to 100 eV . Find its wavelength associated with a beam, if mass of electron $= 9.1 \times 10^{-31} \text{ kg}$ and $1\text{ eV} = 1.6 \times 10^{-19} \text{ J/eV}$. (Plank's constant $= 6.6 \times 10^{-34} \text{ Js}$)
 (A) $24.6 \text{ \AA}$ (B) $0.12 \text{ \AA}$ (C) $1.2 \text{ \AA}$ (D) $6.3 \text{ \AA}$
19. An electron of mass m and charge e is accelerated from rest through a potential difference V in vacuum. Its final velocity will be :
 (A) $\sqrt{\frac{2eV}{m}}$ (B) $\sqrt{\frac{eV}{m}}$ (C) $\frac{eV}{2m}$ (D) $\frac{eV}{m}$
20. An electron of mass m , when accelerated through a potential difference V , has de-Broglie wavelength λ . The de-Broglie wavelength associated with a proton of mass M accelerated through the same potential difference, will be :
 (A) $\lambda \frac{M}{m}$ (B) $\lambda \frac{m}{M}$ (C) $\lambda \sqrt{\frac{M}{m}}$ (D) $\lambda \sqrt{\frac{m}{M}}$
21. The time taken by a photoelectron to come out after the photon strikes is approximately :
 (A) 10^{-4} s (B) 10^{-10} s (C) 10^{-16} s (D) 10^{-1} s
22. The wavelength of a 1 keV photon is $1.24 \times 10^{-9} \text{ m}$. What is the frequency of 1 MeV proton ?
 (A) 1.24×10^{15} (B) 2.4×10^{20} (C) 1.24×10^{18} (D) 2.4×10^{23}
23. The work function of a substance is 4.0 eV . The longest wavelength of light that can cause photoelectron emission from this substance is approximately :
 (A) 540 nm (B) 400 nm (C) 310 nm (D) 220 nm
24. The momentum of a photon of an electromagnetic radiation is $3.3 \times 10^{-29} \text{ kg ms}^{-1}$. What is the frequency of the associated waves? [$h = 6.6 \times 10^{-34} \text{ Js}$; $c = 3 \times 10^8 \text{ ms}^{-1}$]
 (A) $1.5 \times 10^{23} \text{ Hz}$ (B) $7.5 \times 10^{12} \text{ Hz}$ (C) $6 \times 10^3 \text{ Hz}$ (D) $9 \times 10^{-32} \text{ Hz}$

25. Hydrogen (${}_1\text{H}^1$), deuterium (${}_1\text{H}^2$), singly ionized helium (${}_2\text{He}^4$)⁺ and doubly ionized lithium (${}_3\text{Li}^8$)⁺⁺ all have one electron around the nucleus. Consider an electron transition from $n = 2$ to $n = 1$. If $\lambda_1, \lambda_2, \lambda_3$ and λ_4 respectively for four elements, then approximately which one of the following is correct?
- (A) $4\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$ (B) $\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$
 (C) $\lambda_1 = \lambda_2 = 4\lambda_3 = 9\lambda_4$ (D) $\lambda_1 = 2\lambda_2 = 3\lambda_3 = 4\lambda_4$
26. The electric potential between a proton and an electron is given by $V = V_0 \ln \frac{r}{r_0}$, where r_0 is a constant. Assuming Bohr's model to be applicable, what is the variation of r_n with n , (n being the principal quantum number) ?
- (A) $r_n \propto n$ (B) $r_n \propto \frac{1}{n}$ (C) $r_n \propto n^2$ (D) $r_n \propto \frac{1}{n^2}$
27. Radiations of wavelength 180 nm ejects photo-electrons from a plate whose work functions is 2.0 eV . If a uniform magnetic field of flux density $5.0 \times 10^{-5} \text{ tesla}$ is applied parallel to the plate, what should be the radius of the path followed by electrons ejected normally from the plate with maximum energy ?
- (A) 0.2 m (B) 0.148 m (C) 0.048 m (D) 0.418 m
28. A photocell is illuminated by a small bright source placed 1 m away. When the same source of light is placed $\frac{1}{2} \text{ m}$ away, the number of electrons emitted by photocathode would :
- (A) Decrease by a factor of 4 (B) Increase by a factor of 4
 (C) Decrease by a factor of 2 (D) Increase by a factor of 2
29. The de-Broglie wave corresponding to a particle of mass m and velocity v has a wavelength associated with it :
- (A) $\frac{h}{mv}$ (B) hmv (C) $\frac{mh}{v}$ (D) $\frac{m}{hv}$
30. In the spectrum of hydrogen, the ratio of the longest wavelength in Lyman series to the longest wavelength in the Balmer series is :
- (A) $\frac{27}{5}$ (B) $\frac{5}{27}$ (C) $\frac{4}{9}$ (D) $\frac{9}{4}$
31. In radius of the ${}_{13}^{27}\text{Al}$ nucleus is taken to be R_{Al} , then the radius of ${}_{53}^{125}\text{Te}$ nucleus is nearly :
- (A) $\frac{3}{4} R_{Al}$ (B) $\left(\frac{23}{53}\right)^{1/3} R_{Al}$ (C) $\left(\frac{53}{13}\right)^{1/3} R_{Al}$ (D) $\frac{5}{3} R_{Al}$
32. Consider 3^{rd} orbit of He^+ (Helium), using non-relativistic approach, the speed of electron in this orbit will be :
 [Given $K = 9 \times 10^9$ constant, $Z = 2$ and h (Plank's Constant) = $6.6 \times 10^{-34} \text{ Js}$] :
- (A) $0.73 \times 10^6 \text{ m/s}$ (B) $3.0 \times 10^8 \text{ m/s}$ (C) $2.92 \times 10^6 \text{ m/s}$ (D) $1.46 \times 10^6 \text{ m/s}$
33. Sodium and copper have work functions 2.3 eV and 4.5 eV respectively. Then the ratio of the threshold wavelength is nearest to
- (A) 1:2 (B) 4:1 (C) 2:1 (D) 1:4
34. The bonding energy per nucleon of ${}^7_3\text{Li}$ & ${}^4_2\text{He}$ are 5.60 MeV and 7.06 MeV respectively. In the nuclear reaction ${}^7_3\text{Li} + {}^1_1\text{H} \rightarrow {}^4_2\text{He} + {}^4_2\text{He} + Q$ the value of energy Q released is :
- (A) 19.6 MeV (B) -2.4 MeV (C) 8.4 MeV (D) 17.3 MeV

35. In a hydrogen like atom electron make transition from an energy level with quantum number n to another with quantum number $(n - 1)$. If $n \gg 1$, the frequency of radiation emitted is proportional to
 (A) $\frac{1}{n}$ (B) $\frac{1}{n^2}$ (C) $\frac{1}{n^3/2}$ (D) $\frac{1}{n^3}$
36. If the nuclear radius of ^{27}Al is 3.6 fermi, the approximate nuclear radius of ^{64}Cu in fermi is :
 (A) 2.4 (B) 1.2 (C) 4.8 (D) 3.6
37. A charged oil drop is suspended in uniform field of $3 \times 10^4 \text{ Vm}^{-1}$ so that it neither falls nor rises. The charge on the drop will be (Take the mass of the charge = $9.9 \times 10^{-15} \text{ kg}$ and $g = 10 \text{ ms}^{-2}$) :
 (A) $3.3 \times 10^{-18} \text{ C}$ (B) $3.2 \times 10^{-18} \text{ C}$ (C) $1.6 \times 10^{-18} \text{ C}$ (D) $4.8 \times 10^{-18} \text{ C}$
38. The half life of a radioactive nucleus is 50 days. The time interval $(t_2 - t_1)$ between the time t_2 when $2/3$ of it has decayed and the time t_1 when $1/3$ of it had decays is :
 (A) 30 days (B) 50 days (C) 60 days (D) 15 days
39. Two radioactive nuclei P and Q , in a given sample decay into a stable nucleus R . At time $t = 0$, number of P species are $4N_0$ and that of Q are N_0 . Half-life of P (for conversion to R) is 1 minute where as that of Q is 2 minutes. Initially there are no nuclei of R present in the sample. When number of nuclei of P and Q are equal, the number of nuclei of R present in the sample would be :
 (A) $2N_0$ (B) $3N_0$ (C) $\frac{9N_0}{2}$ (D) $\frac{5N_0}{2}$
40. Formation of covalent bonds in compounds exhibits :
 (A) Wave nature of electron (B) Particle nature of electron
 (C) Both wave and particle nature of electron (D) None of these
41. The activity of a radioactive sample is measured as N_0 counts per minute at $t = 0$ and N_0/e counts per minute at $t = 5$ minutes. The time (in minutes) at which the activity reduces to half its value is :
 (A) $\log_e \frac{2}{5}$ (B) $\frac{5}{\log_e 2}$ (C) $5 \log_{10} 2$ (D) $5 \log_e 2$
42. Hydrogen atom is excited from ground state to another state with principal quantum number equal to 4. Then, the number of spectral lines in the emission spectra will be :
 (A) 2 (B) 3 (C) 5 (D) 6
43. An alpha nucleus of energy $\frac{1}{2}mv^2$ bombards a heavy nuclear target of charge Ze . Then the distance of closest approach for the alpha nucleus will be proportional to :
 (A) $\frac{1}{Ze}$ (B) v^2 (C) $\frac{1}{m}$ (D) $\frac{1}{v^4}$
44. A uniform electric field and a uniform magnetic field are acting along the same direction in a certain region. If an electron is projected along the direction of the fields with a certain velocity then :
 (A) Its velocity will decrease
 (B) Its velocity will increase
 (C) It will turn towards right of direction of motion
 (D) It will turn towards left of direction of motion

45. If $M(A; Z)$, M_p and M_n denotes the masses of the nucleus A_ZX , proton and neutron respectively in unit of u ($1u = 931.5 \text{ MeV} / c^2$) and BE represents its bonding energy in MeV , then :
- (A) $M(A, Z) = ZM_p + (A - Z)M_n - BE$ (B) $M(A, Z) = ZM_p + (A - Z)M_n + BE / c^2$
 (C) $M(A, Z) = ZM_p + (A - Z)M_n - BE / c^2$ (D) $M(A, Z) = ZM_p + (A - Z)M_n + BE$
46. Two radioactive materials X_1 and X_2 have decay constant 5λ and λ respectively. If initially they have the same number of nuclei, then the ratio of the number of nuclei of X_1 to that X_2 will be $1/e$ after a time :
- (A) $1/4\lambda$ (B) e/λ (C) λ (D) $1/2\lambda$
47. The threshold frequency for a metallic surface corresponds to an energy of 6.2 eV and the stopping potential for a radiation incident on this surface 5 V . The incident radiation lies in :
- (A) Ultra-violet region (B) Infra-red region
 (C) Visible region (D) X-ray region
48. Radiation of wavelength λ , is incident on a photocell. The fastest emitted electron has speed v . If the wavelength is changed to $\frac{3\lambda}{4}$, the speed of the fastest emitted electron will be :
- (A) $> v\left(\frac{4}{3}\right)^{1/2}$ (B) $< v\left(\frac{4}{3}\right)^{1/2}$ (C) $= v\left(\frac{4}{3}\right)^{1/2}$ (D) $= v\left(\frac{3}{4}\right)^{1/2}$
49. Half-lives of two radioactive elements A and B are 20 minutes and 40 minutes, respectively. Initially, the samples have equal number of nuclei. After 80 minutes, the ratio of decayed numbers of A and B nuclei will be :
- (A) 1:16 (B) 4:1 (C) 1:4 (D) 5:4
50. A hydrogen atom makes a transition from $n = 2$ to $n = 1$ and emits a photon. This photon strikes a doubly ionized lithium atom ($z = 3$) in excited state and completely removes the orbiting electron. The least quantum number for the excited state of the ion for the process is :
- (A) 2 (B) 4 (C) 5 (D) 3
51. When photons of wavelength λ_1 are incident on an isolated sphere, the corresponding stopping potential is found to be V . When photons of wavelength λ_2 are used, the corresponding stopping potential was thrice that of the above value. If light of wavelength λ_3 is used then find the stopping potential for this case :
- (A) $\frac{hc}{e} \left[\frac{1}{\lambda_3} + \frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right]$ (B) $\frac{hc}{e} \left[\frac{1}{\lambda_3} + \frac{1}{2\lambda_2} - \frac{1}{\lambda_1} \right]$
 (C) $\frac{hc}{e} \left[\frac{1}{\lambda_3} - \frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right]$ (D) $\frac{hc}{e} \left[\frac{1}{\lambda_3} + \frac{1}{2\lambda_2} - \frac{3}{2\lambda_1} \right]$
52. A neutron moving with a speed ' v ' makes a head on collision with a stationary hydrogen atom in ground state. The minimum kinetic energy of the neutron for which inelastic collision will take place is :
- (A) 20.4 eV (B) 10.2 eV (C) 12.1 eV (D) 16.8 eV
53. A photoelectric surface is illuminated successively by monochromatic light of wavelength λ and $\frac{\lambda}{2}$. If the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case, the work function of the surface is :
- (A) $\frac{hc}{2\lambda}$ (B) $\frac{hc}{\lambda}$ (C) $\frac{hc}{3\lambda}$ (D) $\frac{3hc}{\lambda}$

54. Match List-1 (Fundamental experiment) with List –II (its conclusion) and select the correct option form the choices given below the list. Match the following :

	List -1		List – II
P	Franck – Hertz	(i)	Particle nature of light
Q	Photo – electric	(ii)	Discrete energy levels of atom
R	Davison – Germer Experiment.	(iii)	Wave nature of electron
		(iv)	Structure of atom

	P	Q	R		P	Q	R
(A)	ii	i	iii	(B)	iv	iii	ii
(C)	i	iv	iii	(D)	ii	iv	iii

55. De-Broglie wavelength of an electron accelerated by a voltage of 50 V is code to :
 $(|e| = 1.6 \times 10^{-19} \text{ C}, m_e = 9.1 \times 10^{-31} \text{ kg}, h = 6.6 \times 10^{-34} \text{ Js})$
(A) 0.5 Å **(B)** 1.2 Å **(C)** 1.7 Å **(D)** 2.4 Å
56. The de-Broglie wavelength associated with the electron in then $n = 4$ level is :
(A) two times the de-Broglie wavelength of the electron in the ground state
(B) four times the de-Broglie wavelength of the electron in the ground state
(C) half of the de-Broglie wavelength of the electron in the ground state
(D) $1/4^{\text{th}}$ of the de-Broglie wavelength of the electron in the ground state.
57. As An electron makes a transition from an excited state to the ground state of hydrogen – like atom/ion
(A) kinetic energy decreases, potential energy increased but total energy remains same
(B) kinetic energy and total energy decrease but potential energy increases
(C) its kinetic energy increase but potential energy and total energy decrease
(D) kinetic energy, potential energy and total energy decrease
58. If one were to apply Bohr model to a particle of mass m and charge q moving in a plane under the influence of a magnetic field B , the energy of the charged particle in the n^{th} level will be :
(A) $n \left(\frac{hqB}{2\pi m} \right)$ **(B)** $n \left(\frac{hqB}{4\pi m} \right)$ **(C)** $n \left(\frac{hqB}{8\pi m} \right)$ **(D)** $n \left(\frac{hqB}{\pi m} \right)$
59. Let N_β be the number of β particles emitted by 1 gram of Na^{24} radioactive nuclei (half life = 15 hrs) in 7.5 hours, N_β is close to (Avogadro number = $6.023 \times 10^{23} / \text{g mole}$)
(A) 6.2×10^{31} **(B)** 7.5×10^{21} **(C)** 1.25×10^{22} **(D)** 1.75×10^{22}

INTEGER ANSWER TYPE QUESTIONS

The Answers to the following questions are positive integers of 1/2/3 digits or zero.

60. Suppose alpha particles of speed $2.0 \times 10^7 \text{ m/s}$ are shot at lead atoms ($Z = 82$). If minimum separation between alpha particle and lead nucleus is 2.82×10^{2n} then the value of n is .
61. The energy of the photon in eV , when absorbed by a hydrogen atom, causes an electronic transition form the initial state $n = 2$ to the final state $n = 5$ is .

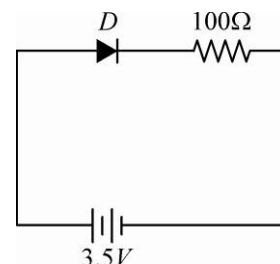
62. A tiny bottle of radon gas contains 8.0×10^{12} atoms of radon. The half life of radon is 3.8 days. If disintegrations per second in the bottle is 1.01×10^n then the value of n is .
63. If a radioactive sample initially contains 8.00×10^{14} parent nuclei with a half life of 7.00 h, if parent nuclei remain 24.0 h later is 7.43×10^n then the value of n is .
64. It is observed that after 3h only 0.25 mg of a pure radioactive material that was originally 2 mg is left. The half life of the material in hours is .
65. If the work function of the target material is 1.24 eV and wavelength of incident light is $4.36 \times 10^{-7} \text{ m}$. Then the regarding potential necessary to stop the emission of photoelectrons is .
66. Position emission by ^{15}O is the first stage in a medical scanning technique called PET (Position Emission Tomography). The half life is 122s. If the initial decay rate of a sample is $0.2 \mu\text{Ci}$, and 1.30×10^n nuclei are present then the value of n .
67. The electron in the hydrogen atom is in the $n = 2$ state. Then the ratio of modulus of potential energy to kinetic energy is .
68. The photoelectric work function of a metal is 1 eV. Light of wavelength $\lambda = 3000 \text{ \AA}$ falls on it. The velocity of ejected photoelectrons is approximately 10^n then the value of n is .
- $$3.1 \times 10^{-19} \times 1.6 = \frac{1}{6} \times 9.31 \times 10^{-31} \times v^2$$
- $$v \approx 10^6 \text{ m/s}$$
69. Iodine-131 is used to treat thyroid disorders because, when ingested, it localizes in the thyroid gland. Its half life is 8.1 days. The activity of $0.80 \mu\text{g}$ of ^{131}I in mCi is .
70. Light of two different frequencies, whose photons have energies 1 and 2.5 eV respectively successively illuminate a metal whose work function is 0.5 eV. If the ratio of the maximum speeds of the emitted electrons is $1 : n$ then value of n is .
71. In an effort to date a piece of bone, its count rate due to ^{14}C is computed and found to be only 0.048 that of a similar fresh piece of bone. If age of the bone is 2.50×10^{2n} then the value of n is . (half life of ^{14}C is 5700 years)
72. If de Broglie wavelength of an electron accelerated to a potential difference of V volt is $\sqrt{\frac{n}{V}} \text{ \AA}$ then the value of n is .
73. The radioactive isotope $^{80}_{27}\text{Co}$ is used in the treatment of tumors. It undergoes β^- decay with a half life of 5.25 years. The initial decay rate of a 0.01 g sample is $n \times 10^{11}$ then the value of n is .
74. The radioisotope $^{90}_{38}\text{Sr}$ has a half life of 29 year. A sample has an initial decay rate of $24 \mu\text{Ci}$. If nuclei decay in the first year is $n \times 10^{13}$ then the value of n is .

Semiconductor & Communication

Choose the correct alternative. Only One Choice is Correct. However, questions marked '*' May have More than One Correct option.

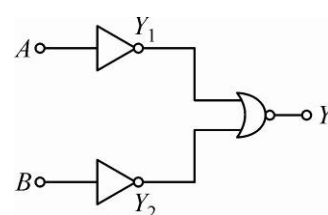
1. In the given figure, a diode D is connected to an external resistance $R = 100\Omega$ and an e.m.f. of $3.5V$. If the barrier potential developed across the diode is $0.5V$, the current in the circuit will be :

(A) 20 mA (B) 35 mA
(C) 30 mA (D) 40 mA

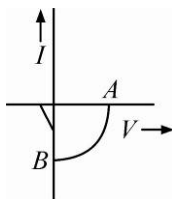


2. Which logic gate is represented by the following combination of logic gates ?

(A) AND
(B) NOR
(C) OR
(D) NAND



3. The given graph represents V - I characteristic for a semiconductor device.



Which of the following statement is correct ?

- (A) It is V - I characteristic for solar cell where, point A represents open circuit voltage and point B short circuit current
(B) It is for a solar cell and points A and B represent open circuit voltage and current, respectively
(C) It is for a photodiode and points A and B represent open circuit voltage and current, respectively
(D) It is for a LED and points A and B represent open circuit voltage and short circuit current, respectively

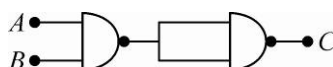
4. The barrier of a p - n junction depends on :

I. type of semiconductor material II. amount of doping III. temperature

Which one of the following is correct

- (A) I and II only (B) II only (C) II and III only (D) I, II and III

5. The output from a NAND gate is divided into two in parallel and fed to another NAND gate. The resulting gate is a:



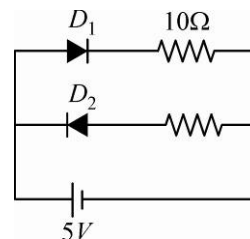
- (A) AND gate (B) NOR gate (C) OR gate (D) NOT gate

6. One way in which the operation of a n - p - n transistor differs from that of a p - n - p :

- (A) The emitter junction injects minority carriers into the base region of a p - n - p
(B) The emitter injects holes into the base of the p - n - p and electrons into the base region of n - p - n
(C) The emitter injects holes into the base of a n - p - n
(D) The emitter junction is reversed biased in n - p - n

7. Two ideal diodes are connected to a battery as shown in the circuit. The current supplied by the battery is :

(A) 0.75 A
 (B) Zero
 (C) 0.25 A
 (D) 0.5 A

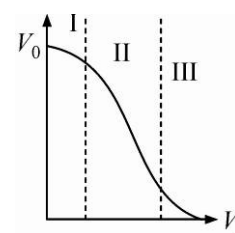


8. C and Si both have same lattice structure, having 4 bonding electrons in each. However, C is insulator whereas Si is intrinsic semiconductor. This is because :

(A) In case of C the valence band is not completely filled at absolute zero temperature.
 (B) In case of C the conduction band is partly filled even at absolute zero temperature.
 (C) The four bonding electrons in the case of C lie in the second orbit, whereas in the case of Si they lie in the third.
 (D) The four bonding electrons in the case of C lie in the third orbit, whereas for Si they lie in the fourth orbit.

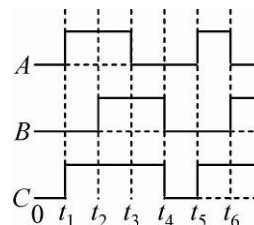
9. Transfer characteristics [output voltage (V_0) vs input voltage (V_i)] for a base biased transistor in CE configuration is as shown in the figure. For using transistor as a switch, it is used :

(A) in region III
 (B) both in region I and III
 (C) in region II
 (D) in region I



10. The figure shows a logic circuit with two inputs A and B and output C . The voltage wave forms across A , B and C are as given. The logic circuit gate is :

(A) OR gate
 (B) NOR gate
 (C) AND gate
 (D) NAND gate

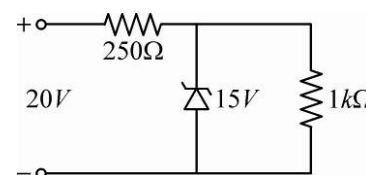


11. The input resistance of a silicon transistor is 100Ω . Base current is changed by $40\mu\text{A}$ which results in a change in collector current by 2 mA . This transistor is used as a common emitter amplifier with a load resistance of $4\text{ k}\Omega$. The voltage gain of the amplifier is :

(A) 2000
 (B) 3000
 (C) 4000
 (D) 1000

12. A Zener diode, having breakdown voltage equal to 15 V , is used in a voltage regulator circuit shown in figure. The current through the diode is :

(A) 5 mA
 (B) 10 mA
 (C) 15 mA
 (D) 20 mA

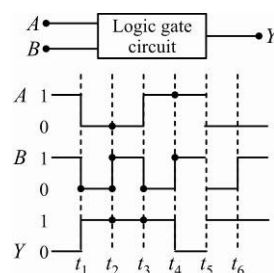


13. Pure Si at 500 K has equal number of electron (n_e) and hole (n_h) concentrations of $1.5 \times 10^{16}\text{ m}^{-3}$. Doping by indium increases n_h to $4.5 \times 10^{22}\text{ m}^{-3}$. The doped semiconductor is of :

(A) p -type having electron concentration $n_e = 5 \times 10^9\text{ m}^{-3}$
 (B) n -type having electron concentration $n_e = 5 \times 10^{22}\text{ m}^{-3}$
 (C) p -type having electron concentration $n_e = 2.5 \times 10^{10}\text{ m}^{-3}$
 (D) n -type having electron concentration $n_e = 2.5 \times 10^{23}\text{ m}^{-3}$

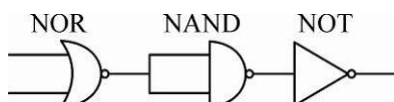
14. A common emitter amplifier has a voltage gain of 50, an input impedance of 100Ω and an output impedance of 200Ω . The power gain of the amplifier is :
 (A) 500 (B) 1000 (C) 1250 (D) 50

15. The following figure shown a logic gate circuit with two inputs A and B and the output Y . The voltage waveforms of A , B and Y are as given :



The logic gate is :

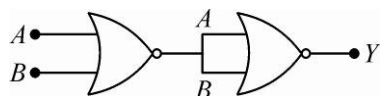
- (A) NOR gate (B) OR gate
 (C) AND gate (D) NAND gate
16. The circuit :



is equivalent to :

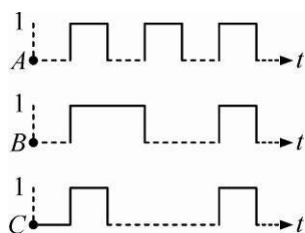
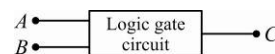
- (A) NOR gate (B) OR gate (C) AND gate (D) NAND gate

17. In the following circuit, the output Y for all possible inputs A and B is expressed by the truth table.



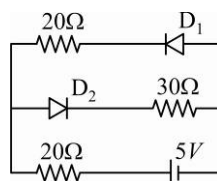
A	B	Y	A	B	Y	A	B	Y	A	B	Y
0	0	1	0	0	1	0	0	0	0	0	0
0	1	1	0	1	0	0	1	1	0	1	0
1	0	1	1	0	0	1	0	1	1	0	0
1	1	0	1	1	0	1	1	1	1	1	1

18. The following figure shows a logic gate circuit with two inputs A and B and the output C . The voltage waveforms of A , B and C are as shown below :



The logic circuit gate is :

- (A) OR gate (B) AND gate (C) NAND gate (D) NOR gate
19. The current in the circuit will be :



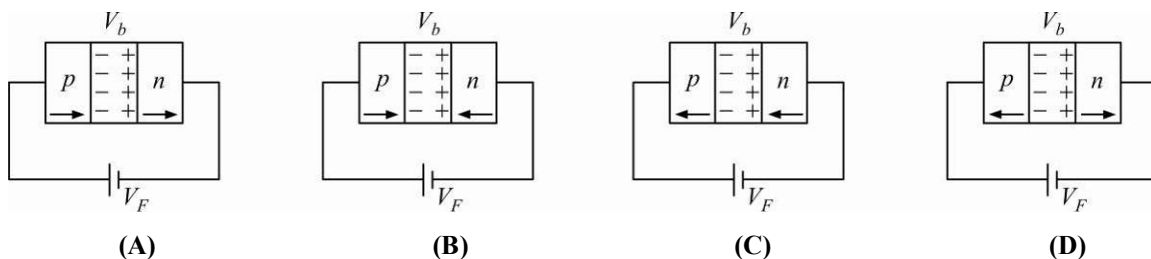
- (A) $5/40 A$ (B) $5/50 A$ (C) $5/10 A$ (D) $5/20 A$
20. For a common base circuit if $\frac{I_C}{I_E} = 0.98$ then current gain for common emitter circuit will be :

- (A) 49 (B) 98 (C) 4.9 (D) 25.5

21. The correct relation for α , β for a transistor :

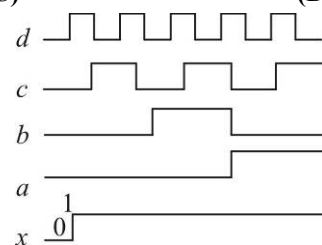
(A) $\beta = \frac{1-\alpha}{\alpha}$ (B) $\beta = \frac{\alpha}{1-\alpha}$ (C) $\alpha = \frac{\beta-1}{\beta}$ (D) $\alpha\beta = 1$

22. In the case of forward biasing of p - n junction, which one of the following figures correctly depicts the direction of flow of carriers ?

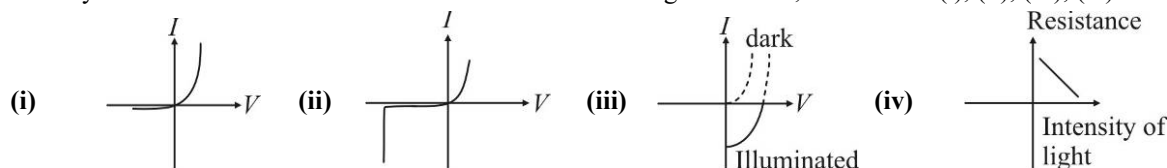


23. If a, b, c, d are inputs to a gate and x is its output then, as per the following time graph, the gate is :

- (A) NOT
(B) AND
(C) OR
(D) NAND



24. Identify the semiconductor devices whose characteristics are given below, in the order (i), (ii), (iii), (iv) :



- (A) Simple diode, Zener diode, Solar cell, Light dependent resistance
(B) Zener diode, simple diode, light dependent resistance, solar cell
(C) Solar cell, Light dependent resistance, Zener diode, simple diode
(D) Zener diode, Solar cell, simple diode, Light dependent resistance

- *25. For a common emitter configuration, if α and β have their usual meanings, the incorrect relationship between α and β is :

(A) $\frac{1}{\alpha} = \frac{1}{\beta} + 1$ (B) $\alpha = \frac{\beta}{1-\beta}$ (C) $\alpha = \frac{\beta}{1+\beta}$ (D) $\alpha = \frac{\beta^2}{1+\beta^2}$

26. An unknown transistor needs to be identified as a nnp or $pnnp$ type. A multimeter, with +ve and -ve terminals, is used to measure resistance between different terminals of transistor. If terminal 2 is the base of the transistor then which of the following is correct for a $pnnp$ transistor?

- (A) +ve terminal 2, -ve terminal 3, resistance low
(B) +ve terminal 2, -ve terminal 1, resistance high
(C) +ve terminal 1, -ve terminal 2, resistance high
(D) +ve terminal 3, -ve terminal 2, resistance high

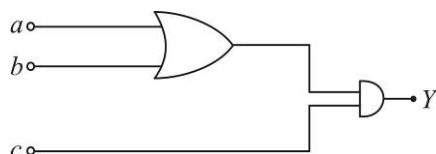
27. An experiment is performed to determine the I - V characteristics of a Zener diode, which has a protective resistance of $R = 100\Omega$, and a maximum power of dissipation rating of $1W$. The minimum voltage range of the DC source in the circuit is :

- (A) $0-5V$ (B) $0-24V$ (C) $0-12V$ (D) $0-8V$

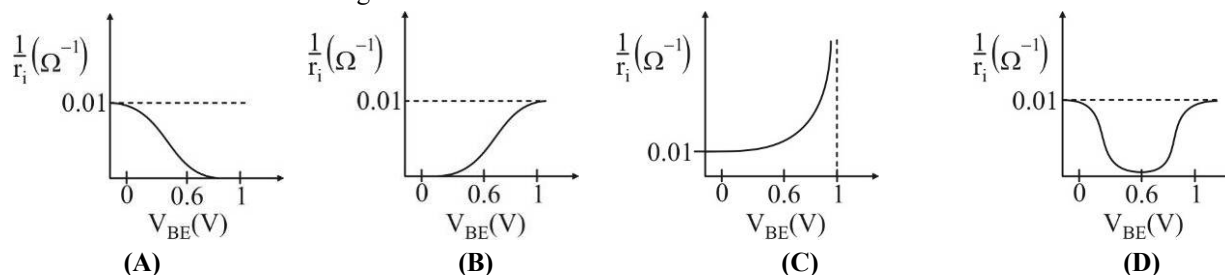
28. The truth table in figure represents :

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

- (A) OR –Gate (B) NAND –Gate (C) AND –Gate (D) NOR –Gate
29. The ratio (R) of output resistance r_o , and the input resistance r_i in measurements of input and output characteristics of a transistor is typically in the range:
- (A) $R \sim 10^2 - 10^3$ (B) $R \sim 1 - 10$ (C) $R \sim 0.1 - 1.0$ (D) $R \sim 0.1 - 0.01$
30. To get an output of 1 from the circuit shown in figure the input must be :



- (A) $a = 0, b = 0, c = 1$ (B) $a = 1, b = 0, c = 0$
- (C) $a = 1, b = 0, c = 1$ (D) $a = 0, b = 1, c = 0$
31. A realistic graph depicting the variation of the reciprocal of input resistance in an input characteristics measurement in a common emitter transistor configuration is :



32. Choose the correct statement:
- (A) In amplitude modulation the amplitude of the high frequency carrier wave is made to vary in proportion to the amplitude of the audio signal.
- (B) In amplitude modulation the frequency of the high frequency carrier wave is made to vary in proportion to the amplitude of the audio signal.
- (C) In frequency modulation the amplitude of the high frequency carrier wave is made to vary in proportion to the amplitude of the audio signal.
- (D) In frequency modulation the amplitude of the high frequency carrier wave to vary in proportion to the frequency of the audio signal.
33. An audio signal consists of two distinct sounds: one a human speech signal in the frequency band of 200 Hz to 2700 Hz, while the other is a high frequency music signal in the frequency band of 10200 Hz to 15200 Hz. The ratio of the ratio of the AM signal bandwidth required to send both the signals together to the AM signal bandwidth required to send just the human speech is :
- (A) 2 (B) 5 (C) 6 (D) 3

34. A modulated signal $C_m(t)$ has the form $C_m(t) = 30 \sin 300\pi t + 10(\cos 200\pi t - \cos 400\pi t)$. The carrier frequency f_c , the modulating frequency (message frequency) f_ω , and the modulation index μ are respectively given by :

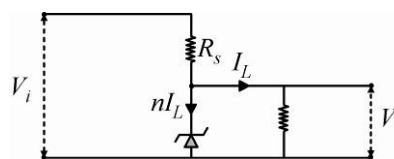
- (A) $f_c = 200\text{Hz}; f_\omega = 50\text{Hz}; \mu = \frac{1}{2}$ (B) $f_c = 150\text{Hz}; f_\omega = 50\text{Hz}; \mu = \frac{2}{3}$
 (C) $f_c = 150\text{Hz}; f_\omega = 30\text{Hz}; \mu = \frac{1}{3}$ (D) $f_c = 200\text{Hz}; f_\omega = 30\text{Hz}; \mu = \frac{1}{2}$

35. In an unbiased n - p junction electrons diffuse from n -region to p -region because

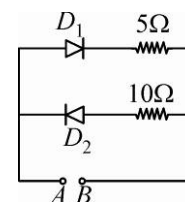
- (A) holes in p -region attract them
 (B) electrons travel across the junction due to potential difference
 (C) electron concentration in n -region is more as compared to that in p -region
 (D) only electrons move from n to p region and not the vice-versa.

36. The value of the resistor, R_s , needed in the dc voltage regulator circuit shown here, equals

- (A) $(V_i - V_L)/nI_L$ (B) $(V_i + V_L)/nI_L$
 (C) $(V_i - V_L)/(n+1)I_L$ (D) $(V_i + V_L)/(n+1)I_L$



37. A $2V$ battery is connected across AB as shown in the figure. The value of the current supplied by the battery when in one case battery's positive terminal is connected to A and in other case when positive terminal of battery is connected to B will respectively by :



- (A) 0.2 A and 0.1 A (B) 0.4 A and 0.2 A (C) 0.1 A and 0.2 A (D) 0.2 A and 0.4 A

38. A red LED emits light at 0.1 watt uniformly around it. The amplitude of the electric field of the light at a distance of 1 m from the diode is :

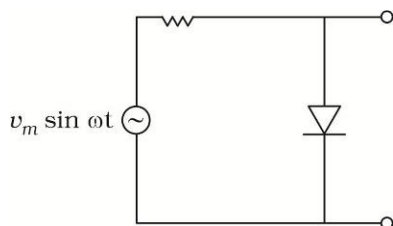
- (A) 5.48 V/m (B) 7.75 V/m (C) 1.73 V/m (D) 2.45 V/m

39. A signal of 5 kHz frequency is amplitude modulated on a carrier wave of frequency 2 MHz . The frequencies of the resultant single is/are

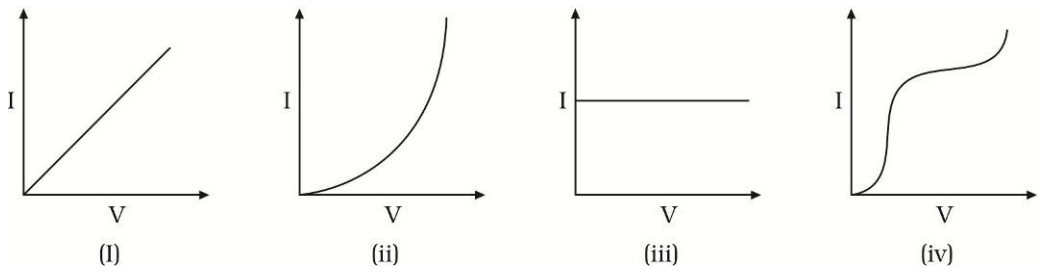
- (A) 2005 kHz , 2000 kHz and 1995 kHz (B) 2000 kHz and 1995 kHz
 (C) 2 MHz only (D) 2005 kHz and 1995 kHz

NCERT Exemplar

40. The output of the given circuit in figure.



- (A) would be zero at all times
 (B) would be like a half wave rectifier with positive cycles in output
 (C) would be like half wave rectifier with negative cycles in output
 (D) would be like that of a full rectifier

41. Three waves A, B and C of frequencies and 60 MHz , respectively are to be transmitted from one place to another. Which of the following is the most appropriate mode of communication :
- (A) A is transmitted via space wave while B and C are transmitted via sky wave
 (B) A is transmitted via ground wave, B via sky wave and C via space wave
 (C) B and C are transmitted via ground wave while
 (D) would be like that of a full rectifier
42. A 1 KW signal is transmitted using a communication channel which provides attenuation at the rate -2 dB per km . If the communication channel has a total length of 5 km , the power of the signal received is : $\left[\text{gain in dB} = 10 \log \left(\frac{P_0}{P_i} \right)\right]$
- (A) 900 W (B) 100 W (C) 990 W (D) 1010 W
43. $I - V$ characteristics of four devices are shown in figure.
- 
- Identify devices that can be used for modulation :
- (A) i and iii (B) only iii
 (C) ii and some regions of iv (D) all the devices can be used
44. A male voice after modulation-transmission sounds like that of a female to the receiver. The problem is due to :
- (A) poor selection of modulation index (selected $0 < m < 1$)
 (B) poor bandwidth selection of amplifiers
 (C) poor selection of carrier frequency
 (D) loss of energy in transmission
45. A basic communication system consists of :
- (a) transmitter (b) information source (c) user of information
 (d) channel (r) receiver
- Choose the correct sequence in which these are arranged in a basic communication system:
- (A) ABCDE (B) BADEC (C) BDACE (D) BEADC

INTEGER ANSWER TYPE QUESTIONS

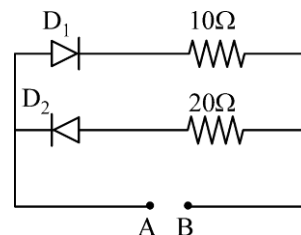
The Answers to the following questions are positive integers of $1/2/3$ digits or zero.

46. A doped semiconductor has impurity levels 30 meV below the conduction band. In a thermal collision, an amount kT of energy is given to the extra electron loosely bound to the impurity ion and this electron is just able to jump into the conduction band. What is the temperature T .
47. The energy of a photon a sodium light ($\lambda = 589\text{ nm}$) equals the band gap of a semi conducting material. Find the value of E/kT at a temperature of 300 K .

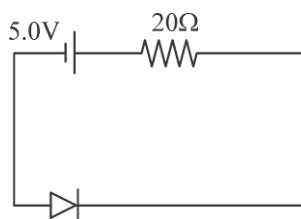
48. The band gap in germanium is $\Delta E = 0.68 \text{ eV}$. Assuming that the number of hole-electron pairs is proportional to $e^{-\Delta E / 2kT}$, find the percentage increase in the number of charge carriers in pure germanium as the temperature is increased from 300 K to 320 K.

49. The concentration of hole-electron pairs in pure silicon at $T = 300 \text{ K}$ is 7×10^{15} per cubic metre. Antimony is doped into silicon in a proportion of 1 atom in 10^7 atoms. The number of silicon atoms per cubic meter is 5×10^{28} . Assuming that half of the impurity atoms contribute electrons in the conduction band, the factor by which the number of charge carriers increases due to doping is $\boxed{} \times 10^7$.

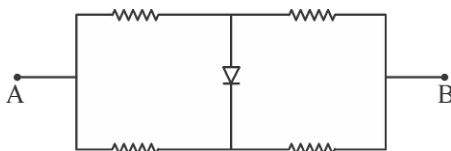
50. A 2 V battery may be connected across the points A and B as shown in figure. Assume that the resistance of each diode is zero in forward bias and infinity in reverse bias. Find the ratio of currents supplied by the battery if the positive terminal of the battery is connected to (a) the point A (b) the point B.



51. Calculate the potential difference across the diode shown in figure. The drift current for the diode is $20 \mu\text{A}$ $\boxed{} \text{ V}$.



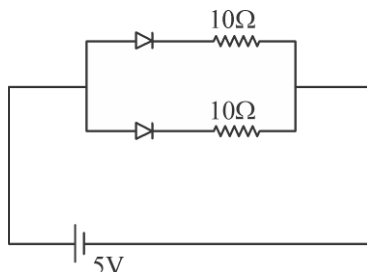
52. Each of the resistances shown in figure has a value of 20Ω . Find the equivalent resistance between A and B $\boxed{} \Omega$.



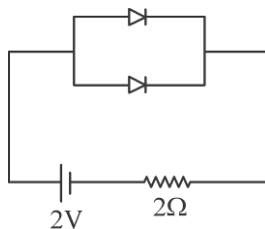
53. When the base current in a transistor is changed from $30 \mu\text{A}$ to $80 \mu\text{A}$, the collector current is changed from 1.0 mA to 3.5 mA. Find the current gain β .

54. A load resistor of $2 \text{ k}\Omega$ is connected in the collector branch of an amplifier circuit using a transistor in common-emitter mode. The current gain $\beta = 50$. The input resistance of the transistor is 0.50Ω . If the input current is changed by $50 \mu\text{A}$, by what amount does the output voltage change $\boxed{} \text{ V}$.

55. Find the current through the battery in each of the circuits shown in figure $\boxed{} \text{ A}$.



56. Find the currents through the resistances in the circuit shown in figure A.



57. The drift current in a p-n junction is $20.0\mu A$. Estimate the number of electrons crossing a cross-section per second in the depletion region is $\times 10^{12}$.

Electromagnetic Waves

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED '*' MAY HAVE MORE THAN ONE CORRECT OPTION.

- The energy of the *em* waves if of the order of 15 keV . To which part of the spectrum does it belong ?
 (A) Ultraviolet (B) γ - rays (C) X-rays (D) Infra-red rays
- A radiation of energy ' E ' falls normally on a perfectly reflecting surface. The momentum transferred to the surface is (C = velocity of light).
 (A) $\frac{2E}{C^2}$ (B) $\frac{E}{C^2}$ (C) $\frac{E}{C}$ (D) $\frac{2E}{C}$
- Light with an energy flux of $25 \times 10^4 \text{ Wm}^{-2}$ falls on a perfectly reflecting surface at normal incidence. If the surface area is 15 cm^2 , the average force exerted on the surface is :
 (A) $1.25 \times 10^{-6} \text{ N}$ (B) $2.50 \times 10^{-6} \text{ N}$ (C) $1.20 \times 10^{-6} \text{ N}$ (D) $3.0 \times 10^{-6} \text{ N}$
- The condition under which a microwave oven heats up a food item containing water molecules most efficiently is :
 (A) Microwaves are heat waves, so always produce heating.
 (B) Infra - red waves produce heating in a microwave oven.
 (C) The frequency of the microwaves must match the resonant frequency of the water molecules.
 (D) The frequency of the microwaves has no relation with natural frequency of water molecules.
- An electromagnetic wave of frequency $\nu = 3.0 \text{ MHz}$ passes from vacuum into a dielectric medium with relative permittivity $\epsilon = 4.0$ then :
 (A) Wavelength is doubled and frequency becomes half
 (B) Wavelength is halved and frequency remains unchanged
 (C) Wavelength and frequency both remain unchanged
 (D) Wavelength is doubled and frequency unchanged
- The electric field associated with an *em* wave in vacuum is given by $E = \hat{i} 40 \cos(kz - 6 \times 10^8 t)$ where E , z and t are in volt/m, meter and seconds respectively. The value of wave vector k is :
 (A) 2m^{-1} (B) 0.5m^{-1} (C) 6m^{-1} (D) 3m^{-1}
- The ratio of amplitude of magnetic field to the amplitude of electric field for an electromagnetic wave propagating in vacuum is equal to :
 (A) The speed of light in vacuum (B) Reciprocal of speed of light in vacuum
 (C) The ratio of magnetic permeability to the electric susceptibility of vacuum (D) Unity
- The electric and the magnetic field, associated with an *em.* wave, propagating along the $+z$ -axis, can be represented by :
 (A) $[\vec{E} = E_0 \hat{i}, \vec{B} = B_0 \hat{j}]$ (B) $[\vec{E} = E_0 \hat{k}, \vec{B} = B_0 \hat{i}]$
 (C) $[\vec{E} = E_0 \hat{j}, \vec{B} = B_0 \hat{i}]$ (D) $[\vec{E} = E_0 \hat{j}, \vec{B} = B_0 \hat{k}]$
- The decreasing order of wavelength of infrared microwave, ultraviolet and gamma rays is :
 (A) Microwave, infrared, ultraviolet, gamma rays (B) Gamma rays, ultraviolet, infrared microwaves
 (C) Microwaves, gamma rays, infrared, ultraviolet (D) infrared, microwave, ultraviolet, gamma rays

10. Which of the following statement is false for the properties of electromagnetic waves ?
 (A) Both electric and magnetic field vectors exist at same time.
 (B) The energy in electromagnetic wave is divided equally between electric and magnetic vectors.
 (C) Both electric and magnetic field vectors are parallel to each other and perpendicular to the direction propagation of wave.
 (D) These waves do not require any material medium for propagation.
11. The electric field of an electromagnetic wave in free space is given by: $\vec{E} = 10 \cos(10^7 t + kx) \hat{j} \text{ V/m}$, where t and x are in seconds and meters respectively. It can be inferred that
 (1) The wavelength λ is 188.4 m . (2) The wave number k is 0.33 rad/m .
 (3) The wave amplitude is 10 V/m (4) The wave is propagating along $+x$ direction.
 Which one of the following pairs of statements is correct ?
 (A) (3) and (4) (B) (1) and (2) (C) (2) and (3) (D) (1) and (3)
12. The electric field part of an electromagnetic wave in a medium is represented by $E_x = 0$;
 $E_y = 2.5 \frac{N}{C} \cos \left[\left(2\pi \times 10^6 \frac{\text{rad}}{\text{m}} \right) t - \left(\pi \times 10^{-2} \frac{\text{rad}}{\text{s}} \right) x \right]$ $E_z = 0$. The wave is :
 (A) Moving along x direction with frequency 10^6 Hz and wavelength 100 m
 (B) Moving along x direction with frequency 10^6 Hz and wavelength 200 m
 (C) Moving along $-x$ direction with frequency 10^6 Hz and wavelength 200 m
 (D) Moving along y direction with frequency $2\pi \times 10^6 \text{ Hz}$ and wavelength 200 m
13. The velocity of electromagnetic radiation in a medium of permittivity ϵ_0 and permeability μ_0 is given by :
 (A) $\frac{1}{\sqrt{\mu_0 \epsilon_0}}$ (B) $\sqrt{\frac{\mu_0}{\epsilon_0}}$ (C) $\sqrt{\frac{\epsilon_0}{\mu_0}}$ (D) $\sqrt{\mu_0 \epsilon_0}$
14. The electric and magnetic field of an electromagnetic wave are :
 (A) In opposite phase and perpendicular to each other
 (B) In opposite phase and parallel to each other
 (C) In phase and perpendicular to each other
 (D) In phase and parallel to each other
15. If λ_y, λ_x , and λ_m represent the wavelengths of visible light, X-rays and microwaves respectively, then :
 (A) $\lambda_m > \lambda_x > \lambda_y$ (B) $\lambda_m > \lambda_y > \lambda_x$ (C) $\lambda_y > \lambda_x > \lambda_m$ (D) $\lambda_y > \lambda_m > \lambda_x$.
16. We consider the radiation emitted by the human body. Which one of the following statements is true ?
 (A) The radiation emitted is in the infra-red region
 (B) The radiation is emitted only during the day
 (C) The radiation is emitted during the summers and absorbed during the winters
 (D) The radiation emitted lies in the ultraviolet region and hence is not visible
17. Which of the following rays are not electromagnetic waves ?
 (A) X-rays (B) γ - rays (C) β - rays (D) heat rays
18. The velocity of electromagnetic wave is parallel to :
 (A) $\vec{B} \times \vec{E}$ (B) $\vec{E} \times \vec{B}$ (C) $\vec{E}$ (D) $\vec{B}$

19. What is the cause of green house effect ?
 (A) infra-red rays (B) ultra violet rays (C) X-rays (D) radio waves.
20. Biological importance of ozone layer is :
 (A) It stops ultraviolet rays (B) Ozone layer reduces green house effect
 (C) Ozone layer reflects radio waves (D) Ozone layer controls O_2 / H_2 ratio in atmosphere.
21. The frequency order for γ – rays (B), X-rays (A), UV rays (C) is :
 (A) $B > A > C$ (B) $A > B > C$ (C) $C > B > A$ (D) $A > C > B$
22. Ozone layer blocks the radiations of wavelength
 (A) More than $3 \times 10^{-7} m$ (B) equal to $3 \times 10^{-7} m$
 (C) less than $3 \times 10^{-7} m$ (D) all of these
23. Wavelength of light of frequency $100 Hz$
 (A) $4 \times 10^6 m$ (B) $3 \times 10^6 m$ (C) $2 \times 10^6 m$ (D) $5 \times 10^{-5} m$
24. If ϵ_0 and μ_0 are the electric permittivity and magnetic permeability in a free space, ϵ and μ are the corresponding quantities in medium, the index and refraction of the medium is :
 (A) $\sqrt{\frac{\epsilon_0 \mu_0}{\epsilon \mu}}$ (B) $\sqrt{\frac{\epsilon \mu}{\epsilon_0 \mu_0}}$ (C) $\sqrt{\frac{\epsilon_0 \mu}{\epsilon \mu_0}}$ (D) $\sqrt{\frac{\epsilon}{\epsilon_0}}$
25. Which of the following electromagnetic radiations have the smaller wavelength?
 (A) X-rays (B) γ – rays (C) UV waves (D) microwaves
26. A signal emitted by an antenna from a certain point can be received at another point of the surface in the form of :
 (A) Sky wave (B) ground wave (C) sea wave (D) both (A) and (B)
27. The structure of solids is investigated by using.
 (A) Cosmic (B) X- rays (C) γ – rays (D) infra-red radiations
28. The frequency of electromagnetic wave, which best suited to observe a particle of radius $3 \times 10^{-4} cm$ is of the order of
 (A) 10^{15} (B) 10^{14} (C) 10^{13} (D) 10^{12}
29. Which of the following electromagnetic radiations have the longest wavelength ?
 (A) X – rays (B) γ – rays (C) Microwaves (D) Radiowaves
30. Arrange the following electromagnetic radiations per quantum in the order of increasing energy:
 (a) Blue light (b) Yellow light (c) X-ray (d) Radiowave
 The correct choice is :
 (A) d, b, a, c (B) a, b, d, c (C) c, a, b, d (D) b, a, d, c
31. Microwave oven acts on the principal of
 (A) giving rotational energy to water molecules
 (B) giving translational energy to water molecules
 (C) giving vibrational energy to water molecules
 (D) Transferring electrons from lower to higher energy levels in water molecule

32. Consider an electromagnetic wave propagating in vacuum. Choose the correct statement.
- (A) For an electromagnetic wave propagating in $+y$ direction the electric field is $\vec{E} = \frac{1}{\sqrt{2}} E_{yz}(x, t) \hat{z}$ and the magnetic field is $\vec{B} = \frac{1}{\sqrt{2}} B_z(x, t) \hat{y}$
- (B) For an electromagnetic wave propagating in $+y$ direction the electric fields is $\vec{E} = \frac{1}{\sqrt{2}} E_{yz}(x, t) \hat{y}$ and the magnetic field is $\vec{B} = \frac{1}{\sqrt{2}} B_z(x, t) \hat{z}$
- (C) For an electromagnetic wave propagating in $+x$ direction the electric field is $\vec{E} = \frac{1}{\sqrt{2}} E_{yz}(y, z, t)(\hat{y} + \hat{z})$ and the magnetic field is $\vec{B} = \frac{1}{\sqrt{2}} B_{yz}(y, z, t)(\hat{y} + \hat{z})$
- (D) For an electromagnetic wave propagating in $+x$ direction the electric field is $\vec{E} = \frac{1}{\sqrt{2}} E_{yz}(x, t)(\hat{y} - \hat{z})$ and the magnetic field is $\vec{B} = \frac{1}{\sqrt{2}} B_{yz}(x, t)(\hat{y} + \hat{z})$
33. An electromagnetic wave travelling in the x -direction has frequency of $2 \times 10^{14} \text{ Hz}$ and electric field amplitude of 27 V m^{-1} . From the options given below, which one describes the magnetic field for this wave?
- (A) $\vec{B}(x, t) = (3 \times 10^{-8} \text{ T}) \hat{j} \sin[2\pi(1.5 \times 10^{-8} x - 2 \times 10^{14} t)]$
- (B) $\vec{B}(x, t) = (9 \times 10^{-8} \text{ T}) \hat{k} \sin[2\pi(1.5 \times 10^{-6} x - 2 \times 10^{14} t)]$
- (C) $\vec{B}(x, t) = (9 \times 10^{-8} \text{ T}) \hat{i} \sin[2\pi(1.5 \times 10^{-8} x - 2 \times 10^{14} t)]$
- (D) None of these
34. For plane electromagnetic wave propagating in the z direction, which one of the following combination gives the correct possible direction for $\vec{E}$ and $\vec{B}$ field respectively?
- (A) $(\hat{i} + 2\hat{j})$ and $(2\hat{i} - \hat{j})$
- (B) $(-2\hat{i} - 3\hat{j})$ and $(3\hat{i} - 2\hat{j})$
- (C) $(2\hat{i} + 3\hat{j})$ and $(\hat{i} + 2\hat{j})$
- (D) $(3\hat{i} + 4\hat{j})$ and $(4\hat{i} - 3\hat{j})$

INTEGER ANSWER TYPE QUESTIONS

The Answers to the following questions are positive integers of 1/2/3 digits or zero.

35. Plane electromagnetic waves of intensity S are incident normally on a flat surface. Only a fraction f of the incident energy is absorbed. The radiation pressure is given by $\frac{S}{c}(\lambda - f)$. Find the value of λ .
36. The number of photons per second enter one eye if you look directly at a 100 W light bulb 2.00 m away (Assume a pupil diameter of 4.00 mm and a wavelength of 600 nm) is 7.55×10^n . Find n
37. The number of photons of yellow light ($\lambda = 575 \text{ nm}$) that must be absorbed by 1.0 cm^2 of human skin to raise the skin temperature by 1.0°C is (Assume that the photon energy is absorbed by 1.0 cm^3 of tissue. As rough estimates of the tissue's density and specific heat, use the values for water) 1.2×10^n . Find n .

38. A leaky capacitor with circular plates of radius $R = 12 \text{ cm}$ has a capacitance of $5 \mu\text{F}$ and an effective resistance of $4 \times 10^5 \Omega$. At $t = 0$, the potential difference across the plates is zero but is increasing at the constant rate of 2000 V/s . The displacement current I_D is $1 \times 10^{-n} \text{ A}$. Find n .
39. A plane electromagnetic wave with an intensity of 220 W/m^2 is incident normal to a flat plate of radius 30 cm . If the plate absorbs 60% and reflects 40% of the incident radiation, the momentum transferred to it in 5 min is $8.71 \times 10^{-n} \text{ kg m/s}$. Find n .
40. At a distance of 6 m from a monochromatic (single wavelength) point source, the amplitude of the electric field is 10 V/m . The amplitude of the magnetic field is nearly $\lambda \times 10^{-8} \text{ T}$. Find λ .
41. The distance from a 100 W isotropic power source at which electric field of electromagnetic wave will be $E_m = 15 \text{ V/m}$ is approximately $x \text{ cm}$. Find x .
42. Energy from the sun falls on the Earth at a rate of 1353 W/m^2 which is known as the solar constant. The rms value of the electric field of sunlight is $E_{rms} \frac{V}{m}$. Find E_{rms} .
43. A plane wave of intensity $I = 0.70 \text{ W/cm}^2$ illuminates a sphere with ideal mirror surface of radius 5 cm . The force experienced by the sphere is $P \times 10^{-2} \text{ N}$. Find P .
44. An isotropic point source emits light of wavelength 589 nm . The power of the source is 10 W . The mean density of the flow of photons at a distance 2 m from the source is $6 \times 10^P \text{ cm}^{-2} \text{ s}^{-1}$. Find P .
45. The energy of visible light corresponding to its mean wavelength is about xeV . The value of x is:
46. The electric field on an e.m. wave is given by

$$E = 2.5 \cos(2\pi \times 10^{-6} t - \pi \times 10^{-2} x) \text{ Vm}^{-1}$$
 where t and x are in second and metre respectively. The wavelength of wave is found to be $100x$. The value of x is:
47. An electromagnetic wave is represented as

$$E = 38.8 \sin(1.2 \times 10^7 z - 3.6 \times 10^{15} t) \text{ Vm}^{-1}$$
 The average intensity of the beam in watt/m^2 is:
48. A radio can tune to a station broadcasting at 7.5 MHz . The wavelength of the wave is $10x \text{ metre}$. The value of x
49. The electric field amplitude of e.m. waves is 120 N/C . The peak value of magnetic field is $n \times 10^{-7} \text{ tesla}$. The value of n is :